

Volume-uniform electric-shell isolation, exact shared-link hopping, and a homological carrier in strong-coupling $SU(N)$ Hamiltonian lattice gauge theory

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Abstract

We study the Kogut–Susskind Hamiltonian for pure $SU(N)$ gauge theory on a periodic cubic spatial lattice at strong coupling. In the normalised charge-odd fundamental-plaquette sector, an explicit order-two Weingarten calculation gives the exact coefficient of hopping between adjacent plaquettes that share one link,

$$t_N = \frac{2N(N^2 - 4)}{(N^2 - 1)(2N^2 - 1)(4N^2 - 9)} > 0, \quad N \geq 3.$$

The oriented cellular boundary $B = \partial_2^\dagger$ fixes the corresponding local geometry. A Casimir and short-cycle argument proves the volume-uniform electric window

$$\text{spec}(H_E|_{\text{trivial flux}}) \cap [0, 5C_F/2) = \{0, 2C_F\},$$

with the $2C_F$ eigenspace exactly the oriented one-plaquette shell and an external margin at least $C_F/2$. A link-centre classification then proves that the four shared-link fusion channels exhaust every second-order off-diagonal process between distinct plaquettes in this charge-odd shell for the untruncated plaquette perturbation. Consequently the global order- u^2 symbol is $t_N u^2 B(k) B(k)^\dagger$. Its spectrum is $\{0, t_N u^2 q, t_N u^2 q\}$ up to a common scalar, with $q = 4 \sum_j \sin^2(k_j/2)$, and its flat carrier $\ker \partial_2$ has dimension $L^3 + 2$. A normalized Fourier sum of radius-one cube boundaries is the exact carrier at every nonzero allowed momentum; on nested volumes its order- u^2 splitting is L -independent, although one local cube seed has only $O(L^{-3})$ weight at a fixed momentum. A supplied $SU(3)$ two-cube calculation reports the same coefficient channel by channel and localises the $\mathcal{B}$ -cutoff dependence to one symmetry orbit; those finite-precision contractions are corroboration, not an independent derivation. A later detached replay on the same open prism yields nested radius-two finite- u Ritz spectra and the leading incremental order- u^4 and order- u^5 effects of entering Q_2 and restoring its retained magnetic block. These are finite-volume nested-model differences, not complete fourth- or fifth-order coefficients. Exact integer geometry also gives the closed-cube shell $A_1^{-} \oplus T_1^{+-} \oplus E^{--}$ with Gram eigenvalues 0, 4, 6.

New in this revision, the same three inputs — the fusion table, the plaquette-loop energies and the centre-charge census — give the second-order *scalar* in closed form at every rank: the on-site coefficient is the reduced same-face route plus $1/C_F + 12\ell_N$, with $\ell_N = A_N + B_N + 1/C_F$, the torus size cancelling exactly, and the flat scalar subtracts $4t_N$. For $N \geq 5$ the charge-odd tower is $-12N/((N^2 - 1)(N^2 - 9))$; $N = 3$ and $N = 4$ are exceptional and stated separately, and the $N = 3$ specialisation reproduces every registered $SU(3)$ value and an independent

engine’s second-order table. The same bookkeeping proves that the charge-even second-order Bloch symbol is ℓ_N times the *unsigned* incidence: the vacuum-mediated route, which reaches every pair of plaquettes, is cancelled on every link-disjoint pair by the doubly-excited exchange route. The identification of the unsigned comparison operator with the charge-even sector, a hypothesis in earlier revisions, is therefore a theorem at order u^2 and conditional only at order u^3 ; the charge-even Γ -point splitting $12\ell_N$, $-22/51$ at $N = 3$, is unconditional and is what Hamer’s 0^{++} series corroborates. The even split of the link-resolved two-cube census, previously conjectured, is proved at every rank. The organized upstream corpus records a cold exact replay of the retained $SU(3)$ third-order chain, although that microscopic replay is not bundled here.

At fourth order the disputed off-axis coefficient C_{shp} is now located exactly: the saved kernel is a polynomial in the two cellular Laplacians plus one operator, whose weight is C_{shp} , so the dispute is two shared-link amplitudes. Three amplitudes have been recomputed from outside both pipelines by cluster expansion: the two-hop weight and the in-plane and normal shared-link amplitudes agree with the historical kernel exactly, the cold rival is wrong on all three, and the rotation amplitude agrees with neither, which puts a third value of C_{shp} on record without promoting any. The obstruction theorem stands: no Γ -point or axial datum can decide it. The rational and polynomial identities the paper rests on are collected as ninety-five Lean 4 theorems, proof-checked with no omitted proof. The exact coefficient results are finite-volume and finite-order. In addition, for $SU(3)$ the periodic-boundary version explicitly permitted by a published spectral-localisation theorem gives a volume-uniform, nonperturbative finite-volume Riesz island for the complete neutral charge-odd shell at sufficiently small local coupling; the threshold is existential and the result remains at fixed lattice spacing. No continuum glueball or Yang–Mills mass-gap claim is made.

1 Introduction

Hamiltonian strong-coupling expansions organise lattice gauge dynamics around the electric vacuum and its flux excitations [4, 5]. In three spatial dimensions a fundamental plaquette excitation carries one of three face orientations, and at second order neighbouring faces communicate through a shared link, so the projected problem resembles a three-orbital tight-binding model. The signs of those matrix elements are fixed by the orientation of the plaquette boundary and by charge conjugation, and the resulting interference is the central object here.

Flat bands generated by incidence or Gram matrices have a long history: local topology and signed or line-graph constructions [3, 18, 22, 26, 28], compact localised states and their failure to span singular bands on a torus [25, 27], and supersymmetric incidence factorisations [31] are all established. Closely related closed-membrane structures also occur in two-form spin liquids [32]. A recent classification of exactly flat bands protected by local graph topology [36] reaches extensive face-graph degeneracy from an *unoriented* face–vertex incidence, where rank–nullity alone gives $|F| - |V|$ flat bands. That is a different mechanism from the one here, and the difference is measurable: on the signed face–edge operator of Section 3 the corresponding index is exactly 0 and bounds nothing, while both kernels equal $L^3 + 2$ — homology, not counting. It is recorded as a distinction, not an identification. We do not claim that incidence flatness, cellular compact localised states, or homological counting are new in isolation.

The new dynamical content is local and exact. Starting from non-Abelian $SU(N)$ Hamiltonian lattice gauge theory, the shared-link matrix element in the charge-odd fundamental-plaquette sector factorises into a colour coefficient and an oriented edge–face Gram entry. Representation theory

fixes the former and the cubical chain complex fixes the latter,

$$\begin{aligned} \text{colour dynamics} &\longrightarrow t_N, \\ \text{cellular geometry} &\longrightarrow BB^\dagger. \end{aligned}$$

What is proved, and what remains conditional

The principal second-order statements are self-contained here, in both charge sectors; where a higher-order operator's identification with a physical sector is conditional, that condition is stated separately. Every rational or polynomial identity displayed below is also proof-checked: a statement carrying a `[Lean: name]` tag is proved in Lean 4 from explicit definitions, and Appendix H lists all ninety-five such theorems. The tag certifies the arithmetic of the statement it sits beside and nothing about the physics that produced it.

1. For every $N \geq 3$ the shared-link channel weights are d_ρ/N^2 , derived from the order-two Weingarten values (Theorem 15). This was previously an isotropy *hypothesis*, and everything in Section 4 rests on it.
2. For every $N \geq 3$ and $L \geq 3$, the trivial-flux electric spectrum below $5C_F/2$ is exactly $\{0, 2C_F\}$; the charge-odd part of the $2C_F$ eigenspace is exactly the $3L^3$ -dimensional one-plaquette shell and its external free margin is at least $C_F/2$ (Theorem 1). Every off-diagonal second-order intermediate is then either a vanishing same-face route or one of the four adjacent shared-link channels (Lemma 22). Theorem 23 gives their explicit projector cross matrix elements and coefficient t_N , so the global symbol $t_N u^2 BB^\dagger$ follows without a retained-shell or process-completeness hypothesis (Corollary 24).
3. The Bloch spectrum of $BB^\dagger$ is $\{0, q, q\}$ (Theorem 8), and on the periodic L^3 complex the zero-mode count is $L^3 + 2$ (Theorem 9). The two are the same statement, by a sum rule over the momentum grid (Theorem 11). At fixed nonzero nested momentum the carrier and its splitting are volume-independent, with the exact locality–momentum tradeoff given in Proposition 13.
4. The truncation that keeps only the singlet and $\Lambda^2 F$ routes projects the hopping to $w_{\Lambda^2 F} - w_1 = -1/12$, and what it omits is $w_{\text{Sym}^2 F} - w_{\text{Adj}} = 14/153$; the two sum to t_3 (Corollary 33). This is arithmetic on (35) and needs no input beyond Theorem 15.
5. The defined unsigned-incidence comparison operator has the cubic $\mu(\mu - p)^2 = 4\hat{a}_1\hat{a}_2\hat{a}_3$ with $p + q = 12$ (Theorem 40); its range is exactly $[-4, 12]$, the top attained only at Γ and the floor on the three planes $k_j = \pi$, triple only at R (Proposition 41). Conditional on identifying this comparison operator with the complete linked charge-even symbol, its Γ -point splitting is $12t_{r,+}$ (Proposition 43). That identification is not derived here; the vacuum-mediated all-pairs term must first be reorganised or cancelled.
6. On the closed cube surface the charge-odd shell is $A_1^{--} \oplus T_1^{+-} \oplus E^{--}$, with G -eigenvalues 0, 4, 6; the labels follow from the rotation action rather than being assigned, the A_1^{--} level is the fundamental class of S^2 , and it sits at the scalar for every hopping coefficient (Proposition 36). This is integer linear algebra and takes no input from any delivery.
7. The retained Γ /axis corpus cannot identify the disputed fourth-order off-axis coefficient (Theorem 50), and an explicit second witness exhibits the non-identifiability by construction (Proposition 52).

8. For every $N \geq 3$ the second-order scalar is closed form (Theorem 29): the on-site coefficient is $\sigma_N + 1/C_F + 12\ell_N$ with σ_N the reduced same-face route, the torus size cancelling, and the charge-odd tower $\sigma_N^- + 1/C_F$ is $-12N/((N^2 - 1)(N^2 - 9))$ for $N \geq 5$. Revision 5 gave this scalar at $N = 3$ only, as an upstream identification.
9. For every $N \geq 3$ the charge-even second-order Bloch symbol is $\ell_N \mathcal{N}(k) \mathcal{N}(k)^\dagger$ plus a scalar, with $\mathcal{N}$ the unsigned incidence (Theorem 27): the vacuum-mediated route is cancelled on every link-disjoint pair by the doubly-excited exchange route — the $N = 3$ cancellation is the corpus's, the all-rank operator statement is new. The symbol identification of Section 6 is therefore a theorem at order u^2 and a hypothesis only at order u^3 .
10. The even split of the link-resolved census is the orientation bookkeeping of Theorem 23 (Proposition 32), at every rank; revision 5 stated it as a conjecture.
11. The two recorded fourth-order kernels are one Hodge polynomial plus one operator whose weight is C_{shp} (Reported result 53), so the dispute is two shared-link amplitudes; three amplitudes recomputed from outside both pipelines (Reported result 54) decide the standing of the two records without deciding the coefficient, and put a third value on record.

Four families of statements depend on computational artifacts. The radius-two parents and detached verifier are bundled with this edition; the earlier raw class contractions, symbolic history certificates and microscopic rank-sweep artifacts are not. Repeated assertions in synthesis documents are not treated as independent reproductions. The organized upstream authority corpus records a cold exact 251/251 replay for the retained third-order operator chain; that is stronger than an unchecked delivery, but it is still cited here as external computational evidence because this release does not replay the raw chain.

12. A supplied two-cube delivery reports six link-irrep channel coefficients at $N = 3$, as rational reconstructions of finite-precision contractions. Conditional on those six numbers, they are the four fusion-channel weights individually — channel by channel and not merely in the sum (Reported result 31) — and the $\mathcal{B} = 4$ truncation of that space retains exactly the adjacent shared-link channels the published link cutoff of item 4 retains, so the two land on the same projected hopping — the same channels, not the same scheme. Conditional on the delivered diagonals, the symmetry analysis of Proposition 35 applies to them: the orbit structure itself is exact and takes no delivered input. The $\mathcal{B} = 6$ two-cube build was not reproduced here.
13. A later hash-sealed release on the same open prism reports nested charge-odd and charge-even radius-two Ritz operators. Its detached verifier reconstructs the spectra and the leading $K_{2,\star} - K_1$ and $K_{2,\text{full}} - K_{2,\star}$ coefficients from the sealed parity-sector parents. This establishes numerical consistency relative to those inputs, not convergence in action radius or a physical glueball gap (Reported result 39).
14. For $SU(3)$, the frozen upstream corpus records that all candidate third-order lifter classes vanish and gives the scalar and shared-link coefficients of Section 7. The census is three lifter classes evaluated over 32 five-trace numerator patterns each, 96 contractions in total, within a retained chain that passed 251/251 cold exact gates upstream. This compact verifier checks the resulting rational assembly but not those microscopic contractions. If the census is exhaustive and the linked operator ledger complete, the effective Hamiltonian stays in the incidence algebra through order u^3 .
15. Supplied ledgers report the all-rank cube formula $\alpha_N = 640/[N(N^2 - 1)^3]$ and its equality with the complete axial coefficient for $N = 3, \dots, 12$. Conditional on those calculations, the

$SU(3)$ carrier acquires nonzero axial dispersion at order u^4 . The complete off-axis operator remains open — and by Theorem 50 it cannot be closed by any Γ or axial datum.

These coefficient statements concern the effective Hamiltonian projected to the degenerate one-plaquette manifold. Virtual multi-plaquette states are retained in the resolvents, but they do not by themselves solve the external spectral problem. At $k = 0$ the three incidence branches touch and the finite-volume separation collapses as L^{-2} . Theorem 4 later proves nonperturbative finite-volume persistence of the *complete* three-component cluster in an existential small- $|u|$ domain; it does not prove convergence of the displayed coefficient series, persistence of one internal sheet, identification with the lightest physical glueball, or a continuum mass gap.

2 Hamiltonian and projected sector

Let $\Lambda_L = (\mathbb{Z}/L\mathbb{Z})^3$, $L \geq 3$, be a cubic spatial lattice with periodic boundary conditions. In units of the electric coupling the Kogut–Susskind Hamiltonian is

$$H_\beta = \frac{1}{2} \sum_{\ell} C_2(\ell) + \beta_N \sum_p \left(1 - \frac{1}{N} \operatorname{Re} \operatorname{Tr} U_p\right). \quad (1)$$

Dropping the additive constant,

$$H = H_E - uV, \quad V = \sum_p (\chi_p + \bar{\chi}_p), \quad u = \frac{\beta_N}{2N}, \quad (2)$$

with $\chi_p = \operatorname{Tr} U_p$. For $SU(3)$, $u = \beta_3/6$. The definition $u = \beta_N/(2N)$, not the $SU(3)$ -specific equality, is the all-rank convention; the archived line $Y = 2\beta/3 = 4u$ is a definition-label erratum and the printed coefficients are already in u . Reading it as a coordinate would demand dividing the order- r coefficient by 4^r , which misses the printed tables by exactly 16 at u^2 and 64 at u^3 .

All quoted excitation energies use the linked, vacuum-subtracted operator $H_{\text{exc}}(u) = H(u) - E_{\text{vac}}(u)I$. This changes only the scalar ledger; incidence hopping and carrier shape are unaffected.

For a finite-volume analytic family $A_L(u)$, write $[u^r]A_L$ for its r th Taylor coefficient at $u = 0$ and $T_{\leq r}A_L(u) = \sum_{j=0}^r u^j [u^j]A_L$. Every statement below written with $[u^r]$ or $T_{\leq r}$ is an equality of finite-volume Taylor coefficients. It asserts no remainder estimate uniform in L unless such an estimate is stated separately.

The electric vacuum $|0\rangle$ carries the trivial representation on every link. For $N \geq 3$ the normalised one-plaquette states of definite charge conjugation are

$$|p, \pm\rangle = \frac{\chi_p \pm \bar{\chi}_p}{\sqrt{2}} |0\rangle, \quad (3)$$

of unit norm by character orthogonality. Each of the four boundary links carries the fundamental or antifundamental representation, so

$$E_{F,N} = 4\left(\frac{1}{2}C_F\right) = 2C_F = \frac{N^2 - 1}{N}, \quad C_F = \frac{N^2 - 1}{2N}. \quad (4)$$

This external energy is the input to the resolvent denominators of Section 4, where it is checked together with them. For $SU(2)$ complex conjugation is an inner gauge transformation and the charge-odd state vanishes; this is why the construction begins at $N = 3$. The same fact appears again in Section 4: $\Lambda^2 F$ is the singlet exactly at $N = 2$, which is where t_N vanishes. The analytic continuation of the unsigned-channel coefficient is nonzero there, which locates the $N^2 - 4$ zero in the

channel difference rather than in the incidence. Because the charge-conjugation split degenerates for $SU(2)$, this continuation is not asserted to be an $SU(2)$ charge-even spectrum.

For each coordinate direction choose a dual noncontractible cut Σ_j , and fix one orientation for every geometric edge. For $z \in \mathbb{Z}_N$, let $Z_j(z)$ act by

$$U_e \longmapsto z^{\epsilon(e, \Sigma_j)} U_e,$$

where the signed intersection number $\epsilon(e, \Sigma_j) \in \{-1, 0, +1\}$. These unitaries commute with H_E , while charge conjugation obeys $\mathcal{C}Z_j(z)\mathcal{C}^{-1} = Z_j(z^{-1})$. The simultaneous $+1$ eigenspace of every $Z_j(z)$ is therefore charge-conjugation invariant; call it the trivial electric one-form sector. A loop of N -ality k and winding vector w has eigenvalue z^{kw_j} , so contractible plaquettes lie in this sector while a once-winding fundamental loop does not.

Theorem 1 (Uniform first electric spectral window). *Let $N \geq 3$ and $L \geq 3$. Let the pure-gauge physical Hilbert space be*

$$\mathcal{H}_{\text{phys}} = L^2\left(SU(N)^{E(\Lambda_L)}\right)^{SU(N)^{V(\Lambda_L)}},$$

and define the trivial electric one-form sector and its charge-odd part by

$$\mathcal{H}_{\text{phys}}^{(0)} = \{ \Psi \in \mathcal{H}_{\text{phys}} : Z_j(z)\Psi = \Psi \ \forall j, z \}, \quad \mathcal{H}_{\text{phys}}^{(0,-)} = \{ \Psi \in \mathcal{H}_{\text{phys}}^{(0)} : \mathcal{C}\Psi = -\Psi \}.$$

Normalise the electric Hamiltonian on the Peter–Weyl spin-network basis by

$$H_E \Psi_{\{R_e\}, \{\iota_v\}} = \frac{1}{2} \sum_e C_2(R_e) \Psi_{\{R_e\}, \{\iota_v\}}.$$

Then, with $E_{F,N} = 2C_F$,

$$\text{spec}\left(H_E|_{\mathcal{H}_{\text{phys}}^{(0)}}\right) \cap \left[0, \frac{5}{2}C_F\right) = \{0, 2C_F\}. \quad (5)$$

The zero eigenspace is the vacuum line, while the $2C_F$ eigenspace is the direct sum, over elementary plaquettes, of their two oriented fundamental Wilson-loop lines. Consequently

$$\text{Ran}\left(\mathbf{1}_{\{E_{F,N}\}}(H_E)\Big|_{\mathcal{H}_{\text{phys}}^{(0,-)}}\right) = \text{span}\{|p, -\rangle : p \in F_2(\Lambda_L)\}. \quad (6)$$

and this charge-odd eigenspace has dimension $3L^3$. In particular the full electric shell is separated from the rest of the trivial-flux spectrum by

$$\text{dist}\left(2C_F, \text{spec}(H_E|_{\mathcal{H}_{\text{phys}}^{(0)}}) \setminus \{2C_F\}\right) \geq \frac{1}{2}C_F, \quad (7)$$

uniformly in L .

Proof. Normalise the invariant form so every root has squared length 2, and write a nonzero dominant highest weight as $\lambda = \sum_{i=1}^{N-1} a_i \omega_i$. If $a_i > 0$, put $\lambda = \omega_i + \mu$, with μ dominant. Positivity of the inverse A_{N-1} Cartan matrix gives

$$C_2(\lambda) - C_2(\omega_i) = \frac{1}{2}((\mu, \mu) + 2(\omega_i + \rho, \mu)) \geq 0,$$

with equality only for $\mu = 0$. Moreover

$$C_2(\omega_i) = \frac{i(N-i)(N+1)}{2N}, \quad C_2(\omega_i) - C_F = \frac{N+1}{2N}(i-1)(N-i-1) \geq 0.$$

Thus every nontrivial irrep satisfies $C_2(R) \geq C_F$, with equality only for F and $\bar{F}$.

We need one further all-rank shelf. If a dominant weight λ is neither ω_1 nor ω_{N-1} , then one of three cases occurs: an interior Dynkin coefficient is nonzero; an endpoint coefficient is at least two; or both endpoint coefficients are nonzero. Monotonicity of the preceding Casimir difference therefore bounds $C_2(\lambda)$ below by, respectively, $C_2(\omega_2)$ (for $N \geq 4$), $C_2(2\omega_1)$ or its conjugate, or $C_2(\omega_1 + \omega_{N-1})$. The three comparisons with the required shelf are

$$C_2(\omega_2) - \frac{5}{4}C_F = \frac{(N+1)(3N-11)}{8N} > 0 \quad (N \geq 4), \quad (8)$$

$$C_2(2\omega_1) - \frac{5}{4}C_F = \frac{(N-1)(3N+11)}{8N} > 0, \quad (9)$$

$$C_2(\omega_1 + \omega_{N-1}) - \frac{5}{4}C_F = \frac{3N^2 + 5}{8N} > 0. \quad (10)$$

For $N = 3$ there is no interior coefficient, so the final two cases are exhaustive. Hence every irrep other than $\mathbf{1}, F, \bar{F}$ satisfies

$$C_2(R) > \frac{5}{4}C_F. \quad (11)$$

We also need the centre-neutral refinement. Zero N -ality is equivalent to λ lying in the A_{N-1} root lattice. Write $\lambda = \sum_i m_i \alpha_i$, $m_i \in \mathbb{Z}$, and $a_i = (\lambda, \alpha_i) \geq 0$. Since $m = A^{-1}a$ and every entry of A^{-1} is strictly positive, a nonzero dominant λ has $m_i \geq 1$ for every i . Hence $(\lambda, \rho) = \sum_i m_i \geq N-1$. Every nonzero root-lattice vector has squared norm at least 2, so

$$C_2(\lambda) = \frac{1}{2}((\lambda, \lambda) + 2(\lambda, \rho)) \geq N. \quad (12)$$

Consider a non-vacuum spin-network component with energy $E < 5C_F/2$. If S is the set of links carrying nontrivial irreps, then

$$\sum_{e \in S} C_2(R_e) = 2E < 5C_F,$$

so $|S| \leq 4$. Gauge invariance forbids support degree one, because a single nontrivial irrep has no invariant vector. The torus graph is simple for $L \geq 3$; therefore a nonempty connected support has at least a three-cycle. Two components would require at least six links, and a connected simple graph of minimum degree two with at most four edges is a single C_3 or C_4 .

A three-cycle occurs only for $L = 3$ and is a straight loop winding once around one coordinate direction. Indeed, lift its three unit steps to $\mathbb{Z}^3$: their sum is Lw , and the length and parity constraints force $L = 3$, $w = \pm e_j$. Degree-two intertwiners transport one irrep R around the loop. Trivial one-form flux forces its N -ality to vanish, and (12) gives

$$H_E = \frac{3}{2}C_2(R) \geq \frac{3N}{2} > \frac{5(N^2-1)}{4N} = \frac{5}{2}C_F,$$

a contradiction.

The support is therefore a four-cycle. Degree-two intertwiners transport one irrep R (with the dual on oppositely oriented links) around it, so its energy is $2C_2(R)$. Lift its four unit steps to

$\mathbb{Z}^3$. For $L \geq 5$ the winding is zero; at $L = 3$ a nonzero winding violates step parity; and at $L = 4$ the only nonzero possibility is four collinear steps, a once-winding loop. Trivial flux then forces R to have zero N -ality, so (12) gives $H_E = 2C_2(R) \geq 2N > 5C_F/2$, again a contradiction. With zero winding, simplicity forces, up to cyclic permutation, reversal, coordinate relabelling, and independent sign choices, the cyclic step sequence $\varepsilon_i e_i, \varepsilon_j e_j, -\varepsilon_i e_i, -\varepsilon_j e_j$, with $i \neq j$: the boundary of one elementary plaquette. Equation (11) now forces $R \in \{F, \bar{F}\}$, and its energy is $2C_F$. This proves (5); the empty support gives the unique vacuum.

At each degree-two vertex Schur's lemma makes the intertwiner unique. The two orientations are $\chi_p|0\rangle$ and $\bar{\chi}_p|0\rangle$, exchanged by charge conjugation, so their negative eigenspace is the line $|p, -\rangle$. Conversely every such state is physical, contractible, and has energy $4(C_F/2)$. Peter–Weyl orthogonality separates distinct plaquettes, of which there are $3L^3$. Finally, the vacuum lies $2C_F$ below the shell and (5) places every higher eigenvalue at least $C_F/2$ above it, proving (7). $\square$

The preceding theorem is enough to identify the physical content of a small-coupling spectral patch. The volume-uniform persistence of that patch is not obtained by bounding the norm of the extensive perturbation. Instead we use the local spectral enclosure of Yarotsky [1].

Proposition 2 (Periodic kinematic enclosure and physical-sector descent). *Let*

$$\mathcal{K}_L = \bigotimes_{x \in \Lambda_L} \mathcal{K}_x, \quad \mathcal{K}_x = \bigotimes_{i=1}^3 L^2(SU(3), dU_{x,i}),$$

be the unconstrained link Hilbert space, grouped by the three positively oriented links issuing from each vertex. On the common domain of the electric Hamiltonian let

$$H_L^{\text{kin}}(u) = H_{E,L}^{\text{kin}} - uV_L, \quad \tilde{H}_L^{\text{kin}}(u) = H_L^{\text{kin}}(u) - E_{0,L}^{\text{kin}}(u)I.$$

Invoke Theorem 1 of [1] with the periodic boundary condition explicitly permitted immediately after its equation (3), and let $c_1, c_2 > 0$ be its constants for the fixed interaction support below. If

$$27|u| < c_1, \quad \lambda = 27c_2|u|,$$

then the kinematic ground state is nondegenerate and

$$\text{spec } \tilde{H}_L^{\text{kin}}(u) \subset \bigcup_{a \in \text{spec } H_{E,L}^{\text{kin}}} [a(1 - \lambda), a(1 + \lambda)], \quad (13)$$

with constants independent of L . The unique kinematic ground state is gauge invariant, trivial under the electric one-form symmetries, and charge-conjugation even. Hence

$$E_{0,L}^{\text{kin}}(u) = E_{0,L}^{\text{phys}}(u),$$

and every Gauss, one-form and charge-conjugation sector is reducing for the Hamiltonian and its resolvent. In particular, (13) remains a valid spectral enclosure after restriction to $\mathcal{H}_{\text{phys}}^{(0,-)}$.

Proof. Rescale the kinematic Hamiltonian by $3/2$. Its free part is a tensor-product sum

$$\hat{H}_{E,L} = \sum_x \hat{h}_x, \quad \hat{h}_x = \frac{3}{4} \sum_{i=1}^3 C_2(x, i).$$

The ground vector of each $\widehat{h}_x$ is the tensor product of three constant functions and is nondegenerate. Since the smallest nonzero $SU(3)$ Casimir is $C_F = 4/3$, its local gap is $\frac{3}{4}C_F = 1$. The cited theorem allows infinite-dimensional cell spaces and unbounded onsite Hamiltonians.

Assign to x the three positively oriented plaquettes based at x and put

$$\widehat{\phi}_x(u) = -\frac{3u}{2} \sum_{1 \leq i < j \leq 3} (\chi_{p(x;i,j)} + \bar{\chi}_{p(x;i,j)}).$$

Each summand is supported on the three cells $x, x + e_i, x + e_j$; the union of these supports is fixed, independent of L . Multiplication by a plaquette character obeys

$$\|\chi_p + \bar{\chi}_p\| = \sup_{U \in SU(3)} |\mathrm{Tr} U + \overline{\mathrm{Tr} U}| \leq 6,$$

and therefore

$$\sup_x \|\widehat{\phi}_x(u)\| \leq \frac{3}{2}|u| 36 = 27|u|.$$

The periodic finite-volume version of the cited theorem now gives the unique ground state and its multiplicative spectral enclosure in the rescaled units. Dividing all energies by $3/2$ proves (13).

Every local gauge transformation commutes with $H_L^{\mathrm{kin}}(u)$ and hence acts on its unique ground-state line. The group $SU(3)^{V(\Lambda_L)}$ has no nontrivial one-dimensional unitary representation, so this line is gauge invariant. Along $t \mapsto tu$, $0 \leq t \leq 1$, the same smallness condition holds and the rank-one ground projection is norm-continuous. The one-form and charge-conjugation eigenvalues lie in finite sets and therefore cannot change along that path. At $t = 0$ the electric vacuum has trivial one-form charge and is charge-conjugation even, so the interacting ground state does as well. It is thus an admissible physical vector and the kinematic and physical ground energies coincide.

Finally, Gauss invariance, trivial one-form charge and charge parity define closed reducing subspaces. On a reducing subspace the restricted resolvent is the restriction of the full resolvent whenever the spectral parameter is outside the full spectrum. This proves the sector descent. $\square$

Remark 3 (Exact periodicity input). No theorem for arbitrary bounded-degree quotient graphs is inferred here. The only periodic input is the explicit statement after equation (3) of [1] that other boundary conditions may be used; the rest of Proposition 2 is the model-specific tensor-product and symmetry reduction proved above.

Theorem 4 ($SU(3)$ nonperturbative finite-volume Riesz island). *For periodic L^3 tori with $L \geq 3$, let*

$$\tilde{H}_L(u) = H_L(u) - E_{0,L}(u)I, \quad \lambda = 27c_2|u|,$$

where $c_1, c_2 > 0$ are the range-dependent constants in the spectral localisation theorem of [1]. If

$$27|u| < c_1, \quad \lambda < \frac{1}{33}, \tag{14}$$

then the part of $\mathrm{spec} \tilde{H}_L(u)$ descending from the free energy $E_F = 8/3$ is separated from every other free-energy patch by

$$\delta_-(u) \geq \frac{1 - 15\lambda}{3}, \quad \delta_+(u) \geq \frac{1 - 33\lambda}{6}, \tag{15}$$

uniformly in L . The circle

$$\Gamma_F : \quad |z - 8/3| = \rho, \quad \rho = \frac{1 - \lambda}{12}, \tag{16}$$

encloses that patch and obeys

$$\sup_{z \in \Gamma_F} \|(\tilde{H}_L(u) - z)^{-1}\| \leq \frac{12}{1 - 33\lambda}. \quad (17)$$

Restricted to $\mathcal{H}_{\text{phys}}^{(0,-)}$, its Riesz projection has rank exactly $3L^3$.

Proof. Apply Proposition 2. Its enclosure is first obtained on the genuine tensor-product space and only then restricted to the physical neutral charge-odd sector; no tensor-product structure of the Gauss-invariant Hilbert space is assumed.

For an $SU(3)$ irrep (p, q) , its one-link electric energy in units of $1/6$ has numerator $p^2 + q^2 + pq + 3p + 3q$. The nonzero one-link numerators below 18 are 4, 9, 10, 16, so their additive semigroup has consecutive attainable values 14, 16, 17 around the target 16. Consequently the lower and upper separations between the target interval and its nearest competitors are exactly the right-hand sides of (15); the upper one is positive precisely under the second condition in (14). The target interval has half-width $8\lambda/3$. Direct subtraction from (16) gives the minimum contour clearance $(1 - 33\lambda)/12$, proving (17).

The contour projection commutes with gauge transformations, charge conjugation and the electric one-form symmetries by the proposition. The same contour works along $t \mapsto tu$, $0 \leq t \leq 1$. In finite volume the electric Hamiltonian has compact resolvent and V_L is bounded, so the restricted contour projection is finite-rank and norm-continuous. Its rank is therefore constant. At $u = 0$ Theorem 1 identifies that rank as $3L^3$. $\square$

Remark 5 (Exact boundary of the nonperturbative result). Theorem 4 transports the complete three-component shell, not one internally isolated band. The constants c_1, c_2 are not numerical, the result is confined to fixed-spacing strong coupling, and the separate one-particle theorem of [1] is not invoked: its nondegenerate one-site, nonresonance hypothesis fails for this composite, degenerate plaquette shell. The theorem therefore supplies an interacting finite-volume Riesz cluster, not an infinite-volume particle or Wilson-source residue.

Remark 6 (Proof-checked arithmetic). The finite arithmetic the two theorems delegate to is proof-checked (Appendix H): the three shelf comparisons (8)–(10) as identities of rational functions [Lean: casimir_shelf]; the factorisation $C_2(\omega_i) - C_F = \frac{N+1}{2N}(i-1)(N-i-1)$ [Lean: casimirWeight_sub_fundamental]; the window clearances of the zero- N -ality loops with denominators cleared, $8N^2 - 5(N^2 - 1) = 3N^2 + 5$ and $6N^2 - 5(N^2 - 1) = N^2 + 5$ [Lean: zero_nality_clears_window]; the enumeration of the $SU(3)$ one-link numerators below 18 [Lean: su3_numerators_below_eighteen]; the semigroup gap at 15 with 14 and 17 attained [Lean: su3_semigroup_gap]; the two separations (15) [Lean: riesz_separations]; and the three contour clearances $(1 - 33\lambda)/12$, $(1 - 33\lambda)/12$, $(3 - 27\lambda)/12$ behind (17) [Lean: riesz_contour_clearances]. The representation theory, the cycle census and the operator statements are not formalised, and the cited enclosure theorem is not either.

Remark 7 (The two small-torus coincidences). At $L = 4$ a once-winding fundamental loop has energy $2C_F$ for every $N \geq 3$; trivial one-form flux is therefore load-bearing at every rank, not only at $SU(4)$. The genuinely rank-four coincidence occurs instead at $L = 3$: an $SU(4)$ loop in $\Lambda^2 F$ has $C_2 = 5/2$ and energy $3C_2/2 = 15/4 = 2C_F$, but its N -ality is two, so the same flux condition excludes it (and its character is charge-even). These are volume effects, not a continuum statement.

Set P_- equal to the projector in (6) and $Q_- = 1_- - P_-$, where 1_- is the identity on $\mathcal{H}_{\text{phys}}^{(0,-)}$. Both H_E and V preserve this space: H_E commutes with $\mathcal{C}$ and every $Z_j(z)$, while every contractible

plaquette term in V is charge-even and centre-neutral. No degenerate $E_{F,N}$ direction is omitted, and degenerate perturbation theory gives

$$H_{\text{eff},-} = P_- H_E P_- + u H_1 + u^2 H_2 + u^3 H_3 + \cdots \quad (18)$$

with every H_r linked, vacuum-subtracted, and carrying the appropriate Q -space resolvents.

3 Oriented incidence and the exact carrier

The periodic cubical complex has the cellular chain sequence $C_3 \xrightarrow{\partial_3} C_2 \xrightarrow{\partial_2} C_1$ with $\partial_2 \partial_3 = 0$. One-plaquette amplitudes live in C_2 and their signed leakage into links is $\partial_2 c$: a cellular closed-surface condition, not the microscopic vertex Gauss law, which the Wilson-loop basis already satisfies through $\partial_1 \partial_2 = 0$.

Writing $d_j = e^{ik_j} - 1$ and $q(k) = \sum_j |d_j|^2 = 4 \sum_j \sin^2(k_j/2)$, the face-to-link Bloch incidence matrix is

$$B(k) = \partial_2(k)^\dagger = \begin{pmatrix} d_2 & -d_1 & 0 \\ d_3 & 0 & -d_1 \\ 0 & d_3 & -d_2 \end{pmatrix}. \quad (19)$$

Theorem 8 (Incidence factorisation). *With $w(k) = (\bar{d}_3, -\bar{d}_2, \bar{d}_1)^\top$,*

$$BB^\dagger = qI - ww^\dagger, \quad B^\dagger w = 0, \quad \|w\|^2 = q,$$

and therefore $\text{spec}(BB^\dagger) = \{0, q, q\}$. With $S = BB^\dagger - 4I$ the signed shared-link adjacency, $\text{spec } S(k) = \{-4, q - 4, q - 4\}$.

Proof. Multiplying (19) by its adjoint gives the first identity entry by entry. Since $\|w\|^2 = q$, the rank-one operator $qI - ww^\dagger$ annihilates w and acts as qI on its orthogonal complement. Subtracting $4I$ gives the adjacency spectrum. $\square$

The factorisation, $B^\dagger w = 0$ and $\|w\|^2 = q$ are proof-checked over $\mathbb{C}$ for arbitrary d_1, d_2, d_3 , entry by entry [Lean: bloch_factorisation], [Lean: carrier_in_kernel], [Lean: carrier_norm].

For $k \neq 0$ the carrier is one-dimensional and represented by $w(k)$. At Γ , $B = 0$ and all three branches meet; the normalised Bloch eigenvector has no direction-independent limit there, which is the familiar singular-flat-band mechanism behind the incompleteness of a purely compact localised basis.

Theorem 9 (Finite-torus carrier). *For the periodic cubic cellulation of T^3 , $\ker \partial_2 = \text{im } \partial_3 \oplus H_2$ with $\dim \text{im } \partial_3 = L^3 - 1$ and $\dim H_2 = 3$, so $\dim \ker \partial_2 = L^3 + 2$.*

Proof. There are L^3 three-cells and no four-cells, and $b_3(T^3) = 1$, so $\text{rank } \partial_3 = L^3 - 1$ by rank-nullity. Discrete Hodge decomposition gives $\ker \partial_2 = B_2 \oplus H_2$ with $\dim H_2 = b_2(T^3) = 3$. $\square$

The count is verified over $\mathbb{Z}$ rather than re-derived from the formula: the complex is built from the boundary formulas of Appendix E; the exhibited cycles — every elementary cube boundary together with three wrapping sheets — bound the kernel from below, and rank over $\mathbb{F}_p$ bounds it from above, since rank can only drop under reduction. The two bounds meet at $L = 1, \dots, 5$.

Remark 10. $L^3 + 2$ is the nullity of the down boundary Laplacian, not a Betti number; the Betti contribution is the final 3. The restriction $L \geq 3$ belongs to the twelve-neighbour Bloch adjacency, where $x + e$ and $x - e$ coincide at $L = 2$. The chain-level count has no such restriction and holds at $L = 1, 2$ as well.

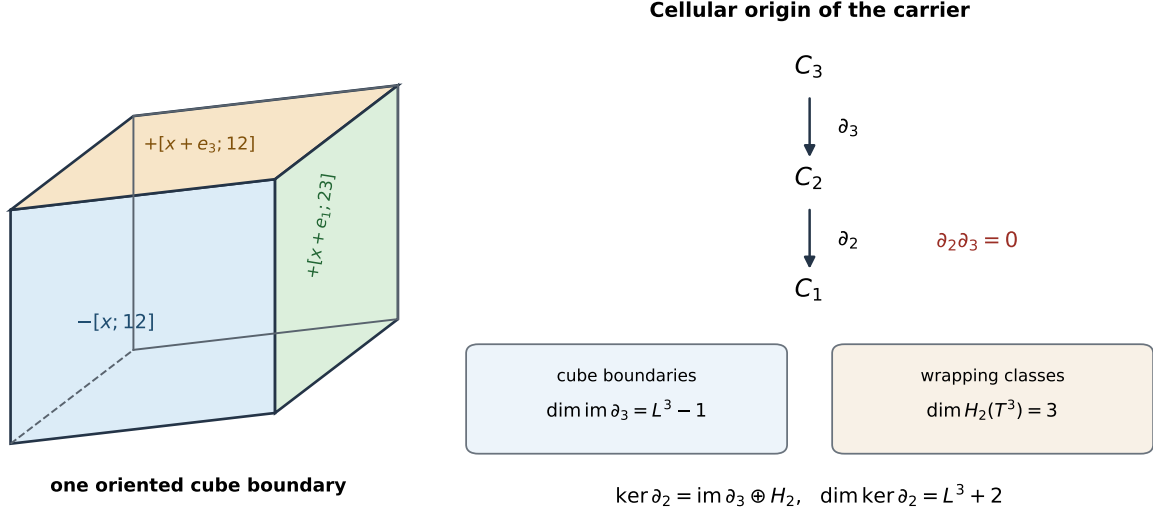


Figure 1: The carrier is a cellular cycle space. The left panel shows three visible terms in the outward-oriented boundary of one cube; the complete six-face formula is given in Appendix E. Cube boundaries span $\operatorname{im} \partial_3$, while three noncontractible wrapping classes complete $\ker \partial_2$ on the three-torus.

The following sum rule joins Theorems 8 and 9, which are otherwise two unrelated statements: a 3×3 spectrum and an $(L^3 + 2)$ -dimensional space.

Theorem 11 (Bloch–chain bridge). *On the periodic L^3 complex,*

$$\sum_k (3 - \operatorname{rank} B(k)) = L^3 + 2,$$

the sum running over the L^3 allowed momenta, with nullity profile 3 at Γ exactly once and 1 at each of the other $L^3 - 1$ momenta.

Proof. $B(0) = 0$ contributes 3. For $k \neq 0$ some $d_j \neq 0$, and Theorem 8 gives $\operatorname{rank} B(k) = 2$, contributing 1 each. Summing gives $3 + (L^3 - 1) = L^3 + 2$. $\square$

Remark 12. The two decompositions produce the same two numbers from different objects. The 3 is the triple degeneracy of $B(0) = 0$, not $b_2(T^3)$; the $L^3 - 1$ counts non- Γ momenta, not cubes. Their agreement is a consistency check on the whole construction, not a restatement.

Proposition 13 (Fixed-momentum carrier and the locality tradeoff). *Let $b_x = \partial_3[x; 1, 2, 3]$ be the six-face boundary of the elementary cube at x , and use the unitary discrete Fourier transform*

$$\widehat{b}_L(k) = L^{-3/2} \sum_{x \in \Lambda_L} e^{-ik \cdot x} b_x.$$

For every allowed $k \neq 0$, $\widehat{b}_L(k)$ has norm $\sqrt{q(k)}$ and, up to the Fourier phase convention, is the vector $w(k)$ of Theorem 8. Hence

$$\phi_L(k) = \frac{\widehat{b}_L(k)}{\sqrt{q(k)}}$$

is exactly the unique unit carrier vector in that momentum fibre. If $k_0 = 2\pi(r_1, r_2, r_3)/L_0 \neq 0$, then k_0 is an allowed momentum on every nested volume $L = mL_0$, and the order- u^2 internal separation from each of the two incidence branches is

$$t_N u^2 q(k_0), \quad (20)$$

independent of m . More generally it is bounded below by $t_N u^2 q_*$ on any set of allowed momenta with $q(k) \geq q_* > 0$.

This exact sharp-momentum state is not local: it is the Fourier sum of all cube boundaries. Conversely, for the normalized radius-one seed $b_x/\sqrt{6}$,

$$\left| \langle \phi_L(k), b_x/\sqrt{6} \rangle \right|^2 = \frac{q(k)}{6L^3}, \quad (21)$$

and $\hat{b}_L(0) = 0$.

Proof. Fourier transforming the six terms of the cubical boundary gives, in the $(12, 13, 23)$ face basis, $(d_3, -d_2, d_1)^\top$, with conjugates if the opposite Fourier sign is chosen. Its squared norm is $q(k)$ and $\partial_2 \partial_3 = 0$ puts it in $\ker B(k)^\dagger$; uniqueness for $k \neq 0$ follows from Theorem 8. The nested-volume statement is the integer identity $k_0 = 2\pi(mr)/(mL_0)$, and (20) follows from Corollary 24. Inverting the unitary Fourier transform gives $b_x = L^{-3/2} \sum_k e^{ik \cdot x} \sqrt{q(k)} \phi_L(k)$, which proves (21); the $k = 0$ coefficient vanishes. $\square$

Remark 14 (What the L^{-2} collapse does and does not say). The minimum of $q(k)$ over all nonzero finite-volume momenta is $4 \sin^2(\pi/L)$, so a contour isolating the single flat sheet from the other two sheets uniformly over the entire Brillouin zone cannot follow from the second-order symbol. This is an *internal* sheet separation. It does not erase the volume-uniform *external* electric-shell gap (7), and it does not affect the fixed- k coefficient (20). Neither fact by itself controls the interacting Riesz projection or supplies transfer-matrix overlap; those remain the substantive part of G18.

3.1 The wrapping sheets are cycles, not harmonic

Theorem 9 exhibits its three H_2 generators explicitly as wrapping sheets $s_{(ij)} = \sum_{x: x_m=0} [x; i, j]$. These satisfy $\partial_2 s = 0$ exactly, so they are cycles and they span the quotient $\ker \partial_2 / \text{im } \partial_3$. They are *not* harmonic: $\partial_3^\dagger s \neq 0$, and

$$\frac{\langle s, \partial_3 \partial_3^\dagger s \rangle}{\|s\|^2} = 2 \quad \text{exactly, for every } L \geq 2. \quad (22)$$

The distinction matters because H_2 is used in two senses: the quotient in Theorem 9, which the sheets span, and the harmonic subspace $\ker \partial_2 \cap \ker \partial_3^\dagger$, which they do not lie in. Nothing downstream breaks — Proposition 48 rests on $B^\dagger w = 0$, which covers all of $\ker \partial_2$ — but a real-space statement phrased for the harmonic subspace does not cover the generators this construction exhibits.

4 Exact adjacent shared-link dynamics and global assembly

Two plaquettes sharing a link leave six nonshared fundamental links after the external contractions. On the shared link the fusion channels depend on the relative induced orientation,

$$F \otimes \bar{F} = \mathbf{1} \oplus \text{Adj}, \quad F \otimes F = \Lambda^2 F \oplus \text{Sym}^2 F, \quad (23)$$

with dimensions and quadratic Casimirs

$$\begin{aligned}
d_1 &= 1, & C_1 &= 0, \\
d_{\text{Adj}} &= N^2 - 1, & C_{\text{Adj}} &= N, \\
d_{\Lambda^2 F} &= \frac{N(N-1)}{2}, & C_{\Lambda^2 F} &= \frac{(N+1)(N-2)}{N}, \\
d_{\text{Sym}^2 F} &= \frac{N(N+1)}{2}, & C_{\text{Sym}^2 F} &= \frac{(N-1)(N+2)}{N}.
\end{aligned} \tag{24}$$

The six nonshared half-links contribute $3C_F$ and the fused link $C_\rho/2$, against an external one-plaquette energy $2C_F$, so the resolvent denominator is $C_F + C_\rho/2$ exactly.

4.1 The channel weights are a theorem

Earlier editions assigned the channel projector squared norm d_ρ/N^2 by an isotropy hypothesis, and every statement in this section rests on that assignment. It is not a hypothesis.

Theorem 15 (Shared-link channel weights). *In the normalised two-plaquette state, the weight of the shared-link channel ρ is exactly d_ρ/N^2 , for both orientation families and every $N \geq 3$.*

Proof. Two steps. First the six nonshared links collapse. Each plaquette contributes a product of three independent Haar links, and a product of independent Haar matrices is Haar, so the amplitude is $\text{Tr}(AU) \text{Tr}(BU^{\pm 1})$ with A, B independent Haar and U the shared link. Integrating A and B contributes δ/N each and leaves a pure degree-(2, 2) moment of U .

Second, two such moments settle both families. In the Weingarten calculus, whose asymptotic origin is [6] and whose finite- N closed forms are [24], the order-two values are $\text{Wg}(e) = 1/(N^2 - 1)$ and $\text{Wg}((12)) = -1/(N(N^2 - 1))$ — the inverse of the S_2 Gram matrix $\begin{pmatrix} N^2 & N \\ N & N^2 \end{pmatrix}$ — the index sums give

$$\begin{aligned}
M_{\text{direct}} &= \sum_{ijkl} \langle |U_{ij}|^2 |U_{kl}|^2 \rangle = N^2, \\
M_{\text{cross}} &= \sum_{ijkl} \langle U_{ij} U_{kl} \bar{U}_{il} \bar{U}_{kj} \rangle = N.
\end{aligned} \tag{25}$$

The standard formula is often written for $U(N)$. The present degree-(2, 2) integrands are invariant under $U \mapsto zU$, so disintegrating Haar measure over the central $U(1)$ shows that their normalised $U(N)$ and $SU(N)$ integrals coincide. No determinant-sector term occurs at this degree for $N \geq 3$. For the like family, $\text{Tr}(AU) \text{Tr}(BU)$ decomposes into $\text{Sym}^2 \oplus \Lambda^2$ by symmetrising the two index pairs jointly, and the two squared norms are $(M_{\text{direct}} \pm M_{\text{cross}})/2$. Normalising,

$$n_{\text{Sym}^2} = \frac{N+1}{2N} = \frac{d_{\text{Sym}^2 F}}{N^2}, \quad n_{\Lambda^2} = \frac{N-1}{2N} = \frac{d_{\Lambda^2 F}}{N^2}.$$

For the mixed family the singlet component of $U_{ij} \bar{U}_{lk}$ is $\delta_{il} \delta_{jk}/N$, of squared norm 1 against $M_{\text{direct}} = N^2$, so $n_1 = 1/N^2 = d_1/N^2$ and $n_{\text{Adj}} = 1 - 1/N^2 = d_{\text{Adj}}/N^2$. $\square$

Remark 16. The derivation imports nothing: the Weingarten pair is the inverse of the S_2 Gram matrix and the index sums in (25) are explicit, so the reader can check it without leaving this page. That the pair is not two fitted constants is separately checked — the identity-permutation value times $N^2 - 1$ is 1 identically in N , so the $SU(3)$ numbers are a specialisation of a rank-generic formula. Together these make the *local shared-link coefficient* self-contained. By themselves they do not enumerate every second-order process; that separate step is supplied by Lemma 22. Appendix B records the index sums.

Remark 17 (External corroboration). The same numerator appears in the quantum-simulation literature from an unrelated direction. Ciavarella, Burbano and Bauer [33] publish a finite-rank plaquette matrix element $|M_\rho|^2 = \dim \rho / (\dim A \dim R)$; for two fundamental factors $\dim A = \dim R = N$ and this is d_ρ/N^2 identically in N . That is provenance for a chain already checked here, not a second derivation. The exhaustion of the shared-link routes is proved independently in Lemma 22.

Remark 18 (A falsified zero backend). The organized authority corpus contains an independent exact diagnostic at $N = 3$:

$$\text{Wg}(e) = \frac{1}{8}, \quad \text{Wg}((12)) = -\frac{1}{24}, \quad \int_{SU(3)} |U_{11}|^4 dU = \frac{1}{6} \neq 0.$$

It therefore falsifies the archived stranded-flux rule that discards this balanced Haar sector. This is useful negative evidence for retaining the contractions above; it is not a second derivation of t_N .

4.2 The off-diagonal process list is complete

Write $X_q = \chi_q + \bar{\chi}_q$, so $V = \sum_q X_q$, and let $Q_0 = 1 - |0\rangle\langle 0|$.

Lemma 19 (Small-support cellular two-cycles). *Fix one orientation for every elementary face of Λ_L , with $L \geq 3$. If*

$$z = \sum_f a_f f \in C_2(\Lambda_L; \mathbb{Z}_N), \quad N \geq 2,$$

satisfies $\partial_2 z = 0$ and $|\text{supp } z| \leq 4$, then $z = 0$.

Proof. Suppose $z \neq 0$ and choose $f \in \text{supp } z$. For every one of the four distinct links $e \subset \partial f$,

$$0 = (\partial_2 z)_e = s(f, e)a_f + \sum_{\substack{g \neq f \\ e \subset \partial g}} s(g, e)a_g.$$

Since $s(f, e)a_f \neq 0$ in $\mathbb{Z}_N$, at least one other supported face g_e contains e . For $L \geq 3$, two distinct elementary faces of the periodic cubic complex share at most one link. The four faces g_e are therefore distinct, giving $|\text{supp } z| \geq 1 + 4 = 5$, a contradiction. $\square$

Remark 20. The hypothesis $L \geq 3$ is sharp for this lemma: at $L = 2$, a coordinate wrapping sheet contains four faces and is a nonzero cellular two-cycle.

Lemma 21 (Reduced same-face route). *For every $N \geq 3$ and distinct plaquettes p, p' ,*

$$\langle p', -|X_{p'}Q_-(E_{F,N} - H_E)^{-1}Q_-X_p|p, -\rangle = 0. \quad (26)$$

For $SU(3)$ the same matrix element, including the diagonal, is

$$\langle p', -|X_{p'}Q_-(E_{F,3} - H_E)^{-1}Q_-X_p|p, -\rangle = -\frac{1}{4}\delta_{p'p}. \quad (27)$$

Proof. Write $\chi_{R,p}$ for the Wilson character in irrep R around p and put $Y_p = X_p|p, -\rangle$. The mixed-orientation terms cancel before any projection:

$$\frac{\chi_p \bar{\chi}_p - \bar{\chi}_p \chi_p}{\sqrt{2}}|0\rangle = 0.$$

Thus the vacuum and adjoint pieces of $F \otimes \bar{F} = \mathbf{1} \oplus \text{Adj}$ are absent, while

$$Y_p = \frac{\chi_p^2 - \bar{\chi}_p^2}{\sqrt{2}} |0\rangle.$$

For $N \geq 4$ this is

$$\frac{\chi_{\text{Sym}^2 F, p} + \chi_{\Lambda^2 F, p} - \chi_{\overline{\text{Sym}^2 F}, p} - \chi_{\overline{\Lambda^2 F}, p}}{\sqrt{2}} |0\rangle,$$

with coincident conjugate terms cancelling when an irrep is self-conjugate. None of the surviving terms belongs to the retained fundamental shell.

For $SU(3)$ the determinant epsilon tensor gives $\Lambda^2 F \simeq \bar{F}$ and

$$\chi_F^2 = \chi_{\mathbf{6}} + \chi_{\bar{F}}, \quad \chi_{\bar{F}}^2 = \chi_{\bar{\mathbf{6}}} + \chi_F.$$

With

$$|p, \mathbf{6}, -\rangle := \frac{\chi_{\mathbf{6}, p} - \chi_{\bar{\mathbf{6}}, p}}{\sqrt{2}} |0\rangle,$$

one obtains

$$X_p |p, -\rangle = |p, \mathbf{6}, -\rangle - |p, -\rangle, \quad (28)$$

and hence

$$P_- X_p |p, -\rangle = -|p, -\rangle, \quad Q_- X_p |p, -\rangle = |p, \mathbf{6}, -\rangle. \quad (29)$$

The determinant-related fundamental component therefore does not cancel; it is removed by the reduced-space projector Q_- .

Every surviving summand of $Q_- Y_p$ is a one-plaquette spin network supported on ∂p and carries a nontrivial irrep on each boundary link. If $p \neq p'$, some link belongs to exactly one of $\partial p, \partial p'$, so Peter–Weyl orthogonality separates the two vectors. The electric Hamiltonian and its reduced resolvent preserve these spin-network sectors, proving (26).

At $N = 3$,

$$H_E |p, \mathbf{6}, -\rangle = 2C_2(\mathbf{6}) |p, \mathbf{6}, -\rangle = \frac{20}{3} |p, \mathbf{6}, -\rangle, \quad E_{F,3} = \frac{8}{3}.$$

The reduced denominator is $(8/3 - 20/3)^{-1} = -1/4$, proving (27). $\square$

Lemma 22 (Second-order process completeness). *For $N \geq 3$, $L \geq 3$, and distinct plaquettes p, p' , define*

$$K_{2,-} = P_- V Q_- (E_{F,N} - H_E)^{-1} Q_- V P_-, \quad (30)$$

$$H_{2,-}^{\text{exc}} = K_{2,-} - E_{\text{vac}}^{(2)} P_-, \quad E_{\text{vac}}^{(2)} = \langle 0 | V Q_0 (0 - H_E)^{-1} Q_0 V | 0 \rangle. \quad (31)$$

If p and p' share no link, then $\langle p', - | K_{2,-} | p, - \rangle = 0$. If they share the unique link e , every nonzero intermediate lies in exactly one of

$$F \otimes \bar{F} = \mathbf{1} \oplus \text{Adj}, \quad F \otimes F = \Lambda^2 F \oplus \text{Sym}^2 F$$

or the conjugate of the second decomposition. Thus these four shared-link channels exhaust every off-diagonal second-order process. The conclusion is preserved by the stated link-Casimir budget cutoff, and more generally by cutoffs that are functions only of the link Casimirs and therefore act identically on all four orientation assignments. A general projection inside the degenerate $2E_{F,N}$ eigenspace need not preserve the non-adjacent cancellation.

Proof. Expand the external states and both insertions in oriented characters. A term $\chi_p^\epsilon \chi_q^\eta$, with $\epsilon, \eta \in \{\pm 1\}$ and $\chi^{-1} = \bar{\chi}$, has linkwise centre charge $\partial(\epsilon p + \eta q)$ in $C_1(\Lambda_L; \mathbb{Z}_N)$. Since H_E commutes with the independent link-centre actions, a common electric intermediate between the p and p' sides requires

$$\partial(\epsilon p + \eta q) = \partial(\epsilon' p' + \eta' r) \pmod{N} \quad (32)$$

for action plaquettes q, r and signs $\epsilon, \eta, \epsilon', \eta'$.

Apply Lemma 19 to

$$z = \epsilon p + \eta q - \epsilon' p' - \eta' r.$$

Equation (32) gives $\partial_2 z = 0$, and the support has at most four faces, so $z = 0$ in $C_2(\Lambda_L; \mathbb{Z}_N)$. For $N \geq 5$, every face coefficient has absolute value at most four, so modular vanishing is ordinary integral cancellation. For $N = 3$, a non-paired coefficient could vanish only by being ± 3 ; this would use three occurrences on one face and, because $p \neq p'$, leave a distinct face with coefficient ± 1 . For $N = 4$, a non-paired coefficient could vanish only by being ± 4 , which would put all four occurrences on one face and again contradict $p \neq p'$. Therefore the four occurrences pair with opposite signs, and the only possibilities are

$$q = p, \quad \eta = -\epsilon, \quad r = p', \quad \eta' = -\epsilon', \quad (33)$$

or

$$q = p', \quad r = p, \quad \eta = \epsilon', \quad \eta' = \epsilon. \quad (34)$$

The alternative (33) is the same-face route of Lemma 21. Its mixed-orientation vacuum and adjoint pieces cancel algebraically. At $N = 3$ the full same-face vector also contains the determinant-induced retained-shell term $-|p, -\rangle$, but (29) shows that Q_- removes it. The surviving sextet route is diagonal by (27). Therefore only (34) can contribute off diagonal. If p, p' are link-disjoint, sharing a vertex being immaterial, linkwise factorisation gives

$$X_{p'}|p, -\rangle = \sqrt{2}|p, -\rangle|p', +\rangle, \quad X_p|p', -\rangle = \sqrt{2}|p, +\rangle|p', -\rangle.$$

Both vectors have electric energy $2E_{F,N}$, so the reduced resolvent is scalar on their span, while their inner product is zero by charge orthogonality.

If p, p' share e , the six nonshared links each carry one fundamental or antifundamental factor and the shared link carries exactly two. The two complete tensor-product decompositions above therefore give all common intermediates. Their energies and gaps are

$$E_\rho = 3C_F + \frac{1}{2}C_\rho, \quad E_\rho - E_{F,N} = C_F + \frac{1}{2}C_\rho > 0.$$

Spectral resolution yields only their four projectors. Vacuum subtraction is diagonal and leaves this cross matrix element unchanged. The two-face term is connected, so the linked Möbius projection leaves it unchanged as well. $\square$

With Theorem 15 the resolvent weight is

$$w_\rho = -\frac{d_\rho/N^2}{C_F + C_\rho/2}, \quad (35)$$

and summing within the two orientation families,

$$A_N = w_{\mathbf{1}} + w_{\text{Adj}} = -\frac{2N^3}{(N^2 - 1)(2N^2 - 1)}, \quad (36)$$

$$B_N = w_{\Lambda^2} + w_{\text{Sym}^2} = -\frac{4N(N^2 - 2)}{(N^2 - 1)(4N^2 - 9)}. \quad (37)$$

4.3 The shared-link law

Theorem 23 (All-rank adjacent shared-link law). *Let $p \neq p'$ be oriented plaquettes sharing the single link e . For every integer $N \geq 3$, let $s = s(p, e)$, $s' = s(p', e)$, and let $\hat{\Pi}_\rho$ be the charge-conjugation-closed electric projector onto the shared-link channel ρ (including the conjugate irrep on the conjugate like-channel subspace). Define*

$$\eta_\rho = \begin{cases} -1, & \rho \in \{\mathbf{1}, \text{Adj}\}, \\ +1, & \rho \in \{\Lambda^2 F, \text{Sym}^2 F\}. \end{cases}$$

Then

$$\hat{\Pi}_\rho X_{p'} |p, -\rangle = \eta_\rho s s' \hat{\Pi}_\rho X_p |p', -\rangle, \quad \|\hat{\Pi}_\rho X_{p'} |p, -\rangle\|^2 = \frac{d_\rho}{N^2}. \quad (38)$$

$$\boxed{\langle p', - | X_p \hat{\Pi}_\rho X_{p'} |p, -\rangle = \eta_\rho s(p', e) s(p, e) \frac{d_\rho}{N^2}}. \quad (39)$$

Consequently the linked order- u^2 contribution through all four channels is

$$\langle p', - | H_2 |p, -\rangle = t_N s(p', e) s(p, e) = t_N (BB^\dagger)_{p'p},$$

where $s(p, e) = \pm 1$ is the cellular boundary incidence and

$$t_N = B_N - A_N = \frac{2N(N^2 - 4)}{(N^2 - 1)(2N^2 - 1)(4N^2 - 9)} > 0.$$

Proof. Because distinct cubical faces share at most one link, $(BB^\dagger)_{p'p} = ss'$. Expand

$$|p; \epsilon\rangle_{\text{or}} := \chi_p^\epsilon |0\rangle, \quad \chi_p^{+1} := \chi_p, \quad \chi_p^{-1} := \bar{\chi}_p, \quad |p, -\rangle = 2^{-1/2} \sum_{\epsilon=\pm 1} \epsilon |p; \epsilon\rangle_{\text{or}}.$$

Put $\sigma = ss'$. On the shared link the two oriented factors are ϵs and $\epsilon' s'$. The complete orientation bookkeeping is

ρ	shared-link family	$\epsilon\epsilon'$	η_ρ	$\eta_\rho ss' d_\rho / N^2$
$\mathbf{1}$	$F \otimes \bar{F}$ or $\bar{F} \otimes F$	$-\sigma$	-1	$-ss' / N^2$
Adj	$F \otimes \bar{F}$ or $\bar{F} \otimes F$	$-\sigma$	-1	$-ss'(N^2 - 1) / N^2$
$\Lambda^2 F$ and its conjugate	$F \otimes F$ or $\bar{F} \otimes \bar{F}$	σ	$+1$	$ss'(N - 1) / (2N)$
$\text{Sym}^2 F$ and its conjugate	$F \otimes F$ or $\bar{F} \otimes \bar{F}$	σ	$+1$	$ss'(N + 1) / (2N)$

(40)

For a fixed monomial $\chi_p^\epsilon \chi_{p'}^{\epsilon'}$, the forward route carries charge-odd coefficient $\epsilon/\sqrt{2}$, whereas the reversed route carries $\epsilon'/\sqrt{2}$. Their ratio is $\epsilon/\epsilon' = \epsilon\epsilon'$. Opposite induced orientations give $\epsilon\epsilon' = -ss'$ and the mixed family; equal induced orientations give $\epsilon\epsilon' = ss'$ and the like family. Hence

$$\hat{\Pi}_\rho X_{p'} |p, -\rangle = \eta_\rho ss' \hat{\Pi}_\rho X_p |p', -\rangle.$$

The two charge-conjugate assignments are orthogonal for $N \geq 3$, so their two external factors combine to the single channel norm d_ρ / N^2 . This proves (38) and (39).

On this channel,

$$(E_{F,N} - H_E)^{-1} \hat{\Pi}_\rho = -\frac{\hat{\Pi}_\rho}{C_F + C_\rho/2}.$$

Its contribution is therefore $\eta_{\rho} ss' w_{\rho}$. Summing gives

$$ss'(w_{\Lambda^2 F} + w_{\text{Sym}^2 F} - w_1 - w_{\text{Adj}}) = ss'(B_N - A_N).$$

Substitution of (36)–(37) gives the displayed rational function. Every denominator factor and $N^2 - 4$ is positive for $N \geq 3$. $\square$

Corollary 24 (Global torus assembly). *On the trivial-flux charge-odd electric shell of Theorem 1, translation and cubic symmetry make the remaining diagonal contribution a scalar. More precisely, there is a scalar $d_{2,N,L}$, computed in closed form and shown L -independent in Theorem 29 below, such that*

$$[u]H_{\text{eff},-}^{(N)}(k, u) = \delta_{N,3}I, \quad [u^2]H_{\text{eff},-}^{(N)}(k, u) = d_{2,N,L}I + t_N B(k)B(k)^{\dagger}.$$

Equivalently, with $E_{\text{flat},N,L}^{[2]}(u) = E_{F,N} + \delta_{N,3}u + d_{2,N,L}u^2$,

$$T_{\leq 2}H_{\text{eff},-}^{(N)}(k, u) = E_{\text{flat},N,L}^{[2]}(u)I + t_N u^2 B(k)B(k)^{\dagger}.$$

Consequently the three band Taylor polynomials through second order are $E_{\text{flat},N,L}^{[2]}(u)$ and $E_{\text{flat},N,L}^{[2]}(u) + t_N u^2 q(k)$ twice. This is a coefficient identity, not a claim of a remainder uniform in L .

Proof. The centre-charge argument of Lemma 22, now with only three plaquette occurrences, gives $\langle p', -|P_-VP_-|p, -\rangle = 0$ for $p \neq p'$. A possible $SU(3)$ determinant balance would require all three occurrences to be the same face and is therefore excluded off diagonal. On the diagonal it occurs only at $N = 3$, where (29) gives

$$P_-VP_- = -\delta_{N,3}P_-, \quad [u]H_{\text{eff},-}^{(N)} = -P_-VP_- = \delta_{N,3}I.$$

At second order Lemma 22 and Theorem 23 assemble every off-diagonal entry. Translations act transitively on faces of a fixed orientation and the cubic group permutes the orientations, so the invariant second-order diagonal is also a multiple of the identity. Thus the adjacency part is $t_N(BB^{\dagger} - 4I)$; absorb $-4t_N$ into the scalar and apply Theorem 8. $\square$

Two consequences,

$$t_3 = \frac{5}{612}, \quad t_N = \frac{1}{4N^3} - \frac{1}{16N^5} - \frac{77}{64N^7} - \frac{1021}{256N^9} + \dots \quad (41)$$

The expansion is governed by a closed form. Writing the planar-normalised coefficient as $4N^3 t_N$,

$$1 - 4N^3 t_N = \frac{2N^4 + 31N^2 - 9}{(N^2 - 1)(2N^2 - 1)(4N^2 - 9)}, \quad (42)$$

an identity of rational functions [Lean: hopping_deficit_numerator], and the right side is positive and the left strictly decreasing for every real $N \geq 2$: substituting $N = M + 2$ makes every numerator coefficient of both the remainder and the derivative nonnegative, so $4N^3 t_N$ increases monotonically to 1. At fixed canonical coordinate u the width of Corollary 24 is $12t_N u^2$, so the entire dispersive content of the projected sector switches off at exactly the rate N^{-3} , with computable subleading coefficient $-1/(4N^2)$. This is consistent with the general suppression of glueball mixing at large N , but no 't Hooft-coupling identification is made here. The channel algebra behind t_N — the four closed-form weights, the two family sums, and the weights summing to one per family — is proof-checked as well [Lean: mixed_family_sum], [Lean: like_family_sum], [Lean: family_weights_sum_to_one].

4.4 The vacuum-mediated route

Lemma 22 is specific to the charge-odd shell. The corresponding assertion is false in general: the charge-even sector admits a vacuum-mediated route. Its cancellation in the odd sector is the charge-conjugation selection rule below.

Proposition 25 (Vacuum-mediated route). *The reduced-resolvent route*

$$\frac{\langle p'|V|0\rangle\langle 0|V|p\rangle}{E_{F,N} - E_0} = \frac{\langle p'|V|0\rangle\langle 0|V|p\rangle}{E_{F,N}}$$

is available for any pair of plaquettes, sharing a link or not. It vanishes identically in the charge-odd sector, because V is charge-even and $|0\rangle$ is charge-even, so $\langle 0|V|p, -\rangle = 0$. Consequently the charge-odd hopping $t_N = B_N - A_N$ carries no such term, while the charge-even shared-link element acquires $1/C_F$: the normalised charge-even matrix-element product is 2 and $E_{F,N} = 2C_F$.

$$\ell_N = A_N + B_N + \frac{1}{C_F} = -\frac{2N(3N^2 - 5)}{(N^2 - 1)(2N^2 - 1)(4N^2 - 9)}.$$

Remark 26. At $N = 3$, $A_3 + B_3 = -481/612$ and $1/C_F = 3/4$, giving $\ell_3 = -11/306$. The first of these is exactly the value once published for the charge-even hopping and later superseded; the correction is precisely the omitted route. Stating the selection rule turns a one-rank erratum into an all-rank formula.

In the charge-odd shell, Lemma 22 and this proposition dispose of non-adjacent pairs: the exchange vectors are orthogonal and the vacuum route vanishes. In the charge-even sector the non-adjacent vacuum route does not vanish, and revision 5 read it as a rank-one all-pairs operator whose cancellation “has not been derived” and “remains a separate requirement”. Those two sentences were stale against this paper’s own corpus: the flat-band manuscript’s lemma on the vacuum-mediated route, and the June 2026 patch to its Section 6, state at $N = 3$ that the route $+ \frac{3}{4}$ “exactly cancels the free two-flux channel sum $2/(8/3 - 16/3) = -\frac{3}{4}$, recovering the vanishing of distant hops”. They are withdrawn. The theorem below is that sentence lifted to every rank and to an operator identity, with the adjacent element’s orientation independence proved monomial by monomial.

Theorem 27 (Charge-even second-order symbol). *For $N \geq 3$, $L \geq 3$ and distinct plaquettes p, p' , with $K_{2,+}$ and $H_{2,+}^{\text{exc}}$ defined as in (30)–(31) with P_+ in place of P_- ,*

$$\langle p', + | H_{2,+}^{\text{exc}} | p, + \rangle = \begin{cases} \ell_N, & p, p' \text{ share a link,} \\ 0, & \text{otherwise,} \end{cases}$$

independently of the boundary orientations. Consequently

$$[u^2]H_{\text{eff},+}^{(N)}(k, u) = d_{2,N}^+ I + \ell_N (\mathcal{N}(k)\mathcal{N}(k)^\dagger - 4I),$$

with $\mathcal{N}(k)$ the unsigned incidence of Section 6 and $d_{2,N}^+$ the scalar of Theorem 29.

Proof. The centre-charge argument of Lemma 22 is blind to charge parity, so the only routes are the same-face route (33) and the exchange route (34).

Link-disjoint pairs. Expanding $X_p|p, +\rangle = (\chi_p + \bar{\chi}_p)^2|0\rangle/\sqrt{2}$ by $\chi_F^2 = \chi_{\text{Sym}^2 F} + \chi_{\Lambda^2 F}$ and $\chi_F \bar{\chi}_F = 1 + \chi_{\text{Adj}}$, its vacuum component is $\sqrt{2}|0\rangle$ and every other component carries flux on ∂p , which $X_{p'}$ cannot remove; so the same-face route contributes exactly the vacuum

route $\langle p', + | X_{p'} | 0 \rangle \langle 0 | X_p | p, + \rangle / (E_{F,N} - 0) = 2/E_{F,N}$. (At $N = 3$ the retained component $|p, +\rangle$ is removed by Q_+ , and could not reach $\langle p', + |$ in any case.) The exchange route goes through $X_{p'} | p, + \rangle = \sqrt{2} | p, + \rangle | p', + \rangle = X_p | p', + \rangle$, one vector at energy $2E_{F,N}$, and contributes $2/(E_{F,N} - 2E_{F,N}) = -2/E_{F,N}$. The two cancel [Lean: disjoint_pair_cancels]. In the charge-odd sector both vanish separately: $\langle 0 | V | p, - \rangle = 0$ and the two exchange vectors are orthogonal.

Adjacent pairs. The vacuum route contributes $2/E_{F,N} = 1/C_F$ as before. In the exchange route, expand both states in oriented characters as in the proof of Theorem 23. Every monomial $\chi_{p'}^{\epsilon'} \chi_p^{\epsilon}$ now carries coefficient $1/\sqrt{2}$ on both sides, and the bra fixes the nonshared-link fluxes, so $\langle p'; \epsilon' | X_p \hat{\Pi}_\rho X_{p'} | p; \epsilon \rangle = d_\rho / N^2$ when ρ lies in the family selected by $(\epsilon s, \epsilon' s')$ and vanishes otherwise. Each family is selected by exactly two of the four monomials, so every channel contributes w_ρ with sign $+1$, whatever s, s' : the total is $A_N + B_N + 1/C_F = \ell_N$. Repeating the count with the charge-odd coefficients $\epsilon \epsilon' / 2$ gives $ss'(B_N - A_N)$, which is Theorem 23.

The off-diagonal is therefore ℓ_N on every adjacent pair, orientation independent, which is the unsigned adjacency $\mathcal{N} \mathcal{N}^\dagger - 4I$. Translation and cubic symmetry make the diagonal a scalar, as in Corollary 24. $\square$

Remark 28 (Regime, and an outside corroboration). This is a finite-volume coefficient identity, exactly as Corollary 24 is: at $k = 0$ the A_1^{++} combination couples to the vacuum with an extensive matrix element, so no remainder uniform in L is claimed. What the theorem changes is the standing of Section 6: the unsigned symbol is the charge-even sector at order u^2 , and its Γ -point splitting $12\ell_N$ and bandwidth $16|\ell_N|$ (for even L , and coefficientwise as $L \rightarrow \infty$) are unconditional. At $N = 3$ the theorem is corroborated from outside. An independent engine, built on the corpus's Haar primitives and reading no register value, returns the charge-even shared-link element $-11/306$ on the coplanar and on the perpendicular pair alike and 0 on a link-disjoint pair (Section 9.6); and Hamer's 0^{++} series matches the Γ -point coefficient the theorem predicts (Section 10).

4.5 The second-order scalar at every rank

Corollary 24 proves that the second-order diagonal is a scalar and, through revision 5, wrote it down only at $N = 3$, as the upstream identification $\text{leak}_2 = \ell_3$ with a within-plaquette tower. It is determined by the same three inputs as the hopping. Write $\sigma_N^\pm$ for the reduced same-face route $\langle p, \pm | X_p Q_\pm (E_{F,N} - H_E)^{-1} Q_\pm X_p | p, \pm \rangle$ of Lemma 21.

Theorem 29 (Second-order scalar ledger). *For $N \geq 3$ and $L \geq 3$ the on-site coefficients are L -independent,*

$$d_{2,N}^\pm = \sigma_N^\pm + \frac{1}{C_F} + 12\ell_N, \quad (43)$$

so that $d_{2,N,L} = d_{2,N}^-$ in Corollary 24, the flat scalar is $d_{2,N}^- - 4t_N$, and in the charge-even sector the A_1^{++} level at Γ is $d_{2,N}^+ + 12\ell_N$ and the $\lambda = -4$ level $d_{2,N}^+ - 4\ell_N$. The same-face routes are, for $N \geq 5$,

$$\sigma_N^- = -\frac{N}{(N-1)(N+3)} - \frac{N}{(N+1)(N-3)} = -\frac{2N(N^2-3)}{(N^2-1)(N^2-9)}, \quad (44)$$

$$\sigma_N^+ = \frac{1}{C_F} - \frac{2N}{N^2+1} + \sigma_N^-, \quad (45)$$

the two denominators of (44) being the two-flux gaps $E_{F,N} - 2C_2(\text{Sym}^2 F) = -(N-1)(N+3)/N$ and $E_{F,N} - 2C_2(\Lambda^2 F) = -(N+1)(N-3)/N$ [Lean: two_flux_gaps], and $-2N/(N^2+1)$ the adjoint

gap $E_{F,N} - 2N$ with the doubled adjoint norm [Lean: `adjoint_gap`]. At the two exceptional ranks

$$\sigma_3^- = -\frac{1}{4}, \quad \sigma_3^+ = -\frac{1}{10}, \quad \sigma_4^- = -\frac{4}{21}, \quad \sigma_4^+ = -\frac{1028}{595} :$$

at $N = 3$ only the sextet survives $Q_\pm$ and the vacuum and adjoint pieces of $\sigma_3^\pm$ are $\frac{3}{4}$ and $-\frac{3}{5}$; at $N = 4$ the real six of $su(4)$ cancels in the odd combination and enters doubled in the even one. Consequently, for $N \geq 5$, the charge-odd tower is

$$\sigma_N^- + \frac{1}{C_F} = -\frac{12N}{(N^2 - 1)(N^2 - 9)}, \quad (46)$$

of order N^{-3} : the same-face route and the vacuum bubble, each of order N^{-1} , cancel at leading order [Lean: `towerOdd_closed`].

Proof. Lemma 22 assumes $p \neq p'$; the diagonal case is its extension. With $p' = p$ the centre-charge chain is $z = (\epsilon - \epsilon')p + \eta q - \eta' r$, of support at most three, so Lemma 19 gives $z = 0$ in $C_2(\Lambda_L; \mathbb{Z}_N)$. If q, r and p are distinct the coefficient $\eta = \pm 1$ of q cannot vanish; if exactly one of q, r equals p the other carries $-\eta' \neq 0$; so $q = r$. For $q = r \neq p$ the coefficients $\epsilon - \epsilon'$ and $\eta - \eta'$ lie in $\{0, \pm 2\}$ and $\pm 2 \neq 0 \pmod N$ for $N \geq 3$, so the four monomials of $X_q|p, \pm\rangle$ are pairwise orthogonal. For $q = r = p$ the single coefficient $\epsilon + \eta - \epsilon' - \eta' \in \{0, \pm 2, \pm 4\}$ can vanish mod 4 at $N = 4$ through $(+, +, -, -)$: that is the real six of $su(4)$ shared by $\chi_p^2|0\rangle$ and $\bar{\chi}_p^2|0\rangle$, and the same-face route is decomposed by irreps below, which covers it. Hence $\langle p|K_2|p\rangle = \sum_q \langle p|X_q Q(E_{F,N} - H_E)^{-1} Q X_q|p\rangle$, with three kinds of q . For $q = p$ this is $\sigma_N^\pm$. For the twelve q sharing a link with p , $X_q|p, \pm\rangle$ is a sum of four mutually orthogonal unit monomials, two in each orientation family, each carrying coefficient squared $\frac{1}{2}$, so the resolvent gives $A_N + B_N$ per neighbour in either sector. For the $3L^3 - 13$ link-disjoint q , $X_q|p, \pm\rangle = \sqrt{2}|p, \pm\rangle|q, +\rangle$ at energy $2E_{F,N}$, worth $-2/E_{F,N} = -1/C_F$ each. The vacuum subtraction is $E_{\text{vac}}^{(2)} = -3L^3 \cdot 2/E_{F,N} = -3L^3/C_F$, since $\|X_q|0\rangle\|^2 = 2$ for every q . Adding, $\sigma_N^\pm + 12(A_N + B_N) + 13/C_F$, which is (43): thirteen bubbles survive the subtraction, one for p itself and one per neighbour, and that is why the per-neighbour leakage is ℓ_N rather than the bare channel sum.

For the same-face route, $X_p|p, \pm\rangle = (\chi_p + \bar{\chi}_p)(\chi_p \pm \bar{\chi}_p)|0\rangle/\sqrt{2}$, which is $(\chi_p^2 - \bar{\chi}_p^2)/\sqrt{2}$ in the odd sector and $(\chi_p^2 + 2\chi_p\bar{\chi}_p + \bar{\chi}_p^2)/\sqrt{2}$ in the even one. With $\chi_F^2 = \chi_{\text{Sym}^2 F} + \chi_{\Lambda^2 F}$ and $\chi_F\bar{\chi}_F = 1 + \chi_{\text{Adj}}$, a single-plaquette loop in irrep R costs $2C_2(R)$, and each distinct character has unit norm. For $N \geq 5$ the four like irreps are distinct and none is fundamental, giving (44) and, with the vacuum at energy 0 (squared norm 2) and the adjoint at $2N$ (squared norm 2), (45). At $N = 4$, $\Lambda^2 F$ is self-conjugate, so $\chi_{\Lambda^2 F} - \bar{\chi}_{\Lambda^2 F}$ vanishes and $\chi_{\Lambda^2 F} + \bar{\chi}_{\Lambda^2 F}$ has squared norm 4. At $N = 3$, $\Lambda^2 F = \bar{F}$ and the term is the retained-shell component $\mp|p, \pm\rangle$ of Lemma 21, removed by $Q_\pm$. Equation (46) is rational arithmetic. $\square$

Remark 30 (The ledger at low rank). At $N = 3$ the theorem returns the register: tower $\frac{1}{2} = -\frac{1}{4} + \frac{3}{4}$, on-site $\frac{7}{102}$, flat $\frac{11}{306}$; charge even, tower $\frac{13}{20}$, on-site $\frac{223}{1020}$, A_1^{++} at $\Gamma - \frac{217}{1020}$, band top $\frac{1109}{3060}$ [Lean: `su3_odd_ledger`], [Lean: `su3_even_ledger`]. Every one of these was recorded upstream as an identification; each is now a consequence, and the independent engine of Section 9.6 returns the two same-face routes $-\frac{1}{4}$ and $-\frac{1}{10}$, the per-neighbour leakage $-\frac{481}{612}$ before and $-\frac{11}{306}$ after the $-\frac{3}{4}$ bubble, and 0 on a link-disjoint pair, from the primitives — the last of these printed in the run's console log rather than its certificate. A second confirmation at $N = 3$ shares nothing with that engine: the two-cube delivery of Section 5, built from Clebsch–Gordan amplitudes and folded at operator level, has a connected diagonal $D_{\mathcal{B}=6}$ that is the per-neighbour ledger face by face. The Möbius weights cancel the same-face route on every face, and what remains is the cross-cell census: a side face has

one shared-link neighbour and four link-disjoint faces in the other cube, $-\frac{481}{612} + 4(-\frac{3}{4}) = -\frac{2317}{612}$; an end cap has five disjoint ones, $-\frac{15}{4}$; the shared face folds to 0. So the diagonal Proposition 35 calls the open boundary is the boundary count of this theorem. (“Linked” here means vacuum-subtracted on the torus, as in (31); the prism’s Möbius diagonal subtracts sub-complex operators instead, and the two prescriptions agree on the cross-cell off-diagonal, which is all Proposition 32 uses.) At $N = 4$ the odd tower is $\frac{12}{35}$, the on-site $\frac{10828}{59675}$ and the flat $\frac{9932}{59675}$ [Lean: su4_ledger]; at $N = 5$ the tower is $-\frac{5}{32}$, the on-site $-\frac{4785}{20384}$ and the flat $-\frac{4945}{20384}$ [Lean: flatOdd_five]. Appendix G tabulates $N = 3$ to 7. In $1/N$ the flat scalar is $-22/N^3 + O(N^{-5})$, the even A_1^{++} level at Γ is $-26/N^3 + O(N^{-5})$, and $\ell_N = -\frac{3}{4}N^{-3} + O(N^{-5})$: every second-order coefficient of the projected sector, scalar or dispersive, is $O(N^{-3})$ in canonical u . One boundary is worth stating plainly. The $N \geq 4$ entries rest on this paper’s algebra and on the verifier’s checks of it, and on nothing else: no engine independent of this derivation has computed a second-order diagonal at any rank above three. A Haar-exact or Clebsch–Gordan second-order run at $N = 4$, where $\Lambda^2 F$ is real and both exceptional mechanisms are exercised at once, would be the first genuine test of the ledger beyond $SU(3)$.

4.6 Finite-volume width

Let $q_{\max}(L)$ be the largest sampled q on the periodic L^3 grid. Then

$$q_{\max}(L) = \begin{cases} 12, & L \text{ even,} \\ 12 \cos^2(\pi/2L), & L \text{ odd,} \end{cases} \quad (47)$$

because only at even L does some k_j reach π exactly. The width of the second-order truncated retained charge-odd manifold is exactly

$$W_{N,L}^{(-),[2]}(u) = q_{\max}(L) t_N u^2.$$

Thus for even L , and coefficientwise as $L \rightarrow \infty$ at fixed perturbative order, its u^2 coefficient is $12t_N \sim 3/N^3$. No volume-uniform third-order remainder is asserted. This concerns the stiffness of the dispersive doublet, not motion of the lowest branch, which is flat at this order.

The smallest nonzero incidence eigenvalue is $q_{\min} = 4 \sin^2(\pi/L)$, attained by one unit of momentum in a single direction.

4.7 The orientation signs are essential

Removing the orientation signs — from the entries of (19) and from the phases, so each $e^{ik} - 1$ becomes $e^{ik} + 1$ — gives an unsigned incidence $\mathcal{N}(k)$. Define its comparison adjacency by $A_+(k) := \mathcal{N}(k)\mathcal{N}(k)^\dagger - 4I$. Its adjacency spectra at Γ and R are $\{12, 0, 0\}$ and $\{-4, -4, -4\}$, which share no eigenvalue, so no level can be momentum independent. The flat branch is therefore not a generic consequence of three face orbitals or cubic symmetry.

Section 6 computes the unsigned comparison spectrum in closed form. The signed span is $8 - (-4) = 12$, giving the charge-odd second-order width $12t_3 = 5/51$ at $N = 3$. The unsigned continuous-zone span is $12 - (-4) = 16$. Interpreting $16|\ell_3| = 88/153$ as a charge-even bandwidth is conditional on deriving the complete linked charge-even symbol; on a finite torus the value -4 is sampled only when L is even.

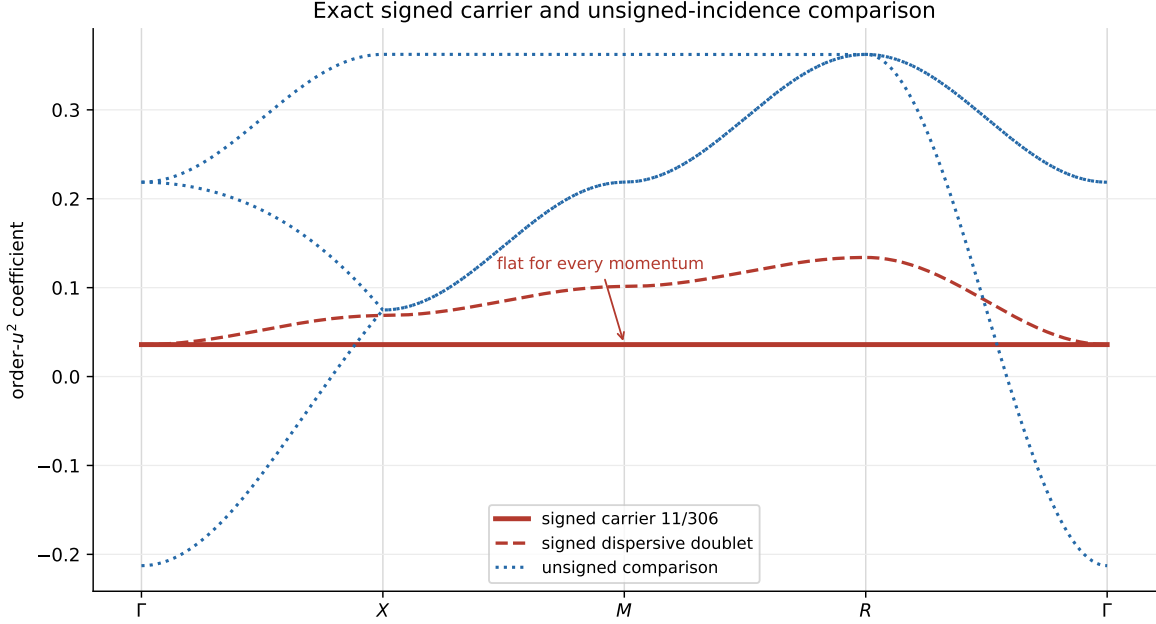


Figure 2: Order- u^2 coefficients along Γ - X - M - R - Γ . The red carrier and dispersive doublet are the signed incidence result; their global assembly is Corollary 24. The blue curves are the exact unsigned-incidence comparison spectrum. Reading the blue curves as physical charge-even bands requires the separate symbol hypothesis of Section 6.

5 The second-order coefficient on a two-cube space

Everything above derives t_N from a resolvent ledger indexed by the fusion channel of one shared link. That ledger is representation theory carried out on a single plaquette pair; it has never been tested against an operator calculation on a space large enough for two cubes to interact. This section reports such a test and, more importantly, states what it settles.

The construction is the open face-sharing $(3, 2, 2)$ prism with its eleven oriented plaquettes, in the exact-Casimir $\mathcal{B} = 6$ $SU(3)$ truncation, of basis dimension 1,590,462. The delivery reports that it reaches the relevant sector through 794 paths and 398 reachable states; it does not assemble a 1,590,462-dimensional magnetic matrix. Its complete charge-odd one-plaquette shell at $E_* = 8/3$ has dimension eleven — a census result and not the plaquette count. The full E_* eigenspace has dimension 22, split $11_{C=+} \oplus 11_{C=-}$ by tensor-derived charge conjugation, and the delivery removes all 22 from the pseudoinverse; because the eleven columns exhaust the odd half, that agrees with the $Q = 1_- - P_-$ prescription of (18). Writing K_{LR} for the raw second-order operator and K_L, K_R, K_F for the left-cube, right-cube and shared-face operators in their own coordinates, the connected operator is defined by the literal Möbius fold

$$K_{\text{conn}} = K_{LR} - J_L K_L J_L^\dagger - J_R K_R J_R^\dagger + J_F K_F J_F^\dagger, \quad (48)$$

and the same fold applied to the geometry gives G_{conn} . The fold is literal in the sense that no eigenvalue or fitted scalar is subtracted: each source operator is folded in its own coordinates and lifted. The delivery corroborates that internally by applying the same weights history by history over its 1,984 ordered two-step histories, of which 1,192 cancel at weight zero, and the two routes agree to 4.9×10^{-15} .

First order settles itself. Equation (29) gives $PVP = -I_{11}$ on the $SU(3)$ charge-odd shell, so the first-order Hamiltonian coefficient is $H_1 = -PVP = +I_{11}$. It shifts all eleven branches equally and creates no branch ambiguity. Its connected fold then vanishes *identically*, and by counting rather than by cancellation — each of the eleven faces is covered exactly once by $\{L, R\}$ once the shared face is restored by F , so every Möbius weight in (48) applied to the identity is $1 - 1 = 0$. Second order is therefore the leading connected term and not a correction to a first-order one.

At second order the result is

$$K_{\text{conn}}^{(2)} = \frac{5}{612} G_{\text{conn}} + D, \quad (49)$$

with D diagonal and delivered. The geometry alone fixes the *splitting* inside each block and nothing else: G_{conn} turns out to be four disjoint 2×2 blocks plus three isolated directions (Appendix F), so each pair splits by $2 \times 5/612 = 5/306$ about its own diagonal entry. Given D , which Appendix F prints, the spectrum is then closed form: 0 once, $-\frac{15}{4}$ twice, $-\frac{34}{9}$ four times, and $-\frac{129}{34}$ four times — exact arithmetic on entries that are themselves reconstructions, a boundary made precise below. The delivery reports the coefficient recovered target-blind — supplied to no builder, graph fit, branching rule or validator — and reports that the payload hashes were produced before the recorded comparison, a chronology it itself calls record-backed and only partly auditable; the value is byte-present in the nested $\mathcal{B} = 4$ package that the $\mathcal{B} = 6$ build seals as an authority object, so the defensible statement is dependency-path nonuse and not corpus-wide absence. This paper re-thresholds those delivered gates rather than re-running the build, including a deliberately wrong-sign control that the held-out scaling rejects: on a 66-dimensional held-out word space the correct second-order residual falls with log-log slope 2.90 while the sign-reversed control falls with slope 2.00, a minimum error ratio of 325. What all this establishes is runtime nonuse of the comparator on the delivered dependency path, which is not a preregistration. The finite- u validations are asymmetric: the $\mathcal{B} = 4$ delivery reports a direct diagonalisation of the full 4,180-dimensional charge-odd space at four small nonzero couplings, two held out. After subtracting $E_* + u + u^2 \text{eig } K_{\text{eff}}^{(2)}$, the observed residuals are reported as consistent with a leading cubic term on that fixed geometry; the delivery supplies neither an exact $[u^3]$ coefficient nor an analytic third-order remainder bound. The $\mathcal{B} = 6$ check is restricted to the 66-dimensional $P + Q_1$ star.

Resolving (49) by the irreducible representation the shared link actually carries in the ranked intermediate state gives six coefficients rather than four, because a link distinguishes ρ from $\bar{\rho}$:

$$\begin{array}{c|cccccc} \rho & \mathbf{1} & \mathbf{3} & \bar{\mathbf{3}} & \mathbf{6} & \bar{\mathbf{6}} & \mathbf{8} \\ \hline c_\rho & \frac{1}{12} & -\frac{1}{12} & -\frac{1}{12} & -\frac{1}{9} & -\frac{1}{9} & \frac{16}{51} \end{array} \quad (50)$$

That these six sum to $5/612$ is weak evidence: six rationals reach one target in many ways, and a fitted set would pass it. The statement worth making is stronger, and it is what makes this more than a coincidence of one sum. First, though, the boundary.

The six coefficients of (50) are not symbolic: the delivery stores them as rational reconstructions of pinned finite-precision Clebsch–Gordan contractions, with a maximum reconstruction residual of 2.2×10^{-15} per channel — 2.1×10^{-14} across the delivery’s reconstructed matrices as a whole — and its own certificate declaring `symbolic_exact_local_amplitudes: false`. What follows is therefore reported, conditional on those six numbers, and not a symbolic identity in the sense of Theorem 15.

Reported computational result 31 (The census is the Weingarten ledger). At $N = 3$ the six reconstructed link-irrep coefficients of (50) are the four fusion-channel weights (35) individually:

$$\begin{aligned} c_1 &= -w_1, & c_8 &= -w_{\text{Adj}}, \\ c_3 &= c_{\bar{3}} = \frac{1}{2}w_{\Lambda^2 F}, & c_6 &= c_{\bar{6}} = \frac{1}{2}w_{\text{Sym}^2 F}. \end{aligned}$$

Substituting (24) into (35) at $N = 3$ gives $w_1 = -1/12$, $w_{\text{Adj}} = -16/51$, $w_{\Lambda^2 F} = -1/6$ and $w_{\text{Sym}^2 F} = -2/9$; comparison with (50) is then arithmetic. Six reconstructed rationals landing on four predicted ones, with the two coincidences the correspondence requires, is a strong consistency statement and not a proof that the underlying contraction is exact.

One further feature of the census is worth separating out, because it is the paper’s central architecture measured rather than assumed. The six link-resolved matrices are not merely six numbers that sum correctly: each is *separately* proportional to the same incidence Gram. The eleven oriented plaquettes of the prism have exactly 56 ordered adjacent pairs, each sharing one link with signed overlap ± 1 ; the delivery reports a constant channel-to-geometry ratio on all 56 of them, for every one of the six channels, with worst residual 2.2×10^{-15} . Colour dynamics fixes a number per channel and the cellular boundary operator fixes the matrix; if the two mixed, the ratio would be constant for the total and not for the parts.

The correspondence is scoped to what the Casimir closure exhausts: adjacent, one-action, shared-link intermediates at order u^2 . Same-face diagonal routes, on-site terms and higher-action intermediates lie outside it — the $\mathcal{B} = 6$ scalar below is exactly a place where one of them is missing — and so does any untruncated-Hilbert-space statement.

It is scoped in one further way, which matters more. The two-cube build draws its local Wilson amplitudes from the same hash-pinned upstream Clebsch–Gordan archive as the single-cube reconstructions of the next subsection, and the squared amplitudes that archive produces are the published d_ρ/N^2 [33] — the numerator of (35). The census therefore does not confirm the weights from an independent source: it tests the *assembly* that carries them, which is the reachable-channel list, the resolvent denominators, the operator-level fold, the even split between conjugates, and the geometry that survives each channel separately. That is the statement worth making, and it is a different statement from confirmation.

Two features of the correspondence carry the content. The mixed family enters negated because $t_N = B_N - A_N$, which is the same relative sign the incidence orientation carries. And each like-family *fusion* weight is split evenly between the irrep and its conjugate — the two orientations of the shared link weighted equally — which the delivery resolves on the two-cube space and which the next proposition proves at every rank.

Proposition 32 (Link-resolved census, all ranks). *For every $N \geq 3$, resolve the adjacent charge-odd cross element of Theorem 23 by the orientation monomial that places $F \otimes F$ or $\bar{F} \otimes \bar{F}$ on the shared link, that is by the irrep ρ or $\bar{\rho}$ the link carries in the intermediate state. Then $c_1 = -w_1$ and $c_{\text{Adj}} = -w_{\text{Adj}}$ on the self-conjugate mixed channels, and the two like monomials carry $c_\rho = c_{\bar{\rho}} = \frac{1}{2}w_\rho$ each for $\rho \in \{\Lambda^2 F, \text{Sym}^2 F\}$, the six summing to t_N . For $N \neq 4$ the two like labels are distinct and this is the link-resolved split; at $N = 4$, where $\Lambda^2 F$ is self-conjugate, its two monomials land on one label and a census keyed by the irrep shows the single entry $w_{\Lambda^2 F}$.*

Proof. In the orientation table (40) the like family is reached by exactly two monomials, $(\epsilon, \epsilon') = (+, +)$ and $(-, -)$ when $ss' = 1$ and the other two when $ss' = -1$, each with coefficient $\epsilon\epsilon'/2 = \frac{1}{2}$; the first puts $F \otimes F$ on the shared link and the second $\bar{F} \otimes \bar{F}$, that is ρ and $\bar{\rho}$, with the same weight d_ρ/N^2 because Haar measure is conjugation invariant. The mixed channels are self-conjugate, and their two monomials add with the sign $\eta_\rho = -1$. Charge conjugation commutes with H_E , with V and with any link-Casimir cutoff, so the statement holds under every such truncation. $\square$

Revision 5 stated this as a conjecture and named a link-resolved census at $N \geq 4$ as its falsifier. No census is needed: the split is the bookkeeping of the shared-link theorem, and the delivered $N = 3$ census (50) is a test of the delivery against it, which it passes channel by channel [Lean: census_three].

5.1 One truncation, two constructions

The two-cube space also isolates why a cutoff can reverse the sign of t_3 . An adjacent shared-link channel ρ is reachable only if the endpoint Casimir budget $C_2(\rho) + 2C_2(\mathbf{3})$ fits inside the truncation: the sextet needs exactly 6 and the adjoint 17/3, so both survive $\mathcal{B} = 6$ and neither survives $\mathcal{B} = 4$.

Corollary 33 (What a link cutoff drops). *Retaining only the singlet and $\Lambda^2 F$ routes projects the second-order hopping to*

$$w_{\Lambda^2 F} - w_{\mathbf{1}} = -\frac{1}{12},$$

opposite in sign to t_3 and 10.2 times larger in magnitude. What is dropped is

$$w_{\text{Sym}^2 F} - w_{\text{Adj}} = \frac{14}{153},$$

and $-\frac{1}{12} + \frac{14}{153} = \frac{5}{612} = t_3$. This is arithmetic on (35) alone.

The published $p + q \leq 1$ link cutoff retains exactly those two routes — a per-link restriction, where $\mathcal{B} = 4$ is a total endpoint budget, so the two schemes agree on the retained channels at this order and are not the same rule — so $-1/12$ is its projected adjacent hopping and $14/153$ is the adjacent-hopping correction that completes it — not $5/612$, which would double-count what it already carries. On the two-cube space the first of those two numbers is delivered rather than inferred: a separately sealed $\mathcal{B} = 4$ run on the identical geometry gives $K_{\text{conn}}^{(2), \mathcal{B}=4} = -\frac{1}{12}G_{\text{conn}} + D_{\mathcal{B}=4}$ outright, and the Casimir budgets of the previous paragraph say without reference to any census that $\mathcal{B} = 4$ can reach only $\{\mathbf{1}, \mathbf{3}, \bar{\mathbf{3}}\}$. Conditional on (50), the same split of the census reproduces it, $c_{\mathbf{1}} + c_{\mathbf{3}} + c_{\bar{\mathbf{3}}} = -1/12$, and gives the omitted half as $c_{\mathbf{6}} + c_{\bar{\mathbf{6}}} + c_{\mathbf{8}} = 14/153$.

Remark 34. Two single-cube reconstructions bracket the same statement. A full $\mathcal{B} = 4$ open cube assembled from the pinned `ymcirc` $d = 3$, $\mathcal{B} = 4$ plaquette data [34, 35] reproduces $-1/12$ and the truncated shell ordering to 4.3×10^{-11} ; a $\mathcal{B} = 6$ cube reproduces $+5/612$ and the inverse ordering to 8.4×10^{-15} . Both locate the missing weight at the same $14/153$.

Neither is an independent replication. Both read tabulated plaquette matrix elements of the same lineage — and the two-cube build draws its Clebsch–Gordan amplitudes from the same hash-pinned `pyclebsch` archive as the $\mathcal{B} = 6$ cube, so that agreement is a cross-check of assembly and not of the amplitudes. The published closed form [33] is d_ρ/N^2 — the very numerator of (35). This is a cross-check of assembly, not a third confirmation of t_3 ; the exact rational identity is Corollary 33, which needs no tolerance. The $p + q \leq 1$ nomenclature and the finite-rank matrix element are both from [33], and single-cube $SU(N)$ diagonalisation in this style goes back to [23].

The correction to the adjacent-hopping coefficient of a published T_1 Hamiltonian is therefore $14/153$ and not $5/612$, which would double-count the two routes T_1 already carries. One coincidence of value is worth naming: $-1/12$ is also the singlet weight $w_{\mathbf{1}}$, and the cutoff value is a *difference* of two weights, not a weight.

The $\mathcal{B} = 6$ cube is channel-complete for the hopping and *not* for the scalar: its scalar is $39/68$ against the bridge’s $11/34$, differing by exactly $1/4$, the local same-face sextet route, which first enters at cutoff $20/3$ — integer label $\mathcal{B} = 7$, so any cutoff above $20/3$ already carries it.

5.2 What the cutoff moves, and where

The connected diagonal D is eleven numbers in both deliveries and is treated as a remainder in both. It is not eleven numbers. The open prism has a symmetry group — the cell exchange $x \mapsto 2 - x$, the two transverse reflections and the $y \leftrightarrow z$ swap, of order 16 — and D is constant on its orbits, which are $8 + 2 + 1$: the eight side faces, the two end caps, the shared face.

Proposition 35 (Where the truncation lives). *The eight-element orbit is exactly the support of the four cross-cell pairs, and it is the only orbit whose diagonal value differs between the two truncations: the end caps carry $-15/4$ and the shared face 0 at both $\mathcal{B} = 4$ and $\mathcal{B} = 6$, while the side value moves from $-7/4$ to $-2317/612$. Together with the four cross-cell blocks carrying the whole off-diagonal change, the entire $\mathcal{B} = 4 \rightarrow \mathcal{B} = 6$ difference is confined to the faces that carry connected transport.*

This also says what the nonuniform D is. Where the faces of a complex form a *single* orbit — the closed cube of the next subsection, or the periodic torus under Corollary 24 — orbit-constancy forces the diagonal to be scalar, and the operator is exactly $\alpha I + tG$. The prism has three orbits because it has a boundary. D is the open boundary showing up in the diagonal, not a failure of the incidence form.

5.3 The one-cube shell, and why its flat level cannot move

Those two single-cube reconstructions are usually read as numerical cross-checks. They are more than that, because the object they diagonalise is the same chain complex. The six faces of a cube are a closed surface, so in the outward orientation $\ker \partial_2$ is one-dimensional and spanned by the sum of all six — the fundamental class, $b_2(S^2) = 1$. That is Theorem 9 on a complex with no three-cells, where the $L^3 - 1$ boundaries are absent and only the class survives.

Proposition 36 (One-cube shell). *On the closed cube surface $G = \partial_2^\top \partial_2$ commutes with all 24 proper rotations. The combinations $b_{+i} - b_{-i}$ carry the defining vector representation and the combinations $b_{+i} + b_{-i}$ the permutation representation of the three unordered axes, so with the charge-odd inversion $Pb_{+i} = -b_{-i}$ the shell is*

$$A_1^{--} \oplus T_1^{+-} \oplus E^{--},$$

with G -eigenvalues 0, 4, 6 and multiplicities 1, 3, 2. Consequently $\alpha I + tG$ has levels α , $\alpha + 4t$, $\alpha + 6t$.

Proof. The rotations permute the six outward faces, and G is built from the boundary map, so it commutes with the permutation action. On the antisymmetric combinations that action is the defining representation by inspection — $b_{+i} - b_{-i}$ transforms as e_i — and on the symmetric combinations it is the permutation representation of $\{\pm e_i\}$, which is $A_1 \oplus E$. Under $Pb_{+i} = -b_{-i}$ the antisymmetric sector is parity even and the symmetric one parity odd. In these sectors G is $4I$ and $4I - 2(J - I)$ respectively, of spectra $\{4, 4, 4\}$ and $\{0, 6, 6\}$; since the three isotypic components have distinct dimensions the eigenvalues attach to them uniquely. $\square$

Three things follow. The cubic labels are derived here rather than assigned: both deliveries assert them and neither builds the symmetry matrices. The A_1^{--} level is Proposition 48's mechanism on a second complex — $Gw = 0$ there, so that level sits at α for every t , and no truncation, no channel and no coefficient in dispute anywhere in this paper can move it. And the reversal the two runs report is therefore a statement about the sign of one number rather than about the shell as a whole: at $\mathcal{B} = 6$ the levels are $\{39/68, 371/612, 127/204\}$ and at $\mathcal{B} = 4$ $\{37/12, 11/4, 31/12\}$, in the opposite order, with A_1^{--} in each case sitting exactly at the scalar. The relative shell $\{0, 4t, 6t\}$ is $\{0, 5/153, 5/102\}$ at $\mathcal{B} = 6$.

The finite-volume fingerprint of the 28 August finite- N bridge certificate — a different release, and named here because a reader tracing it to the two-cube one would find nothing — is worth recording because its most discriminating multiplicity is not a feature of the truncation: on the periodic 3^3 torus the incidence eigenvalues 0, 3, 6, 9 have multiplicities 29, 12, 24, 16, respectively, and $29 = L^3 + 2$ is Theorem 9.

Remark 37 (Scope of the second-order delivery). Everything above in this section is second order in u , finite volume, strong coupling and charge odd. It bears on t_N and decides nothing about the fourth-order coefficient of Section 9. The $\mathcal{B} = 6$ two-cube construction was not re-run here: its arithmetic is audited against geometry rebuilt independently, and its diagonal D is B -truncated and not proved cutoff-stable. The older held-out finite- u validation that rejects the wrong-sign control is a 66-dimensional $P + Q_1$ word space with the Q_1 - Q_1 magnetic block set to zero; it is not the radius-two $K_{2,\star}$ operator defined below. The two branches are not equally covered: at $\mathcal{B} = 4$ the delivery reports a direct diagonalisation of the complete 4,180-dimensional charge-odd block at four small couplings, two of them held out. The residual after $E_* + u + u^2 \text{eig } K_{\text{eff}}^{(2)}$ was described by the delivery as consistent with a leading cubic term; no exact $[u^3]$ coefficient or analytic remainder estimate is certified. At $\mathcal{B} = 6$ no such diagonalisation exists and the 66-dimensional star stands in for it. That $\mathcal{B} = 4$ report carries no check here — its certificate did not travel — and is recorded as the delivery’s rather than as this paper’s. No external group has reproduced that release. The later radius-two release is a distinct construction. It neither upgrades the $\mathcal{B} = 6$ cutoff-stability claim nor supplies a radius-three test.

5.4 A detached-replayed radius-two Ritz extension

A later release on the same open $(3, 2, 2)$ prism supplies charge-odd cumulative dimensions

$$\dim P, \dim(P + Q_1), \dim(P + Q_1 + Q_2) = 11, 77, 773$$

and charge-even dimensions 1, 2, 9. The release calls this construction cutoff-free: here that means no $\mathcal{B}$ cutoff inside the declared one- and two-action frontiers. The radius-two projection itself remains a truncation of the gauge-theory Hilbert space.

To keep action radius distinct from perturbative order, write the three nested matrices as K_1 , $K_{2,\star}$ and $K_{2,\text{full}}$. In $P + Q_1 + Q_2$ block order,

$$H_0 = \text{diag}(E_s I, H_1, H_2), \quad V = \begin{pmatrix} A_0 & B & 0 \\ B^\dagger & G & C \\ 0 & C^\dagger & D_2 \end{pmatrix}. \quad (51)$$

Here $E_s = 8/3$ in the odd seed and $E_s = 0$ in the even seed. The three finite-coupling operators use $K(u) = H_0 + uV = E - uM$, $V = -M$, $u = g^{-4}$:

$$\begin{aligned} K_1(u) &= (H_0 + uV)|_{P+Q_1}, \\ K_{2,\star}(u) &= H_0 + u[V \text{ with only } D_2 = Q_2 V Q_2 \text{ set to zero}], \\ K_{2,\text{full}}(u) &= (H_0 + uV)|_{P+Q_1+Q_2}. \end{aligned} \quad (52)$$

Thus “full” means complete only within the retained Q_2 block.

Proposition 38 (Onset of the nested-radius differences). *Let $R_i = (E_s I - H_i)^{-1}$. If $PVQ_2 = 0$, the leading seed-space operator present in $K_{2,\star} - K_1$ is*

$$\mathcal{I}_4 = BR_1CR_2C^\dagger R_1B^\dagger, \quad \mathcal{I}_5 = BR_1CR_2D_2R_2C^\dagger R_1B^\dagger, \quad (53)$$

where $\mathcal{I}_5$ is the leading term present in $K_{2,\text{full}} - K_{2,\star}$.

Proof. Any contribution that enters Q_2 must follow the shortest closed block walk $P \rightarrow Q_1 \rightarrow Q_2 \rightarrow Q_1 \rightarrow P$, which contains four insertions of V . Distinguishing the full operator from the star operator additionally requires one insertion of D_2 , hence five insertions. The blocks A_0 and G may dress these walks but cannot shorten them. Evaluating the intermediate resolvents at the seed energy gives (53). $\square$

The proposition is exact conditional on the declared frontier and sealed block structure. The following values are direct floating-point evaluations of its closed products, not exact rationals and not eigenvalue extrapolations.

Reported computational result 39 (Detached radius-two replay). The lowest eigenvalue of the odd second-order seed operator is isolated by 0.08438094122012973; the even seed is one-dimensional. Projection of (53) gives

	$C = -$	$C = +$	odd-even gap
$\mathcal{I}_4$	-25.63495194947556	-24.807728943850304	-0.8272230056252567
$\mathcal{I}_5$	-4.933069330883140	-9.116955631684514	4.183886300801373

The last entry is the leading $K_{2,\text{full}} - K_{2,\star}$ contribution to the retained odd-even gap. It is not the complete fifth-order coefficient of either unablated sector Hamiltonian. The weak- u cubic intercept 4.183884987168061 differs by 1.31×10^{-6} ; the direct product, not that fit, defines the reported coefficient.

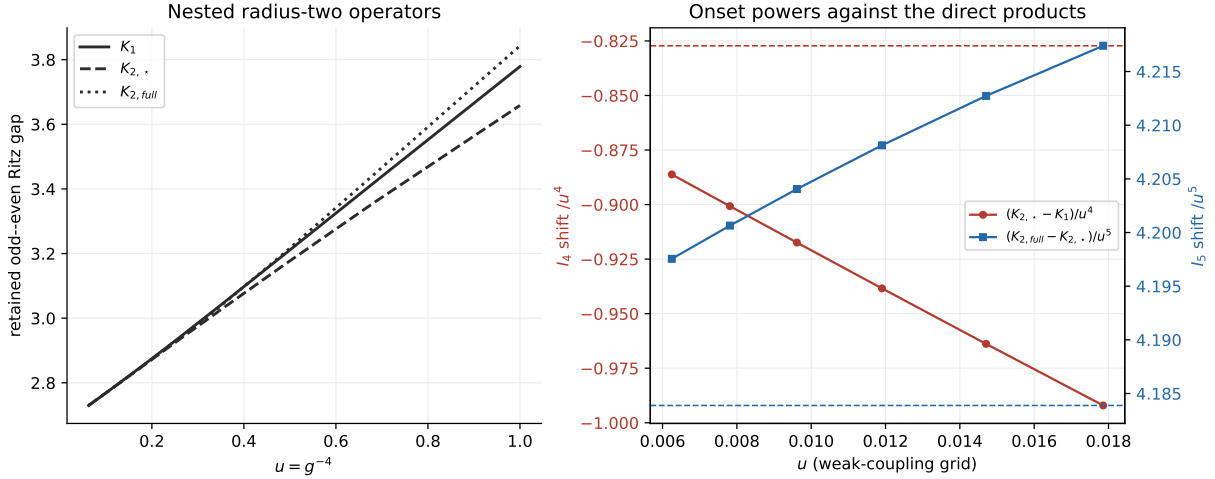


Figure 3: Nested radius-two finite- u evidence. Left: retained odd-even Ritz gaps for (52). Right: Hellmann-Feynman path-integral shifts divided by their predicted onset powers; dashed lines are the direct products in Reported result 39. These are retained-space trajectories, not convergence bands or full-Hilbert-space error estimates.

The hash-pinned detached verifier starts from both sealed parity-sector parent operators and recomputes the two coupling grids, all three nested spectra, residuals, interlacing, direct products, rephasing and adaptive path integrals. It returned zero maximum recomputation error and maximum rephasing discrepancy 4.09×10^{-14} ; all 24 hostile, algebra, numerical-safety and serialization tests passed. This is an independent downstream replay of the sealed parents, not an independent derivation of their microscopic gauge-basis contractions.

The boundary is quantitative. No odd Q_3 matrix is present, so no matched radius-two error enclosure or convergence claim exists. At $g = 1$ the odd and even ground seed weights have fallen to 0.274 and 0.277, while the even exterior residual is 3.784; the retained eigenvalues are still accurately computed, but their accuracy as approximations to a larger space is uncontrolled there. Nor can an open prism determine the periodic off-axis fourth-order shape of Section 9: $\mathcal{I}_4$ and $\mathcal{I}_5$ in (53) do not choose a value of C_{shp} .

6 The charge-even sector and the unsigned incidence

The unsigned incidence $\mathcal{N}(k)$ of Section 4 defines an exact algebraic operator $G_+(k) = \mathcal{N}(k)\mathcal{N}(k)^\dagger$. At order u^2 its identification with the complete linked charge-even effective Hamiltonian is Theorem 27: the all-pairs vacuum route of Proposition 25 is cancelled pair by pair, and the adjacent element is ℓ_N whatever the orientations. At order u^3 the identification remains a hypothesis, because the third-order charge-even coefficients come from the supplied domino ledger and no third-order process census has been carried out here. With that boundary explicit, write

$$\hat{a}_m = 2 + 2 \cos k_m = |1 + e^{ik_m}|^2, \quad p = \sum_m \hat{a}_m.$$

The hat is not decoration: $\hat{a}_m$ is the *complement* of the $a_m = 4 \sin^2(k_m/2)$ used in the charge-odd shape basis (55) below, $\hat{a}_m + a_m = 4$, and that complementarity is the sum rule below.

Theorem 40 (Unsigned-incidence Bloch cubic). *$p + q = 12$ pointwise, and the characteristic polynomial of $\mathcal{N}\mathcal{N}^\dagger$ is*

$$\mu^3 - 2p\mu^2 + p^2\mu - 4\hat{a}_1\hat{a}_2\hat{a}_3 = \mu(\mu - p)^2 - 4\hat{a}_1\hat{a}_2\hat{a}_3,$$

with the unsigned comparison-adjacency eigenvalues $\lambda = \mu - 4$. The signed counterpart of the same computation is $\mu(\mu - q)^2$.

The algebraic closed form is complete rather than merely consistent at sampled momenta: the characteristic polynomial of the 81×81 and 192×192 finite plaquette adjacencies factorises coefficient for coefficient over $\mathbb{Z}$ into this cubic over the momentum grid.

Proposition 41 (Range and attainment). *$\lambda \in [-4, 12]$ exactly. The top $\lambda = 12$ is attained only at Γ ($\hat{a}_1 = \hat{a}_2 = \hat{a}_3 = 4$). The bottom $\lambda = -4$ is attained on the whole of the three planes $k_j = \pi$, where the constant term vanishes, and only at R is it a triple root.*

Proof. Write $f(\mu) = \mu(\mu - p)^2 - 4\hat{a}_1\hat{a}_2\hat{a}_3$. Then $f(0) = -4\hat{a}_1\hat{a}_2\hat{a}_3 \leq 0$ and $\mu(\mu - p)^2 < 0$ for $\mu < 0$, so no root lies below 0; and $f'(\mu) = (3\mu - p)(\mu - p) > 0$ for $\mu > p$, with $\hat{a}_1\hat{a}_2\hat{a}_3 \leq (p/3)^3$ by AM–GM and

$$f(16) \geq 16(16 - p)^2 - \frac{4p^3}{27} = \frac{4}{27}(12 - p)(48 - p)^2 \geq 0.$$

Hence $\mu \in [0, 16]$ and $\lambda = \mu - 4 \in [-4, 12]$. Equality at the top forces $p = 12$ and $\hat{a}_1\hat{a}_2\hat{a}_3 = 64$, which is Γ . Equality at the bottom is $f(0) = 0$, i.e. $\hat{a}_1\hat{a}_2\hat{a}_3 = 0$, i.e. some $k_j = \pi$; there $f(\mu) = \mu(\mu - p)^2$ and the root $\mu = 0$ is simple unless $p = 0$, which is R alone. $\square$

Remark 42. A flat unsigned-comparison branch would need $f(\mu_0) = 0$ identically in k , which the constant term already forbids. Its minimum is attained on three measure-zero planes, against the signed operator's flat band. This contrast is an exact statement about two incidence constructions; at order u^2 the two constructions are the two physical charge sectors, by Corollary 24 and Theorem 27. The range bound itself is proof-checked over the reals: for $a_i \in [0, 4]$ the cubic is strictly negative below $\mu = 0$ and strictly positive above $\mu = 16$, so every root lies in $[0, 16]$ [Lean: even_cubic_neg_below_zero], [Lean: even_cubic_pos_above_sixteen].

6.1 The charge-even assembly, and where it is conditional

The shared-link coefficient at $r = 2$ is all-rank and analytic, and Theorem 27 makes it the charge-even band structure at that order, with the scalar of Theorem 29; nothing at $r = 2$ is supplied. The $r = 3$ values — $t_{3,+} = -6335/249696$ and everything computed from it — come from the same supplied domino ledger as Section 7, and are conditional on it in the same way.

The scalar ledger is not a list of separate facts. Writing λ for the adjacency eigenvalue and $\text{tower}_{r,s}$ for the within-plaquette term in canonical u ,

$$E_s(\lambda, r) = \text{tower}_{r,s} + 12 \text{leak}_{r,s} + \lambda t_{r,s}, \quad (54)$$

with $\lambda \in \{-4, 8\}$ in the signed charge-odd sector (carrier and dispersive top), $\lambda \in \{12, -4\}$ in the unsigned comparison, and $\lambda = 0$ returning the on-site diagonals. This reproduces every registered diagonal and band value — nine of them — across both sectors and both orders. The tower term is the certified coupling conversion $4\Delta(3u/2)$, so (54) takes no unregistered input. The twelve is a count on this lattice, not a citation: the plaquette graph is 12-regular and two distinct faces share at most one link. The two towers are different objects and both are needed: the charge-odd one contributes $8/3 + u + u^2/2 + 7u^3/32$ and the charge-even one $8/3 - u + (13/20)u^2 + (101/200)u^3$, each of them $4\Delta(3u/2)$ on its own sector's series, so $\text{tower}_{2,-} = 1/2$ against $\text{tower}_{2,+} = 13/20$ and $\text{tower}_{3,-} = 7/32$ against $\text{tower}_{3,+} = 101/200$.

Because the supplied ledger has $\text{leak}_{r,+} = t_{r,+}$ at both computed orders, the conditional charge-even assembly collapses to $\text{tower}_{r,+} + (12 + \lambda) t_{r,+}$, a one-parameter family in the charge-even tower. That equality is verified and unexplained: at third order the charge-odd leakage separates from the charge-odd hopping while the charge-even identity survives. It is recorded as a unifying candidate with a falsifier — an order at which the two are computed independently and differ — not as a mechanism.

6.2 Where the rest frame is not blind

Proposition 43 (Charge-even Γ splitting at second order). *At Γ the unsigned spectrum is $\{12, 0, 0\}$, so by Theorem 27 the A_1^{++} level and the E^{++} doublet are distinct at every rank $N \geq 3$, and*

$$E(A_1^{++}) - E(E^{++}) = 12 \ell_N u^2 + O(u^3),$$

which is $-\frac{22}{51}u^2$ at $N = 3$.

Conditional corollary 44 (Third order). If the unsigned symbol is also the complete linked charge-even symbol at order u^3 , the splitting's cubic coefficient is $12t_{3,+} = -6335/20808$.

This is the structural counterpart of the charge-odd blindness recorded in Section 7. The signed spectrum at Γ is $\{-4, -4, -4\}$, one level, so no charge-odd rest-frame series can see the hopping; the unsigned spectrum has two levels, so the charge-even rest frame fixes ℓ_N at second order and, under the third-order symbol hypothesis, $t_{3,+}$. The A_1^{++} branch is isotropic at leading order without assuming it, with curvature $\frac{4}{3}|t_{r,+}|$ in all three directions — $22/459$ at second order and $6335/187272$ at third.

On a finite L^3 grid the conditional charge-even width is

$$W_{r,L}^{(+)} = |t_{r,+}| [12 - \lambda_{\min}^{(+)}(L)], \quad \lambda_{\min}^{(+)}(L) = -4 \iff L \text{ is even.}$$

Thus it equals $16|t_{r,+}|$ for even L and approaches that coefficientwise as $L \rightarrow \infty$; the charge-odd manifold width is $12|t_{r,-}|$ for even L . At any order at which the correction stays in the

incidence algebra — which Proposition 48 warns is not automatic — the second-order ratio is $4(3N^2 - 5)/[3(N^2 - 4)]$, finite and greater than one at every rank $N \geq 3$, decreasing from 88/15 at $N = 3$ towards 4 [Lean: width_ratio_numerator]. So at second order the charge-even band is the wider one at every rank, unconditionally, and that asymmetry is a channel statement rather than a geometric one.

7 $SU(3)$ through third order

The *third-order* content of this section is an externally verified computational input: the frozen upstream corpus reports a 251/251 cold exact replay for the retained chain, while this compact release checks only the assembled rationals. At second order both the hopping t_3 and the flat scalar 11/306 are analytic: Theorem 29 gives the tower $\frac{1}{2} = \sigma_3^- + 1/C_F$ and the leakage $\text{leak}_2 = \ell_3$, which the upstream ledger had recorded as an identification. The within-plaquette tower is $8/3 + u + u^2/2 + 7u^3/32$, its cubic term from the supplied ledger.

Reported computational result 45 (Third-order ledger). The supplied ledgers report, in exact rationals,

$$b_3 = \frac{1975}{124848}, \quad \text{leak}_3 = -\frac{12331}{249696},$$

and the assembled flat scalar $d_3 = \frac{7}{32} + 12\text{leak}_3 - 4b_3 = -\frac{109151}{249696}$.

Conditional corollary 46 (Third-order factorisation). If the lifter census is exhaustive and the linked ledger complete, then

$$T_{\leq 3}H_{\text{eff},-}(k, u) = E_{\text{flat}}^{[3]}(u)I + \tau^{[3]}(u)B(k)B(k)^\dagger,$$

$$\begin{aligned} E_{\text{flat}}^{[3]}(u) &= \frac{8}{3} + u + \frac{11}{306}u^2 - \frac{109151}{249696}u^3, \\ \tau^{[3]}(u) &= \frac{5}{612}u^2 + \frac{1975}{124848}u^3. \end{aligned}$$

This is an equality of Taylor coefficients through order three at each fixed volume; no L -uniform fourth-order remainder is claimed.

The second-order diagonal is the $\lambda = 0$ value $\frac{1}{2} + 12\text{leak}_2 = 7/102$ with $\text{leak}_2 = -11/306$; subtracting $4t_3$ gives the flat scalar 11/306 at $\lambda = -4$. Both are now Theorem 29 at $N = 3$.

Remark 47 (What the external comparison can and cannot do). Hamer’s 1989 rest-frame series [20] agrees with $E_{\text{flat}}^{[3]}$ at every shared order. It cannot test $\tau^{[3]}$. The hopping enters the coefficientwise spectrum only through $\tau^{[3]}(u)q(k)$ and $q(0) = 0$, so every *charge-odd* Γ -point datum is blind to it. The charge-even rest frame is not, by Proposition 43; the one-parameter family $(b_3 + \varepsilon, \text{leak}_3 + \varepsilon/3)$ leaves d_3 and therefore the entire comparison unchanged. What the agreement pins is the combination $12\text{leak}_3 - 4b_3$. The $\lambda = 8$ band top separates them, via (54). Proposition 43 is the charge-*even* analogue and separates leak_+ from t_+ ; it says nothing about b_3 or leak_3 .

8 What the homology protects

Proposition 48 (Boundary-factorised rigidity). *If an order- r Bloch correction has the form $H_r(k) = a_r I + b_r S(k) + B(k)M_r(k)B(k)^\dagger$, then for every $k \neq 0$, $H_r(k)w(k) = (a_r - 4b_r)w(k)$: the correction shifts the carrier without dispersing it.*

Proof. $B^\dagger w = 0$ and $Sw = -4w$, both from Theorem 8. The middle term is annihilated whatever M_r is, which is the content: the protection belongs to the boundary operator and not to the dynamics. $\square$

In real space this is the orthogonality of the two cellular Laplacians, $L_2^\downarrow L_2^\uparrow = L_2^\uparrow L_2^\downarrow = 0$, so any polynomial in them acts on $\ker L_2^\downarrow \cap \ker L_2^\uparrow$ by its constant term. The proposition is not an unconditional all-orders statement: $\{BMB^\dagger\}$ is not a two-sided ideal in $\text{End}(C_2)$, and a general higher-order correction need not factor through links. Cubic symmetry alone permits carrier-projected fourth-order shapes built from

$$1, \quad q, \quad e_2, \quad \frac{4e_2}{q}, \quad \frac{e_3}{q}, \quad (55)$$

with $e_2 = \sum_{i < j} a_i a_j$, $e_3 = a_1 a_2 a_3$ and $a_i = 4 \sin^2(k_i/2)$. The $1/q$ is the carrier projection: $\|w\|^2 = q$ by Theorem 8, so the carrier-projected correction is $\varepsilon_4 = (w^\dagger H_4 w)/q$ and the numerator, not ε_4 , is the polynomial.

9 The fourth-order boundary

Two supplied fourth-order records agree on the axial datum and on the appearance of mobility, and disagree on one off-axis mixed-gradient coefficient C_{shp} . This edition does not adjudicate the coefficient, and no fourth-order band or bandwidth is used above. What it does is three things, each checked. It locates the disagreement exactly, as the weight of the single operator outside the algebra of the two cellular Laplacians (Section 9.5). It reports three of the kernel's amplitudes recomputed from outside both pipelines, which decides the *standing* of the two records without deciding the coefficient, and puts a third value on record (Section 9.6). And it keeps the obstruction theorem, which says why no rest-frame or axial datum can decide it.

After subtracting the rest value, the recorded carrier ansatz is

$$\Delta_4(k) := \varepsilon_4(k) - \varepsilon_4(0) = Aq + Be_2 + C_{\text{shp}} \frac{4e_2}{q} + D \frac{e_3}{q}. \quad (56)$$

The continuous $k \rightarrow 0$ limit is used at $q = 0$. Denote the two recorded mixed-gradient values by C_{old} and C_{new} and set $\Delta_C = C_{\text{new}} - C_{\text{old}}$. A supplied all-rank cube ledger also reports

$$\alpha_N = \frac{640}{N(N^2 - 1)^3}, \quad (57)$$

but this formula and its identification with the complete axial coefficient remain conditional on that ledger. Sealed packages containing other fourth-order scalars use incompatible observable labels; none is promoted here to a canonical rest coefficient.

9.1 An exact theorem for the saved historical kernel

The upstream authority corpus supplies the hash-pinned 189-record historical kernel. Pulling its centered fourth-order operator back to cube amplitudes with $C = \partial_3$, define

$$Q_4 = C^\dagger H_4 C, \quad G = C^\dagger C, \quad \mathcal{Q}_4 = Q_4 - s_4 G,$$

where $s_4 = \text{tr } H_4(\Gamma)/3$. If $L_i = 2I - T_i - T_i^{-1}$, then $G = \sum_i L_i$ and the centered coefficient is the generalized eigenvalue $\mathcal{Q}_4 \phi = \lambda_4 G \phi$. The representative change $(Q_4, G) \sim (Q_4 + \delta G, G)$ is a scalar gauge only when the scalar anchor changes simultaneously; it cannot change an off-axis shape.



Figure 4: Exact dispersion of the saved historical kernel in Reported result 49. The plot is a theorem about that hash-pinned kernel. Its identification with the complete physical linked $SU(3)$ fourth-order operator is disputed, so the curve is not used as a settled physical band.

Reported computational result 49 (Exact saved-kernel Hodge pencil). For the supplied historical kernel,

$$q_{\text{old}}^{(4)} = -\frac{20721577909065127111}{7250590288602460800}, \quad \alpha_{\text{old}} = \frac{5}{12}, \quad \beta_{\text{old}} = \frac{17607806155349}{275331901291200},$$

and

$$\mathcal{Q}_{4,\text{old}} = \frac{5}{48} \sum_i L_i^2 + \frac{17607806155349}{1101327605164800} \sum_{i < j} L_i L_j \succeq 0. \quad (58)$$

Its continuous-zone and even- L width is

$$W_{4,\text{old}} = \alpha_{\text{old}} + \beta_{\text{old}} = \frac{132329431693349}{275331901291200}.$$

These identities are exact for the saved kernel and its cold fixed-kernel checks. They do not establish that this kernel is the unique physical linked fourth-order operator.

The corresponding 25-point numerator stencil has

$$w_0 = \frac{9}{2}\alpha + 3\beta, \quad w_1 = -(\alpha + \beta), \quad w_2 = \frac{1}{4}\alpha, \quad w_d = \frac{1}{4}\beta,$$

and obeys $w_0 + 6w_1 + 6w_2 + 12w_d = 0$ exactly. For $\alpha, \beta > 0$, the generalized dispersion is bounded by $0 \leq \lambda_4 \leq \alpha + \beta$, with the continuous-zone extrema at Γ and R respectively; an odd- L grid does not contain R .

The two records do agree on the axial datum, but the corpus's own quoted tolerance does not cover that agreement: the α gap is 2.43×10^{-13} against a stated bound of 2.3×10^{-13} . The sealed core holds internally — α_{new} and $4A_{\text{new}}$ agree to 2×10^{-15} — which is consistent with the quoted bound being too tight, but two float comparisons do not discriminate that from a shared upstream slip. Recorded as a discrepancy rather than resolved, and the bound is not widened.

9.2 The obstruction is polynomial

Theorem 50 (Obstruction certificate). *The two recorded off-axis coefficients enter (55) only through the shape $4e_2/q$, so the two records' carrier numerators differ by $4\Delta_C e_2$ and their bands by $\Delta_C (4e_2/q)$. On any one-dimensional axial cut two of the three a_i vanish, so e_2 is the zero polynomial there and both differences vanish identically. Consequently no Γ -point or axial datum distinguishes the two records, at any precision. Off axis they do differ: $e_2(M) = 16$ and $e_2(R) = 48$, so the numerators differ by $64\Delta_C$ and $192\Delta_C$ and the bands by $8\Delta_C$ and $16\Delta_C$.*

Remark 51. The vanishing is polynomial, not numerical: this is non-identifiability, not a limitation of the data in hand. Re-anchoring cannot substitute, and neither record is preferred here. An off-axis contraction is the only decider.

On the axial cut itself the invariants collapse to $q = q_{\text{ax}}(k) = 4\sin^2(k/2)$ with $e_2 = e_3 = 0$. Dividing the raw cube-boundary numerator $(\alpha_N/4)q_{\text{ax}}^2$ by $\|w\|^2 = q = q_{\text{ax}}$ leaves one power of q_{ax} with coefficient $\alpha_3/4 = 5/48$.

9.3 A second witness

Non-identifiability is better exhibited than argued from a difference between two records, since a difference could always be a computational error in one of them. The witness below is not itself a candidate for the physical coefficient. It is the balanced continuation obtained in the supplied source by setting an auxiliary bookkeeping parameter ϵ to zero; that source label carries no physical meaning here. Its sole role is to make the $4\Delta_C e_2$ family concrete with an exactly known shift.

Proposition 52 (Explicit second witness). *There is a value C_{alt} with $C_{\text{alt}} - C_{\text{old}} = 25/1024$ exactly, appearing verbatim in the pinned source edition, whose induced band separations are 0 at X , $25/128$ at M and $25/64$ at R . It is identical to C_{old} on every retained datum and distinct from it off-axis.*

9.4 Where the coefficient lives

The four checkpoint deltas are the Γ -subtracted carrier corrections $\Delta_s = \varepsilon_4(k_s) - \varepsilon_4(0)$ at $k_X = (\pi, 0, 0)$, $k_M = (\pi, \pi, 0)$, $k_P = (\pi, \pi/2, 0)$ and $k_R = (\pi, \pi, \pi)$ — P is the off-axis, off-corner point that makes the system invertible, with $a = (4, 2, 0)$. Its algebraic role is explicit here; no additional provenance is inferred for that checkpoint. In the shape basis they are

$$\begin{aligned}\Delta_X &= 4A, \\ \Delta_M &= 8A + 16B + 8C_{\text{shp}}, \\ \Delta_P &= 6A + 8B + \frac{16}{3}C_{\text{shp}}, \\ \Delta_R &= 12A + 48B + 16C_{\text{shp}} + \frac{16}{3}D,\end{aligned}\tag{59}$$

and they invert exactly, so the four shape coefficients are recoverable from four checkpoints. Δ_X contains A alone, which is why the axial cuts agree whatever the dispute does.

Four further measurements locate the coefficient and bound where a disagreement could live. The distinction matters and is easy to lose: showing where a coefficient's *value* is concentrated is not by itself showing where a *discrepancy* can sit. Only the third item below does the second thing.

1. **The shape fit's C row sums to zero**, so no error in the Γ anchoring can move C_{shp} at all; the recorded run's own re-anchoring moved it by 4.6×10^{-15} , as the zero row sum requires. A scalar shift is translation-local and changes nothing observable, which is why the two anchored scalars $q_{\text{band}}^{(4)}$ and $m_{\Gamma}^{(4)}$ are differently anchored coordinates and not rival estimates.

2. **The coefficient's value is concentrated, not spread.** Six of the 189 kernel records — the normal $(0, 0, 1)$ shell — carry 81% of the total weight, while the 120 rotation records carry 6% and are 255 times less sensitive per record. This says where the coefficient lives, not yet where the two records can differ.
3. **The agreed axial coefficient pins the normal sector.** All six normal records share one amplitude h , and it sets both shapes rigidly: $A_{\text{normal}} = -h$ while the raw normal amplitude is $2h$, which in the shape normalisation of (55) is $C_{\text{normal}} = h/2$, so

$$C_{\text{normal}} = -\frac{1}{2}A_{\text{normal}} \quad \text{exactly.}$$

(The factor of four between $2h$ and $h/2$ is the raw-to-shape conversion, not a discrepancy.) Granting $A = 5/48$ — which both records share — together with the block decomposition, which puts $5/48 + 4x$ on the normal block and $-4x$ on the mixed one, therefore fixes h and with it 81% of the coefficient; and everything free to move without disturbing A totals less than half the disputed gap. A rival kernel must differ *structurally* in the A -carrying sector; no redistribution reaches the gap.

4. **The two kernels agree almost everywhere.** Their supports are identical and 144 of 189 records ride one common scale; the divergent cross sector is a single uniform ratio. The dispute is three amplitudes.

Read together these four sharpen the target and prefer no side of it. They say that a rival kernel must differ structurally in the A -carrying sector rather than by redistribution. That is a constraint on candidates built on the recorded 189-record support — which is where the measurement was made; a kernel on a different support is not constrained by it at all. Neither record is promoted here and neither is withdrawn: the interval of *values* between them is not narrowed. What is narrowed is the class of *kernels* that can reach either end, and that constraint bears on both records alike.

9.5 The kernel is a Hodge polynomial plus one operator

Write $L^\downarrow = \partial_2^\dagger \partial_2$ and $L^\uparrow = \partial_3 \partial_3^\dagger$ for the two cellular Laplacians on plaquettes, $S_\square = L^\downarrow - 4I$ for the signed shared-link adjacency of Theorem 8, and R for the cross-plane half of $S_\square$ (equivalently, minus the cross-plane half of $L^\uparrow$). The 189-record historical kernel carries exactly six weight magnitudes, one per cubic orbit of displacements: the on-site σ , the normal ν , the in-plane π , the two-hop u with its doubled orbit $u_2 = 2u$, and the rotation ρ . Over the whole zone the shape coefficients are closed form in them,

$$A = \frac{5}{48}, \quad B = D = 0, \quad C_{\text{shp}} = -\frac{5}{96} - u - \frac{1}{2}(\rho + \pi), \quad (60)$$

with $\nu = -(\frac{5}{48} + 4u)$ forced by A and σ entering the rest value alone [Lean: `nu_from_A_and_u`], [Lean: `C_shp_from_orbits_historical`].

Reported computational result 53 (Hodge normal form of the saved kernel). With $\tilde{\nu} = \nu + 4u$, $\tilde{\pi} = \pi + 2u$ and $\tilde{\sigma} = \sigma - 12u$,

$$H_4 = -\tilde{\nu}(L^\uparrow - 2) + uS_\square^2 - \tilde{\pi}S_\square + \tilde{\sigma}I - 2C_{\text{shp}}R \quad (61)$$

holds on all 189 records of the historical kernel exactly, with $\tilde{\nu} = -5/48$, and on the cold record to 10^{-13} relative with the same $\tilde{\nu}$. The 144 records the two kernels agree on in shape are $uS_\square^2$; the unit sign pattern of the two disputed orbits ρ and π is the off-diagonal of $L^\downarrow$; and the diagonal shadow of $S_\square^2$ (-4 on the normal keys, -2 in plane, $+12$ on site) is what the tilded amplitudes absorb.

On the cube-boundary carrier $L^\downarrow\psi = 0$ and $L^\uparrow\psi = e_1\psi$, so every term of (61) but the last is a scalar there: $A = -\tilde{\nu}$ with no u in it, $B = D = 0$ with nothing to cancel, and $4C_{\text{shp}}e_2$ is the projection of $-2C_{\text{shp}}R$ alone. That is the tier collapse of Section 9.9, as far as algebra takes it: the only operator outside the algebra generated by the two Laplacians is R , and R projects to pure e_2 . The carrier is an exact eigenvector of H_4 if and only if $C_{\text{shp}} = 0$. Two consequences for the dispute. First, symmetry fixes no sign: on plaquette-centred displacements with the orientation character, every orbit is separately Hermitian and invariant under all 48 elements of O_h in both kernels, and no plane-basis regauging other than ± 1 preserves the 144 agreed records, so the flipped signs of ρ and π between the kernels are a disagreement about which operator the theory produces, not about how to write one down. Second, the dispute is four numbers $(u, \rho, \tilde{\pi}, \tilde{\sigma})$, of which u and $\tilde{\sigma}$ reach the carrier band only through its constant; on the carrier C_{shp} is two shared-link amplitudes.

9.6 Three amplitudes computed from outside both pipelines

The lattice fourth-order matrix element between two plaquettes is a finite sum of connected cluster cumulants, and the smallest clusters can be evaluated from the corpus engine's Haar primitives — Wilson-word states, exact Haar inner products, the electric action — by a resolvent that reads neither kernel. Before any fourth-order number is read, the same assembly returns the second-order constants of Section 4 from the primitives: $\mp 5/612$ on a coplanar and a perpendicular pair, the signs of $S_\square$; $-11/306$ for the charge-even hop on both; $-11/306$ for the charge-odd leakage once the neighbour's $-3/4$ vacuum bubble is subtracted; and 0 on a link-disjoint pair in both sectors, the cancellation of Theorem 27 observed.

Reported computational result 54 (Amplitudes from cluster expansion). 1. **The two-hop weight.** On the three-plaquette chain $P \rightarrow Q \rightarrow R$ with Q sharing a link with each endpoint, the charge-odd cumulant is

$$W(P, Q, R) - W(P, R) = s \cdot \frac{360421351}{40327601932800}, \quad s = \pm 1,$$

with s the product of the two signed shared-link incidences, on the coplanar–coplanar, coplanar–perpendicular and perpendicular–perpendicular chains, from two independently written implementations: a block characteristic-polynomial resolvent with the $PVP = 0$ form, and a Krylov minimal-polynomial resolvent with the engine's own baryonic PVP assembly. This is the historical kernel's u , as a rational. The cold record's u is 4.1327437 times it. The charge-even cumulant, $948253471/40327601932800$ on all three chains, is a number neither kernel holds.

2. **The in-plane and normal shared-link amplitudes.** Assembling the pair cluster, its single-contact and shared-link dressings and, where the pair spans a cube, the cube-completion term through the cube's other four faces, the in-plane and normal elements are

$$\pi = -\frac{20535103905179}{1264270320593280}, \quad \nu = -\frac{1050558388351}{10081900483200} = -\left(\frac{5}{48} + 4u\right),$$

the historical kernel's π and ν to the digit, with the primitive cube completion $-5/48$ computed rather than imported. The cold record carries the opposite sign for π .

3. **The rotation amplitude.** The same assembly on the perpendicular pair, whose dressings include two *corner* clusters (three faces at a cube vertex, pairwise sharing a link) and a cube completion through adjacent faces worth $53/768$ rather than $5/48$, gives

$$\rho_{\text{cl}} = \frac{588708011765248393}{14501180577204921600} = +0.0405972,$$

against the historical $+0.0082309$ and the cold -0.0879716 : neither.

9.7 The standing of the records, and a third value

Substituting the assembled amplitudes into (60) gives

$$C_{\text{shp}}^{\text{cl}} = -\frac{54822624038066723}{853010622188524800} = -0.0642696, \quad (62)$$

recorded beside the historical -0.0480864 and the cold -0.0202133 as a third value and promoted no further than they are [Lean: `C_shp_from_orbits_cluster`]. What the three outside computations decide is standing, not the coefficient. The cold pipeline is wrong on every amplitude checked — u , π , ρ — and until its error is located and shown not to reach the shared-link sector its sign-flipped values carry no independent weight; the historical kernel is right on u , π and ν and wrong on ρ , so its C_{shp} is not the fourth-order coefficient either. The assembled ρ rests on the corner cluster, which two implementations of the resolvent agree on but which no record both kernels got right contains; that is the difference between recording (62) and promoting it, and the corner cluster from a third implementation is the calculation that closes the dispute.

The ϵ -sector prediction of an earlier unifying candidate — that the balanced, determinant-free contraction shifts $\rho + \tilde{\pi}$ at $N = 3$ by $-25/512$ — is refuted by its own falsifier. With every unbalanced Haar family dropped, the dressings do not move and each pair cluster moves by exactly $-55/13872$, so the ϵ -sector of $\rho + \tilde{\pi}$ is $-55/6936$ [Lean: `eps_sector_refutes_U5`]. The continuation shift $25/64$ of Proposition 52 is therefore not the ϵ -sector's contribution, and remains what it was there: a witness, not a value.

9.8 What cannot decide it

A large sealed sweep over 609 marked clusters has been the standing candidate for settling this. It cannot: its engine assembles coefficients through a Γ -point scalar and emits no kernel block at all, and by Theorem 50 Γ -point data are structurally incapable of constraining C_{shp} . This is a statement about the instrument, made by reading it, not a report of a failed run. Nor can the two-hop weight decide it, however exactly it is known: u enters the carrier band only through its constant.

9.9 The tier collapse, as two integers

The two shape coefficients B and D vanish in the recorded kernel even though the two-hop enumeration generates both, and the degree count does *not* forbid them. The carrier numerator reaches a -degree three, because the projection contributes one further power of a : for any vector whose components have squared modulus a_i — the carrier is one, in whichever of the two equivalent labellings — the quadratic form $\text{diag}(a_i^2)$ evaluates to $\sum_i a_i^3 = q^3 - 3qe_2 + 3e_3$, an explicit witness carrying a nonzero e_3 . An earlier degree-bound mechanism proposed for the collapse was therefore retracted. What is checked is finer: every degree-three block amplitude is an integer multiple of one raw record weight x , and

$$B = 0 : (+1, +1, -2) \cdot x, \quad D = 0 : (-3, +6, -3) \cdot x$$

over the displacement shells. The kernel does populate the forbidden tier; the dynamics cancels it with small integers. Reported result 53 says what the two vectors are: the diagonal shadow of $S_{\square}^2$, so that in orbit amplitudes $D = -6u + 3u_2$ with $u_2 = 2u$ and $B = (+1, +1, -2) \cdot x$ vanish

identically [Lean: tier_collapse_orbits]. What that leaves for a mechanism is the cube-completion channel, which is $-5/48$ for opposite faces and $53/768$ for adjacent ones; a mechanism has to produce both.

9.10 A near- Γ statement that does not adjudicate

One quantitative fourth-order statement can be made without adjudicating anything, because it depends on the kernel only through $\sqrt{W_4}$. With Jordan’s inequality $q(k) \geq (4/\pi^2)|k|^2$ on the whole zone — tight at $k = 0$ and at the zone corner — and $\tau^{[3]}(u) \geq t_3 u^2$ for $u > 0$, the fourth-order shape terms are dominated by the second-order dispersion outside a ball of radius Ku with $K = (\pi/2)\sqrt{W_4}/(\theta t_3)$. The two records give $K = 17.04$ and $K = 23.66$ — floating point, since one of the two bandwidths exists only as a float — and the larger bounds both, so the exclusion radius can be stated for either record without choosing between them. It is not a bound on the open coefficient: a value of C_{shp} outside the two records carries its own W_4 , and nothing here confines the answer to the interval the records span. The third value (62) is a case in point: with $W_4 = \alpha + \beta$ and $\beta = 8A + 16C_{\text{shp}}$ [Lean: width_eq_alpha_add_beta], [Lean: beta_from_A_and_C], it carries the smallest bandwidth of the three, and the exclusion radius is stated for whichever record is in hand. The statement is non-vacuous only for $u < 0.133$ at $\theta = 1/2$, and the excluded zone fraction grows as u^3 .

10 External comparisons

External results are useful here when their provenance is independent of the present calculation. We use them only within the scope stated below; all normalisation conversions and coefficient matches are calculations of this paper.

10.1 Rigorous strong-coupling particles and effective operators

The priority boundary is important. This paper does *not* present the first strong-coupling glueball, the first rigorous infinite-volume particle statement, the first effective low-energy transfer matrix, or the first momentum-dependent glueball calculation. Schor proved that, at sufficiently large coupling, plaquette correlations possess an isolated simple pole and identified the associated state as an isolated one-particle glueball [7]; his sequel states existence of isolated one-particle states in the full energy-momentum spectrum [8]. A separate spectroscopy paper analyzes the low strong-coupling spectrum in three-dimensional pure lattice gauge theories [9]. In the dimension-matched Euclidean 3+1-dimensional Wilson theory, O’Carroll and Dantas Barbosa prove, in the time-zero plaquette subspace, between one and four mass branches of the form $m(\beta) = -\log \beta + r(\beta)$, with convergent analytic or half-analytic corrections and a finite coefficient algorithm [10]. Bricmont and Fröhlich place such results in a general random-path and random-tube framework for particle structure in lattice field theory [11]. These are genuine infinite-volume Euclidean particle or mass statements and are physically stronger than the finite-volume Riesz-island statement proved here.

The relation is not one of containment in either direction. The comparison located no prior statement of the exact Kogut–Susskind electric census in Theorem 1: simultaneously for every $N \geq 3$ and every periodic $L \geq 3$, in the trivial electric one-form sector, the spectrum below $5C_F/2$ is exactly $\{0, 2C_F\}$ and the external margin of the retained shell is $C_F/2$. Conversely, that free-Hamiltonian classification does not imply the infinite-volume particle conclusions of the Euclidean papers. Its defensible novelty is the all-rank, all-periodic-volume, one-form-resolved classification and constant margin, not “existence of a glueball.”

The general stability input in Theorem 4 is Yarotsky’s, not ours. His spectral-enclosure theorem applies to weak finite-range perturbations of a non-interacting gapped quantum lattice system and explicitly permits boundary conditions other than the displayed open boxes [1]. The stronger one-particle conclusion in the same work requires an isolated non-degenerate single-site excitation and a nonresonance condition. The plaquette shell here is composite, $3L^3$ -fold degenerate and additively resonant, so that one-particle theorem is not invoked. The new input in Theorem 4 is instead model-specific: exact $SU(3)$ additive Casimir arithmetic, the local interaction bound $27|u|$, the periodic gauge-sector descent in Proposition 2, and rank continuation using Theorem 1. The conclusion is a volume-uniform *finite-volume* Riesz island for the complete shell, not an infinite-volume quasiparticle. Yarotsky’s later ground-state-uniqueness paper is a distinct result and is not the source cited for the enclosure [2].

Münster constructed an effective transfer matrix whose low eigenvalues agree order by order with those of the full Euclidean transfer matrix and used it to obtain energy–momentum dispersion relations [17]. He also identifies the $SU(3)$ charge-odd T_1 triplet conventionally denoted 1^{+-} and states that the construction applies in that sector. Priority for an effective matrix, nonzero-momentum spectroscopy, or a charge-odd plaquette triplet therefore belongs to the older literature. What the present Theorems 8–11 add is the exact chain-level normal form: the shared-link operator is the cellular down-Laplacian $BB^\dagger$, its Bloch spectrum is $\{0, q(k), q(k)\}$, and its kernel is $\text{im } \partial_3 \oplus H_2$ with dimension $L^3 + 2$.

Theorem 23 resolves a different, local coefficient not exposed by a rest-mass table. The Haar moment gives the four fusion weights, Lemma 22 proves that these adjacent shared-link routes exhaust the off-diagonal second-order processes, and Eq. (39) fixes the cross-matrix-element magnitude and orientation sign. Their resolvents sum to

$$t_N = \frac{2N(N^2 - 4)}{(N^2 - 1)(2N^2 - 1)(4N^2 - 9)} > 0,$$

and the exact Taylor polynomial through order two is the scalar part plus $t_N u^2 BB^\dagger$. We found no printed all- N fusion-projector derivation of this coefficient or its exact homological assembly. This is a narrow structural novelty claim, not a global priority claim; an exhaustive priority claim would require a broader search than the primary-source comparison made for this edition.

The resulting claim boundary is therefore the conjunction of: (i) the exact all- N , all-periodic-volume free electric window; (ii) the complete charge-odd second-order process census; (iii) the projector cross element and rational t_N ; (iv) the $BB^\dagger$ carrier and homological kernel; and (v) the model-specific finite-volume Riesz island. The older results prevent this paper from claiming the first glueball, first dispersion relation, first effective Hamiltonian, or an already completed infinite-volume particle interpretation.

Hamer 1989, both sectors. Let a_n denote the coefficient of x^n in Hamer’s tabulated dimensionless mass series. The 1^{+-} rest-frame series agrees with E_{flat} at every shared order: $16/3 \div 2 = 8/3$ exactly, $a_1 = +1$, and $2a_2, 4a_3$ matching $11/306$ and d_3 to 9.9×10^{-14} and 1.1×10^{-12} . The 0^{++} series also matches the charge-even A_1^{++} Γ -point coefficients at x^2 and x^3 , with the sign structure ($a_1 = -1$ in the scalar channel where the carrier’s is $+1$) reproduced. At x^2 that coefficient is now the theorem $d_{2,3}^+ + 12\ell_3 = -217/1020$ (Theorems 27 and 29), so the agreement, to the precision of the published table, is an external corroboration of the charge-even symbol at $N = 3$; at x^3 it remains conditional on the domino ledger. The E^{++} doublet is not constrained by it, which is why Proposition 43 uses the splitting rather than the level. The conversion $m_n = 2^{n-1}a_n$ is $x = 2u$ together with Hamer’s normalisation $ma = (g^2/2)M$ — the 2^n from the substitution, the half from

the normalisation — exactly, with no 4^r ambiguity anywhere. Hamer states in print that his x^3 and x^4 terms disagree with the earlier series of [5]. We report Hamer’s comparison and do not independently adjudicate the earlier coefficient table.

The string tension. Let t_n denote the coefficient sequence in the transcribed Kogut–Pearson–Shigemitsu $SU(3)$ table [13], and let σ_n denote the corresponding ledger coefficient after the displayed convention conversion. The transcription satisfies exactly $2^{n-1}t_n = \sigma_n$ for $n = 2, \dots, 5$, in the physical sign convention $\sigma_n^{\text{phys}} = (-1)^n \sigma_n^{\text{raw}}$. The same series reproduces E_{flat} term by term through u^3 , including the discriminating u^3 coefficient.

Euclidean series. Historical Euclidean strong-coupling series by Münster, Seo and Smit [12, 14, 15, 16, 17] provide context only. The accompanying corpus contains a transcription comparison and an apparent eighth-order discrepancy between the 1982 erratum and the 1985 table. Because every original table page has not been independently re-transcribed for this paper, neither the apparent equality nor the discrepancy is used as evidence. No Hamiltonian coefficient is inferred from these Euclidean records.

The overlap obstruction. Schierholz [19] and Brandstaeter, Kronfeld and Schierholz [21] document the degradation of local-operator projection and its critical signal-to-noise scaling as the lattice spacing is reduced. Nothing quantitative is transferred to the present strong-coupling finite-order setting. Read only as a warning, these results motivate posing **G18** in a smeared basis; they supply no bound and do not identify the carrier with a physical glueball.

Cross-regime discipline. Two indexed papers declare an explicit scope firewall and neither supplies a value here: Davoudi–Raychowdhury–Shaw [29], whose Hilbert-space cost comparison is locked to 1+1 dimensions with matter, and Kadam–Raychowdhury–Stryker [30]. The rule is enforced by the literature validator rather than recorded in prose.

11 Scope

This work does not establish a four-dimensional infinite-volume Yang–Mills mass gap, a continuum glueball mass, convergence of the strong-coupling series to the continuum, regulator independence of the first dispersive order, the full $SU(3)$ fourth-order rest coefficient or off-axis shape, or an identification of any level here with a physical glueball. At Γ the three face orbitals transform as an axial vector of the cubic group, which with parity even and charge conjugation odd gives the retained-space label T_1^{+-} ; this is an operator-space assignment, not a spectral one.

The carrier is macroscopically degenerate at finite L . Under the third-order factorisation input, the minimum *internal* separation of one sheet from the incidence sheets satisfies the coefficient identity

$$T_{\leq 3}\Delta_L(u) = 4\tau^{[3]}(u)\sin^2(\pi/L),$$

which falls as L^{-2} at each displayed order; it cannot provide a volume-uniform contour for that single sheet over the entire Brillouin zone. This statement must not be conflated with external isolation of the complete electric shell. Theorem 1 gives the latter a volume-independent free margin $C_F/2$, while Proposition 13 gives an L -independent internal coefficient at every fixed nonzero nested momentum.

Three crossings are named explicitly, and none of them is made here.

Proposition 55 (Exact reduction of the G17 free-energy target). *For the Euclidean $SU(3)$ Wilson plaquette density $W_p = 1 - \frac{1}{3} \text{Re Tr } U_p$, let*

$$Z_\Lambda(b) = \int \exp\left(-\sum_p b_p W_p\right) d\mu, \quad b_p(s) = \begin{cases} s, & p \in \Gamma, \\ \beta, & p \notin \Gamma, \end{cases}$$

and write $Z_{\beta,\alpha,\Gamma} = Z_\Lambda(b(\alpha\beta))$ and $Z_\beta = Z_\Lambda(b(\beta))$. Then

$$\log \frac{Z_{\beta,\alpha,\Gamma}}{Z_\beta} = \int_{\alpha\beta}^{\beta} \sum_{p \in \Gamma} \langle W_p \rangle_{s,\Gamma} ds. \quad (63)$$

Consequently the single local estimate

$$\sup_{\Lambda, \Gamma, p \in \Gamma, s \in [\alpha\beta, \beta]} s \langle W_p \rangle_{s,\Gamma} \leq C \quad (64)$$

would imply the desired animal-uniform bound with

$$\frac{Z_{\beta,\alpha,\Gamma}}{Z_\beta} \leq K_\alpha^{|\Gamma|}, \quad \log K_\alpha \leq C \log(1/\alpha). \quad (65)$$

Proof. Differentiating $\log Z_\Lambda(b(s))$ gives $-\sum_{p \in \Gamma} \langle W_p \rangle_{s,\Gamma}$ and integration from β down to $\alpha\beta$ proves (63). Under (64), each summand is at most C/s , which proves (65). The elementary pointwise estimate $0 \leq W_p \leq 3/2$ gives only $\log K_\alpha \leq \frac{3}{2}(1-\alpha)\beta$; its growth in β is precisely why the uniform C/s estimate remains a substantive open problem. $\square$

- **The specific finite-to-infinite-volume bridge** requires the inhomogeneous Wilson free-energy bound with useful $\log K_\alpha$ and the source-radius reduction formulated for this construction. This is not a claim that infinite-volume strong-coupling particles are unknown: the Euclidean results reviewed in Section 10.1 already establish such particles and mass branches. Proposition 55 turns the present G17 target into the explicit one-plaquette estimate (64), but does not prove it; the source-radius theorem is also still unstated. The Hamiltonian Riesz island belongs to small u and does not prove this inhomogeneous Euclidean bound.
- **This protected carrier to a particle** requires a volume-uniform overlap of a smeared, dressed T_1^{+-} operator built on the exact $BB^\dagger$ carrier with the transfer-matrix spectrum, together with multi-plaquette survival of its protected level (G18). Older Euclidean particle theorems establish existence in their own formulations; they do not identify their one-particle subspace with this carrier or supply the Hamiltonian-to-transfer-map used here. Conversely, this edition's free external gap, finite-volume interacting Riesz island at existentially small $|u|$, fixed- k carrier and locality-momentum tradeoff do not reproduce those infinite-volume theorems. A volume-uniform dressed-source frame, infinite-volume totality and the transfer identification remain open.
- **Lattice to continuum** (G19) is untouched. Small u is strong coupling; the continuum is the opposite end of the axis.

All three crossings are required to promote the *specific carrier constructed here* to the intended continuum interpretation, and none is completed. The new window and Riesz-island theorems remove the free-spectrum obstruction and its small- u finite-volume persistence subproblem from G18, but they do not promote this protected lattice cluster to an infinite-volume physical particle.

The two former charge-odd completeness premises are now theorems. Theorem 1 proves more than shell completeness: below $5C_F/2$ the full trivial-flux electric spectrum is exactly $\{0, 2C_F\}$ for every $N \geq 3$ and $L \geq 3$, and the range of P_- is exactly the charge-odd one-plaquette span. Lemma 22 proves that the only off-diagonal second-order routes are the four adjacent shared-link channels: disjoint exchange routes are orthogonal, while the reduced same-face route is diagonal (at $SU(3)$, Q_- removes its determinant-induced retained-shell component). The explicit projector matrix element (39) then fixes their signs and normalisations. These statements are analytic results; the finite-volume $L = 3, 4, 5$ enumerations in the verifier are regressions, not substitutes for their proofs.

One physical identification remains conditional, at third order only. That the complete linked charge-even Bloch symbol is the unsigned incidence $\mathcal{N}(k)$ — the signed incidence with the boundary orientations dropped — is Theorem 27 at order u^2 . At order u^3 the charge-even coefficients come from the supplied domino ledger, and the third-order splitting and bandwidth of Section 6 rest on it.

12 Evidence, authority, and reproducibility

The accompanying dependency-free program `verify_publication_core.py` uses exact integers and `Fraction` arithmetic. Its 38/38 exact checks include the all-rank Casimir shelf, the $SU(3)$ additive-spectrum and Riesz-contour arithmetic, the coefficient ledger and Weingarten sums, signed incidence and torus homology, closed-cube shell, two-cube connected geometry and reported channel arithmetic, the unsigned cubic, the saved historical Hodge pencil, the fourth-order obstruction, and the coupling convention. It does not reconstruct the finite-precision Clebsch–Gordan inputs, replay the third-order microscopic contractions, or choose C_{shp} .

The float half lives in `verify_radius2_report.py`: six checks at stated tolerances against the sealed radius-two artifact, read with a standard-library NPZ reader and a Jacobi diagonalisation, so no dependency stands between the reader and the numbers; Figure 3 is rendered from the same artifact by `make_figure_radius2.py`. The hash-pinned detached verifier of the radius-two release, which recomputes all three nested spectra and the incremental operators from the sealed parents and returned zero maximum reconstruction error, belongs to that release and not to this repository; its verdict is reported in Section 5.4 as the release’s. The all-rank second-order ledger, the charge-even symbol, the census split, the Hodge normal form and the cluster-assembled amplitudes are checked by the verifier’s registry rather than by the compact script, under the names printed beneath them.

12.1 Claim-level evidence map

12.2 Upstream corpus and downstream verifier

The organised ALL THEORY repository at the user-supplied local path was consulted as the upstream scientific corpus. Its four-document v4.3 authority stack is frozen by byte count and SHA-256 in `export/MAN_GOV_export_manifest.csv`; all four entries were rehashed for this edition and matched. The downstream verifier is [WORKHOUSE](#). Document flow is one-way—frozen corpus to hash manifest to verifier—while verifier findings return to review and status records. The later two-cube results are not in the August 20 corpus and are therefore kept as later WORKHOUSE-local evidence.

This edition is pinned in the verifier at commit `304dbcd15d2deda16dbf5e576c52035ef4e7cdde`, where the registry holds 292 checks in 28 suites, all passing, and the Lean core holds 95 theorems with no omitted proof and no axiom beyond the three standard ones. Every displayed statement

carries, in the source, a `\chk` label naming the registered check that establishes it; the labels are suppressed typographically, and a registered check reads every pinned edition and fails if any label names no passing check, so this edition is covered by that guard. Appendix I prints the labels, and Appendix H the theorems.

The authority files are identified by the following SHA-256 values:

- master v4.3: 1f8295c3a797aeb854b4b0ca82485bd7f4a149617777d2a43fc147a4a53c26dd;
- detailed formulas v3.1: 924d5b26f865e33f530d166735e847b89f60df7fe363995e2788ab24e1ff4254;
- consolidation guide v4.3: 60b8a93c8149049f50ed3ce8e0cfd1b379723b6765f201395a2fda6c4a7e7aa7;
- canonical source manifest v4.3: 730bc99ed8d749de0dffbabfda85cd4512d8f7748782620b694759a03bacde05.

12.3 Data and code availability

This release directory contains the TeX source, generated vector figures, figure generator, exact verifier, build log, rendered-page audit, and a SHA-256 manifest. The historical kernel payload used in Reported result 49 has SHA-256 635d40fa8a5d7da841fd30f36185eb96f14ec4c040678ddd8fb010379afb2900. The pinned ymcirc $d = 3$, $\mathcal{B} = 4$ data blob is identified in Ref. [35] and has SHA-256 4cad9538f2c052488556362e0cf6e4e63363e7f5801597aeab9013ef945c59dc. The exact stranded-flux audit source and result have SHA-256 a42800a91809a77bbd1d339ac604ea80a234eed305377bbabbf890fee98fde55 and d215fa1e9cd0bde0410eec16c687182c30bc62fa3f28c1445fd81e93888e1d60. The two-cube build and its finite-precision Clebsch–Gordan payloads are not reproduced by the compact verifier; the exact statements that depend on them are labelled reported. The radius-two supplement is governed by `two_cube_cutoff_free_radius2_finite_u_spectrum_manifest.json`. That manifest pins twelve members distributed together here: the derivation note, builder, hostile tests, six sealed-parent files, result NPZ, certificate and detached verifier. The result NPZ has SHA-256 52a17f87d3990b234635828a0c92f73f781e9a3abec8bb3c127e990b7a717e6c. A durable tag or archival DOI is still needed before journal submission.

Machine checks certify algebra relative to their definitions and inputs. They do not replace derivations, promote numerical contractions to symbolic ones, or certify an uncompleted off-axis contraction.

13 Conclusion

The strongest exact coefficient result is a finite-order operator theorem built on a new volume-uniform free spectral window: for every $N \geq 3$ and periodic $L \geq 3$, the trivial-flux electric spectrum below $5C_F/2$ is exactly $\{0, 2C_F\}$, the retained-shell theorem identifies the full charge-odd $2C_F$ eigenspace, the process-completeness lemma exhausts its second-order off-diagonal routes, and the four shared-link fusion projectors give the closed coefficient t_N with the oriented cellular boundary fixing its sign and geometry. This assembles without a completeness hypothesis into a homological flat carrier of dimension $L^3 + 2$. The apparent L^{-2} obstruction is now localized: it is the minimum internal sheet separation near Γ , whereas the free external shell margin is at least $C_F/2$ and the coefficient at every fixed nonzero nested momentum is volume-independent. The $SU(3)$ two-cube calculation supplies a nontrivial operator-level assembly check, including channel-by-channel

geometry and orbit localisation, but shares its amplitude lineage and remains numerical at the contraction layer.

For $SU(3)$ the analysis goes one step beyond finite order. Yarotsky’s local spectral enclosure, together with the exact additive Casimir arithmetic and the cited periodic-boundary version, gives a nonperturbative rank- $3L^3$ Riesz island for the complete neutral charge-odd cluster whenever $27|u| < c_1$ and $27c_2|u| < 1/33$. This is a volume-uniform finite-volume statement with existential constants. It neither isolates one internal sheet nor proves a source-carrying thermodynamic particle.

A later detached-replayed radius-two extension adds nested finite- u Ritz spectra and leading incremental order- u^4 and order- u^5 effects. Because the odd exterior action is absent, it supplies neither a matched truncation error nor a complete fourth- or fifth-order physical coefficient.

The second-order scalar is now a closed form at every rank, and the charge-even sector is a theorem at that order: its Bloch symbol is the charge-even hop ℓ_N times the unsigned incidence, because the vacuum-mediated route is cancelled pair by pair, and its Γ -point splitting is $12\ell_N$ unconditionally. The unsigned cubic is therefore the charge-even band at order u^2 and a comparison operator only at order u^3 . At fourth order the saved historical kernel is a polynomial in the two cellular Laplacians plus one operator whose weight is the disputed coefficient; three of its six amplitudes have been recomputed from outside both pipelines, refuting the cold record on all three and the historical record on the rotation amplitude, and a third value of C_{shp} is on record. No scalar reanchoring, Γ datum, or axial datum can decide it. The decisive next object is therefore the corner cluster from a third implementation, or the historical pipeline’s own face-resolved ledger for the eighteen three-clusters. Independently, promoting the finite-volume Riesz island to a thermodynamic, source-carrying multiplet with a volume-uniform Wilson frame and a controlled transfer map is now the precise fixed-spacing G18 target. Proposition 55 also isolates G17’s first missing estimate without pretending to prove it. Until those steps are supplied, the operator and Riesz-island theorems are substantial, but the physical fourth-order kernel, infinite-volume particle interpretation, and continuum limit remain open.

A Channel weights in closed form

Substituting (24) into (35),

$$\begin{aligned} w_1 &= -\frac{2}{N(N^2 - 1)}, & w_{\Lambda^2 F} &= -\frac{N - 1}{(N + 1)(2N - 3)}, \\ w_{\text{Adj}} &= -\frac{2(N^2 - 1)}{N(2N^2 - 1)}, & w_{\text{Sym}^2 F} &= -\frac{N + 1}{(N - 1)(2N + 3)}. \end{aligned}$$

Their family sums are (36) and (37), and the difference has numerator $2N(N^2 - 4)$, which is the $SU(2)$ zero and the positivity for $N \geq 3$ at once. All four closed forms and both sums are proof-checked [Lean: wSinglet_closed], [Lean: wAdjoint_closed], [Lean: wAntisym_closed], [Lean: wSym_closed]. At $N = 3$: $w_1 = -1/12$, $w_{\text{Adj}} = -16/51$, $w_{\Lambda^2 F} = -1/6$, $w_{\text{Sym}^2 F} = -2/9$, which are the four values Reported result 31 matches against (50).

B The Weingarten index sums

With $\text{Wg}(e) = 1/(N^2 - 1)$ and $\text{Wg}((12)) = -1/(N(N^2 - 1))$, and σ, τ running over S_2 ,

$$\begin{aligned} &\langle U_{i_1 j_1} U_{i_2 j_2} \bar{U}_{i'_1 j'_1} \bar{U}_{i'_2 j'_2} \rangle \\ &= \sum_{\sigma, \tau} \text{Wg}(\sigma\tau^{-1}) \prod_a \delta_{i_a i'_{\sigma(a)}} \delta_{j_a j'_{\tau(a)}}. \end{aligned}$$

Summing the delta products over all N^4 index quadruples,

$$M_{\text{direct}} = \text{Wg}(e)(N^4 + N^2) + 2 \text{Wg}((12))N^3 = N^2,$$

$$M_{\text{cross}} = 2 \text{Wg}(e)N^3 + \text{Wg}((12))(N^4 + N^2) = N.$$

Both are confirmed by explicit summation at $N = 3, 4, 5$. The related fourth moment $\int |U_{11}|^4 dU = 2/[N(N+1)]$ is $1/6$ at $N = 3$ and in particular nonzero, which is what refuted a claimed Haar vanishing in the balanced $(2, 2)$ sector.

C The $SU(3)$ coefficient ledger

Order	Quantity	Value
u^0	electric plaquette	$8/3$
u^1	scalar	1
u^2	$BB^\dagger$ coefficient	$5/612$
u^2	flat scalar	$11/306$
u^3	$BB^\dagger$ coefficient	$1975/124848$
u^3	per-neighbour leakage	$-12331/249696$
u^3	flat scalar	$-109151/249696$

The u^1 row is $SU(3)$ -only rather than an $SU(3)$ specialisation: it is $-\langle p, -|V|p, -\rangle = 1$, which needs the ϵ channel and vanishes for $N \geq 4$. Of the rest, the u^0 row and the u^2 $BB^\dagger$ coefficient specialise all-rank statements ($2C_F$ and t_N); the u^2 flat scalar inherits the $SU(3)$ tower term through (54); and the three u^3 rows are $SU(3)$ -only and conditional on the supplied domino ledger, exactly as Section 7 says.

D The charge-even ledger

With Theorem 27 the u^2 rows below are theorems and the u^3 rows are conditional on the domino ledger. The same convention, with $\lambda \in \{12, -4\}$ and (54), gives:

Order	Quantity	Value
u^2	hopping $t_{2,+} = \ell_3$	$-11/306$
u^2	on-site diagonal	$223/1020$
u^2	band bottom ($\lambda = 12$)	$-217/1020$
u^2	band top ($\lambda = -4$)	$1109/3060$
u^2	bandwidth $16 t_+ $	$88/153$
u^3	hopping $t_{3,+}$	$-6335/249696$
u^3	on-site diagonal ($\lambda = 0$)	$52163/260100$
u^3	band bottom ($\lambda = 12$)	$-54049/520200$
u^3	band top ($\lambda = -4$)	$471353/1560600$

Because $t_{r,+} < 0$ at both orders, the $\lambda = -4$ endpoint is the band *maximum* in this sector. A certificate key naming it **bandmin** at both orders is a naming error against its own arithmetic, recorded rather than silently corrected.

E Finite-volume chain calculation

For $i < j$,

$$\partial_2[x; i, j] = [x + e_i; j] - [x; j] - [x + e_j; i] + [x; i],$$

$$\begin{aligned} \partial_3[x; 1, 2, 3] &= [x + e_1; 2, 3] - [x; 2, 3] \\ &\quad - [x + e_2; 1, 3] + [x; 1, 3] \\ &\quad + [x + e_3; 1, 2] - [x; 1, 2]. \end{aligned}$$

Substituting the second into the first cancels every edge pairwise, so $\partial_2\partial_3 = 0$ over $\mathbb{Z}$. On the torus $\ker \partial_3$ is generated by the sum of all oriented cubes, so $\text{rank } \partial_3 = L^3 - 1$.

F The two-cube geometry

The eleven oriented plaquettes of the $(3, 2, 2)$ prism, by base vertex and ordered plane axes, are

$$\begin{aligned} &(000; 12), (000; 13), (000; 23), (001; 12), \\ &(010; 13), (100; 12), (100; 13), (100; 23), \\ &(101; 12), (110; 13), (200; 23), \end{aligned}$$

with left, right and shared-face index sets $I_L = (0, 1, 2, 3, 4, 7)$, $I_R = (5, 6, 7, 8, 9, 10)$, $I_F = (7)$. Building $G = BB^\top$ from the oriented boundaries and applying the geometric Möbius fold (48) leaves exactly four nonzero off-diagonal pairs, $(0, 5), (1, 6), (3, 8), (4, 9)$, each equal to -1 : four disjoint 2×2 blocks and three isolated directions, which is why the spectra in Section 5 are closed form once the diagonals below are supplied. The two connected diagonals are

$$\begin{aligned} D_{\mathcal{B}=6} &= \frac{1}{612} \text{diag}(-2317, -2317, -2295, -2317, -2317, \\ &\quad -2317, -2317, 0, -2317, -2317, -2295), \\ D_{\mathcal{B}=4} &= \text{diag}\left(-\frac{7}{4}, -\frac{7}{4}, -\frac{15}{4}, -\frac{7}{4}, -\frac{7}{4}, \right. \\ &\quad \left. -\frac{7}{4}, -\frac{7}{4}, 0, -\frac{7}{4}, -\frac{7}{4}, -\frac{15}{4}\right), \end{aligned}$$

so both spectra are one line of arithmetic: a pair block contributes $D_{ii} \mp c$ twice and each isolated direction its own D_{ii} . The $\mathcal{B} = 4$ comparator on the same geometry gives 0 once, $-\frac{15}{4}$ twice, $-\frac{11}{6}$ four times, and $-\frac{5}{3}$ four times: the paired levels sit *above* the isolated $-15/4$ pair, where at $\mathcal{B} = 6$ they sit below it. The three isolated levels $\{-\frac{15}{4}, -\frac{15}{4}, 0\}$ are identical in the two truncations, so the entire effect of the cutoff lives in the four cross-cell blocks. That reversal is not the coefficient's sign by itself — within a 2×2 block the pair splits as $D \mp c$, which is invariant under $c \rightarrow -c$ — but the truncation's effect on the connected diagonal as well. What the sign of c alone does reverse is the ordering on a *single* cube, where the kernel is $\alpha I + tG$ with one diagonal and the spectrum of G fixed; that is the shell reversal the one-cube reconstructions report.

G The second-order ledger at every rank

Theorem 29 evaluated at low rank, in canonical u . The charge-odd rows are the tower $\sigma_N^- + 1/C_F$, the on-site coefficient $d_{2,N}^-$ and the flat scalar $d_{2,N}^- - 4t_N$; the charge-even rows are the tower $\sigma_N^+ + 1/C_F$, the on-site coefficient $d_{2,N}^+$ and the A_1^{++} level at Γ , $d_{2,N}^+ + 12\ell_N$. The $N = 3$ column is the register; the others appear nowhere in the corpus.

	$N = 3$	$N = 4$	$N = 5$	$N = 6$	$N = 7$
odd tower	$\frac{1}{2}$	$\frac{12}{35}$	$-\frac{5}{32}$	$-\frac{8}{105}$	$-\frac{7}{160}$
odd on-site	$\frac{102}{11}$	$\frac{10828}{59675}$	$-\frac{4785}{20384}$	$-\frac{1496}{13925}$	$-\frac{206493}{2902240}$
flat scalar	$\frac{102}{306}$	$\frac{10828}{59675}$	$-\frac{4785}{20384}$	$-\frac{1496}{111825}$	$-\frac{214893}{2902240}$
even tower	$\frac{13}{20}$	$-\frac{2132}{1785}$	$-\frac{155}{1248}$	$-\frac{32}{555}$	$-\frac{77}{2400}$
even on-site	$\frac{223}{1020}$	$-\frac{4126292}{3043425}$	$-\frac{12395}{61152}$	$-\frac{46832}{459725}$	$-\frac{2589503}{43533600}$
A_1^{++} at Γ	$-\frac{217}{1020}$	$-\frac{4617524}{3043425}$	$-\frac{17195}{61152}$	$-\frac{201472}{1379175}$	$-\frac{3782303}{43533600}$
ℓ_N	$-\frac{11}{306}$	$-\frac{344}{25575}$	$-\frac{25}{3822}$	$-\frac{412}{111825}$	$-\frac{497}{217668}$
t_N	$\frac{5}{612}$	$\frac{32}{8525}$	$\frac{5}{2548}$	$\frac{128}{111825}$	$\frac{105}{145112}$

H Proof-checked statements

The verifier’s Lean 4 core, `lean/Workhouse/Basic.lean`, proves the statements below from explicit definitions with no `sorry` and only the axioms `propext`, `Classical.choice` and `Quot.sound`; make `lean` rebuilds it. They are the rational and polynomial layer of this paper and nothing else: no perturbative derivation, no operator theory, and nothing that adjudicates the fourth-order dispute is formalised. A `[Lean: name]` tag in the text points here.

- `rank_law_numerator` — The rank law, with denominators cleared. Over the common denominator $(n^2-1)(2n^2-1)(4n^2-9)$ the channel difference has numerator $2n(n^2-4)$, which is the hopping numerator. Stated this way it needs no non-vanishing hypotheses.
- `hopping_three` — $SU(3)$.
- `hopping_two` — $SU(2)$ is excluded at the source; the $n^2 - 4$ factor is its algebraic shadow.
- `hopping_deficit_numerator` — The N^{-3} cancellation leaves a positive deficit, denominators cleared:
- `d3_ledger` — The third-order ledger identity.
- `alphaPen_three`
- `alphaPen_four`
- `alphaPen_five`
- `alphaPen_six`
- `alphaPen_three_eq_four_A` — The axial coefficient is four times the sealed shape coefficient.
- `blind_holdout` — The blind holdout: R is determined by X and M , so it can be withheld from a fit and used as an independent check.
- `stencil_zero_mode` — The 25-point stencil obeys a zero-mode gate: the weights sum to zero.
- `C_from_beta` — The off-axis coefficient is fixed by the diagonal coefficient and the axial one.
- `width_eq_alpha_add_beta` — Bandwidth is the sum of the two pencil coefficients.
- `beta_from_A_and_C` — The tier-collapse relation, valid when $B = D = 0$.
- `even_gram_minors` — Why the linear coefficient is p^2 . The three principal 2×2 minors of the unsigned Gram matrix are $a \cdot p$, $b \cdot p$ and $c \cdot p$: each off-diagonal modulus-squared (bc, ac, ab) cancels the cross term of its minor exactly, and the three then sum to p^2 . This is the step that makes the cubic depend on a , b , c only through p and abc .
- `even_cubic_at_zero` — The band floor. $\mu = 0$, that is $\lambda = -4$, is a root exactly when $abc = 0$, which is exactly the union of the three zone-boundary planes $k_j = \pi$.
- `even_cubic_at_sixteen` — The band ceiling, in the form that carries the bound: at $\mu = 16$ the cubic is $16(16-p)^2 - 4abc$.
- `even_cubic_derivative_factors` — Above p the cubic is strictly increasing, because its derivative factors as $(3\mu - p)(\mu - p)$.
- `newton_three` — Newton’s identity in three variables. This is what lets a degree-3 numerator carry an e_3 component, which is the step that refuted the degree-bound mechanism proposed for the tier collapse.
- `extraction_A` — The four checkpoint-extraction formulas invert the cubic-invariant ansatz.
- `extraction_B`
- `extraction_C`

- `extraction_D`
- `dim_Z2` — The cycle count on the three-torus: cube boundaries plus the harmonic triplet.
- `cPrimTwo_forms` — The law's two printed forms are one identity at the target order: the Casimir form with count S is the $2^{r-1} S / (N(N^2-1)^{r-1})$ form.
- `tetra_from_count` — The tetrahedral count $S_2 = -4$ produces the asserted row $-8/(N(N^2-1))$.
- `tetraCompletion_three` — The $SU(3)$ tetrahedral value.
- `prismCompletion_three` — The $SU(3)$ prism (square-sector) value.
- `cubeCompletion_three` — The $SU(3)$ cube value is minus the sealed shape coefficient.
- `pentCompletion_three` — The $SU(3)$ pentagonal (cap-sector) value.
- `alphaPen_eq_neg_four_cube` — The axial law is -4 times the cube row, at every rank — the bridge between the completion family and the sealed core's `alphaPen`.
- `bandVar_zero`
- `bandVar_pi`
- `bandVar_pi_div_two`
- `q_at_checkpoints` — q at the four high-symmetry points: 0, 4, 8, 12 at Γ , X, M, R. The corpus figure plots this spectrum and asserts it; here it is derived.
- `delta_X` — The checkpoint delta at $X = (\pi, 0, 0)$: $dX = 4A$. X is blind to B, C and D, which is what makes it fix A alone.
- `delta_M` — The checkpoint delta at $M = (\pi, \pi, 0)$: $dM = 8A + 16B + 8C$.
- `delta_P` — The checkpoint delta at $P = (\pi, \pi/2, 0)$:
- `delta_R` — The checkpoint delta at $R = (\pi, \pi, \pi)$: $dR = 12A + 48B + 16C + 16D/3$. R is the only checkpoint that sees D, because it is the only one where all three band variables are nonzero and so `e3` does not vanish.
- `channelWeight_eq` — The weight in cleared form: $w = -2d / (N(N^2 - 1 + Nc))$.
- `wSinglet_closed`
- `wAdjoint_closed`
- `wAntisym_closed`
- `wSym_closed`
- `mixed_family_sum` — The mixed family $F \otimes \bar{F} = 1 \oplus \text{Adj}$ sums to A_N .
- `like_family_sum` — The like family $F \otimes F = \Lambda^2 F \oplus \text{Sym}^2 F$ sums to B_N .
- `channel_gap` — The resolvent gap of a shared-link channel: six nonshared half-links at $3C_F$ plus the fused link at $C_{\rho/2}$, against the external $2C_F$, leave $C_F + C_{\rho/2}$.
- `family_weights_sum_to_one` — The weights of each family sum to one: $(1 + (N^2-1))/N^2 = 1$ and $(N(N-1)/2 + N(N+1)/2)/N^2 = 1$.
- `evenHop_eq` — The vacuum-mediated route adds exactly $1/C_F = 2N/(N^2 - 1)$ to the two family sums.
- `evenHop_three`
- `evenHop_three_from_channels` — The shared-link-only sum at $N = 3$ is the superseded $-481/612$ of C13; the vacuum route $3/4$ is exactly what was omitted.
- `evenHop_two` — The charge-even hop never vanishes where the charge-odd one does:
- `width_ratio_numerator` — The bandwidth ratio $16|\ell_N| / (12 t_N) = 4(3N^2-5)/(3(N^2-4))$, denominators cleared: the charge-even band is the wider one at every rank $N \geq 3$.
- `two_flux_gaps` — The two-flux plaquette gaps: $E_F - 2C_2(\text{Sym}^2 F) = -(N-1)(N+3)/N$ and $E_F - 2C_2(\Lambda^2 F) = -(N+1)(N-3)/N$.
- `adjoint_gap` — The adjoint gap: $E_F - 2N = -(N^2 + 1)/N$.
- `towerOdd_closed` — The same-face route and the vacuum bubble cancel at leading order: the charge-odd tower is $-12N/((N^2-1)(N^2-9))$, of order N^{-3} .
- `towerOdd_five`
- `onsiteOdd_five`
- `flatOdd_five`
- `towerOdd_six`
- `flatOdd_six`
- `towerEven_five`

- `gammaEven_five`
- `su3_odd_ledger` — $N = 3$: the sextet route $-1/4$ plus the vacuum bubble $3/4$ is the tower $1/2$; twelve leakages ℓ_3 give the on-site $7/102$; subtracting $4t_3$ gives the flat $11/306$.
- `su3_even_ledger` — $N = 3$, charge even: vacuum $3/4$, adjoint $-3/5$, sextet $-1/4$ give $\sigma^+ = -1/10$; the tower is $13/20$, the on-site $223/1020$, the Γ -point A_1^{++} level $-217/1020$, and the band top ($\lambda = -4$) $1109/3060$.
- `su4_ledger` — $N = 4$: $\Lambda^2 F$ is self-conjugate, so only $\text{Sym}^2 F$ survives in the odd same-face route ($-4/21$) and it is doubled in the even one ($-8/5$).
- `disjoint_pair_cancels` — The charge-even vacuum route on a link-disjoint pair, $2/E_F$, is cancelled exactly by the doubly-excited exchange route, $2/(E_F - 2E_F)$.
- `census_three` — The link-resolved census at $N = 3$: the self-conjugate mixed channels carry $-w_\rho$, the like channels carry $w_\rho/2$ each for ρ and $\bar{\rho}$, and the six sum to t_3 .
- `casimirWeight_one`
- `casimirWeight_sub_fundamental` — $C_2(\omega_i) - C_F = (N+1)(i-1)(N-i-1)/(2N)$: minimal exactly at the fundamental pair.
- `casimir_shelf` — The three shelf comparisons of the window theorem, as rational functions of N .
- `zero_nality_clears_window` — The one-form flux exclusion at $L = 4$: a once-winding fundamental loop sits at $2C_F$ for every N , while a zero- N -ality loop costs $2C_2(R) \geq 2N$, which clears the window $5C_F/2 = 5(N^2 - 1)/(4N)$; denominators cleared:
- `winding_triangle_clears_window` — ... and the three-cycle at $L = 3$: $(3N/2) \cdot 4N - 5(N^2 - 1) = N^2 + 5 > 0$.
- `su3_numerators_below_eighteen` — The nonzero one-link numerators below 18 are exactly 4, 9, 10, 16 (F , Adj , $\text{Sym}^2 F$ and its conjugate, $(2,1)$ -type).
- `su3_numerator_large` — Any label above 4 already exceeds 18:
- `su3_semigroup_gap` — The additive semigroup generated by $\{4, 9, 10, 16\}$ does not reach 15 with at most four links, while it reaches 14 and 17.
- `riesz_separations` — The two separations of the Riesz island, from the neighbours 14 and 17 of the shell 16 in units of $1/6$, each interval dilated by $1/\lambda$.
- `riesz_contour_clearances` — The contour $|z - 8/3| = (1 - \lambda)/12$ clears the shell interval, the upper neighbour and the lower neighbour by $(1 - 33\lambda)/12$, $(1 - 33\lambda)/12$ and $(3 - 27\lambda)/12$.
- `riesz_min_clearance` — For $\lambda \geq 0$ the lower clearance dominates, so the minimum is $(1 - 33\lambda)/12$ and the resolvent bound on the contour is $12/(1 - 33\lambda)$.
- `kinematic_constants` — The local interaction bound $27|u|$: the $3/2$ rescaling, three plaquettes per cell, and $+ \leq 6$; and the rescaled $SU(3)$ local gap $(3/4) C_F = 1$.
- `bloch_factorisation` — The incidence factorisation $B B^\dagger = q I - w w^\dagger$, entry by entry.
- `carrier_in_kernel` — The carrier is annihilated by $B^\dagger$: the Bloch form of $\partial_2 \partial_3 = 0$.
- `carrier_norm` — $w^2 = q$.
- `evenCubicR_factor`
- `even_cubic_neg_below_zero` — Below $\mu = 0$ the cubic is strictly negative, so $\lambda \geq -4$.
- `even_cubic_pos_above_sixteen` — Above $\mu = 16$ the cubic is strictly positive when each $a_i \in [0, 4]$, so $\lambda \leq 12$:
- `two_cube_spectra` — $B = 6$: the four cross-cell pairs split D c about $-2317/612$ by $c = 5/612$, the end caps sit at $-2295/612 = -15/4$; $B = 4$: about $-7/4$ by $c = -1/12$.
- `cutoff_split` — The link cutoff: $w_\Lambda - w_1 = -1/12$, what it drops is $w_{\text{Sym}} - w_{\text{Adj}} = 14/153$, and the two sum to t_3 .
- `one_cube_shell` — The $B = 6$ cube scalar misses the bridge's by exactly the same-face sextet route $1/4$, and its shell $\{\alpha, \alpha + 4t, \alpha + 6t\}$ is $\{39/68, 371/612, 127/204\}$.
- `nu_from_A_and_u` — The normal amplitude both kernels agree on is $-(5/48 + 4u)$: the cube-completion channel plus the two-hop shadow of S^2 .
- `C_shp_from_orbits_historical` — $C_{\text{shp}} = -5/96 - u - (\rho + \pi)/2$ reproduces the historical value on the historical amplitudes ...
- `C_shp_from_orbits_cluster` — ... and the cluster-assembled third side on the cluster ρ .
- `tier_collapse_orbits` — The tier collapse in orbit amplitudes: $D = -6u + 3u_2$ with $u_2 = 2u$, and $B = (+1 + 1 - 2) \cdot x$, both zero identically.
- `eps_sector_refutes_U5` — The ϵ -sector of $\rho + \pi$ measured by the ϵ -blind assembly, $2 \times (-55/13872) = -55/6936$, is not $U5$'s $-25/512$.

I Machine-check coverage

Every inline `\chk` marker in the source — each naming the registered check that certifies the sentence it sits beside — is extracted verbatim by `make_coverage.py` and printed here, so the coverage claim of Section 12 is auditable in print. A statement absent from this list is not covered; the third-order charge-even identification of Section 6 appears nowhere below, and that is the point.

The source carries 146 inline markers. Each line below is one marker, verbatim, under the section that carries it.

Introduction

- the discrete index theorem gives 0 on the signed face-edge operator, bounding nothing

Hamiltonian and projected sector

- the printed towers are canonical-u: $4\Delta(3u/2)$ reproduces them verbatim
- the 4^*r rescaling breaks the bridge: order 2 off by 16, order 3 by 64
- at $N = 2$ the C-odd hopping vanishes while the unsigned-channel continuation does not
- the retained electric shell is exactly the trivial-flux charge-odd plaquette span
- below $5 C_F/2$ the trivial-flux electric spectrum is exactly 0 and $2 C_F$

Oriented incidence and the exact carrier

- $B B^\dagger = q I - d \operatorname{conj}(d)^T$ for the curl incidence
- $\dim Z_2 = L^3 + 2$ by rank, not by re-arranging the formula
- cube boundaries and three wrapping sheets $\operatorname{SPAN} Z_2$
- the L^3+2 count is chain-level, not the Bloch convention
- the Bloch and chain routes to the carrier agree

The wrapping sheets are cycles, not harmonic

- FINDING: the wrapping sheets are cycles but NOT harmonic

Exact adjacent shared-link dynamics and global assembly

- each channel gap is $C_F + C_R/2$, and the weights sum to one

The channel weights are a theorem

- the shared-link weights are Weingarten, not an isotropy assumption
- the Weingarten route is independent of the corpus
- the published dimension-ratio matrix element is this registry’s weight formula

The off-diagonal process list is complete

- second-order off-diagonal processes are exactly the adjacent shared-link channels
- the four channel weights follow from dimension and Casimir
- A_N and B_N are the channel sums, not transcriptions

The shared-link law

- $t_N = B_N - A_N$
- $t_N > 0$ for $N \geq 3$
- $t_2 = 0$ and $t_3 = 5/612$
- large- N expansion of t_N through $1/N^9$
- deficit identity $1/4 - N^3 t_N$
- $N^3 t_N$ increases monotonically to $1/4$

The vacuum-mediated route

- $\text{ell_N} = A_N + B_N + 1/C_F$, the vacuum-mediated route at every rank
- FINDING: revision 5 called the charge-even vacuum-route cancellation underived; the pinned corpus proves it for distant pairs at $N = 3$
- the charge-even second-order symbol is ell_N times the unsigned incidence: the vacuum route cancels the doubly-excited route on every link-disjoint pair

The second-order scalar at every rank

- the second-order on-site scalar is closed form at every rank: same-face route + $1/C_F + 12 \text{ell_N}$, and L cancels
- the same-face route is exceptional at $N = 3$ and $N = 4$: $\Lambda^2 F$ is the conjugate fundamental, then self-conjugate
- at $N = 3$ the all-rank second-order ledger is the register and an independent engine's table, line by line
- the two-cube $B=6$ connected diagonal is the per-neighbour ledger face by face: one shared-link neighbour and four disjoint faces in the other cube
- the large- N second-order ledger: the towers and flat scalar are $O(N^{-3})$, and the leading same-face and bubble terms cancel

Finite-volume width

- the zone maximum of q is 12 only at even L
- $q_{\min}$ on the L -torus grid is $4 \sin^2(\pi/L)$

The orientation signs are essential

- the two band spans ARE the two incidence spectra

The second-order coefficient on a two-cube space

- connected first order vanishes by inclusion-exclusion, not by cancellation of numbers
- the connected two-cube geometry has exactly four cross-cell pairs, each -1
- the $B=6$ connected kernel is $(5/612) G_{\text{conn}} + \text{diag}$, with the certified spectrum
- the certificate's own gates pass, target-blind, with the wrong-sign control rejected
- the $B=6$ six-channel census sums to the registry's own $t_3 = 5/612$
- the $B=6$ six-channel census IS the Weingarten four-channel ledger, channel by channel
- every shared-link channel is separately proportional to the geometry, on all 56 pairs
- the link-resolved census split is a theorem at every rank: $-w_{\rho}$ on the self-conjugate mixed channels, $w_{\rho}/2$ on each like channel and its conjugate

One truncation, two constructions

- $B=6$ retains every adjacent shared-link channel; $B=4$ provably cannot
- the $B=6$ six-channel census IS the Weingarten four-channel ledger, channel by channel
- FINDING: the $T1$ link cutoff reverses the sign of t_3 , and $14/153$ is what it omits
- FINDING: a full $T1 = B = 4$ cube Hamiltonian reproduces $-1/12$ and the reversed shell
- FINDING: the $B = 6$ cube flips the sign back to $+5/612$, closing the decisive test
- the $B = 6$ scalar misses the bridge's by exactly the same-face sextet route

What the cutoff moves, and where

- the connected diagonal is orbit-constant, and only the transporting orbit moves

The one-cube shell, and why its flat level cannot move

- the one-cube shell is $A1 + T1 + E$, and its flat level is the S^2 fundamental class
- the certificate's finite-volume fingerprints, and $29 = L^3 + 2$ is the Lean cycle count

The charge-even sector and the unsigned incidence

- the unsigned-incidence characteristic polynomial is $\mu(\mu - p)^2 = 4 a_1 a_2 a_3$
- $p + q = 12$: one zone function runs both sectors
- the Bloch cubic IS the finite $L = 3$ and $L = 4$ plaquette spectrum, exactly
- the C-even range $[-4, 12]$ is exact; the top only at Gamma, the floor on three planes
- the C-even band touches its floor exactly on the three planes $k_j = \pi$
- the C-even band touches its floor exactly on the three planes $k_j = \pi$
- the C-even spectra at the four high-symmetry momenta

The charge-even assembly, and where it is conditional

- one assembly formula gives every registered band value
- one assembly formula gives every C-even value at both orders
- the plaquette graph is 12-regular and two faces share at most one link
- both declared coincidences, checked: one is ell_N at all ranks, the other is bare

Where the rest frame is not blind

- the C-even Gamma point pins t_+ , exactly where no C-odd Gamma datum can
- the C-even curvature is $(4/3)|t_+|$ at both orders, and isotropic
- the C-even bandwidth is $16|t_+|$ at every order; the C-odd manifold width is $12|t_-|$
- at $N = 2$ the C-odd hopping vanishes and the C-even one does not

$SU(3)$ through third order

- $d_3 = 7/32 + 12*\text{leak}_3 - 4*b_3$
- leak_3 is assembled from the domino diagonal and the vacuum piece
- E_{flat} and $t(u)$ carry the ledger coefficients
- $d_- = 1/2 + 12*\text{leak}_2$, and $\text{leak}_2 = -11/306$
- FINDING: no Gamma-point datum can constrain the hopping

What the homology protects

- a boundary-factorised correction shifts the carrier without dispersing it, for generic M
- the carrier projection is where the $1/q$ comes from
- clearing the denominator reproduces the five-element numerator basis

An exact theorem for the saved historical kernel

- v10a.26 A, B, D match the sealed rationals within $2.3e-13$
- FINDING: α_{new} falls outside the corpus's own $2.3e-13$ bound

The obstruction is polynomial

- FINDING: the retained Gamma/axis data cannot identify C_{shp}
- Φ_C vanishes at Gamma along every direction
- the crosswalk is exactly scalar on the momentum axes
- on an axial cut the mixed invariants vanish and the norm divides

A second witness

- FINDING: an explicit second witness C_{alt} exhibits the C2 non-identifiability
- the C_{alt} witness IS the balanced eps-free continuation, and $\Delta_C/(A/2) = 15/32$

Where the coefficient lives

- checkpoint values at X, M, P, R
- the four extraction formulas invert the ansatz
- X is blind to B, C, D — it fixes A alone

- the shape fit’s C row sums to zero, so no Gamma-anchor error can move C_shp at all
- a translation-local scalar shift changes nothing observable
- FINDING: C_shp is carried by 6 of 189 records, not spread across the kernel
- FINDING: $A = 5/48$ pins the normal sector’s whole C4 contribution
- $C_normal = -A_normal/2$: the agreed axial coefficient pins the normal channel
- the two 189-record kernels agree everywhere except three amplitudes, and the on-site anchor swap moves C by exactly zero

The kernel is a Hodge polynomial plus one operator

- the 189 records carry exactly six weight magnitudes
- the shape coefficients are solved, not fitted: $A = 5/48$, $B = D = 0$ exactly
- $C_shp = -5/96 - u - (\rho + \pi)/2$, exactly
- $A = 5/48$ forces the normal amplitude: $nu = -(5/48 + 4u)$
- the 144 agreed records are $u S_sq^2$, the shared-link adjacency squared
- $H4 = -nu (L_up - 2) + u S_sq^2 - \pi S_sq + \sigma - 2 C_shp R$ exactly: C_shp is the coefficient of the one non-Hodge operator
- FINDING: the cold kernel has the same Hodge form, with $nu = -5/48$ and its own (u, ρ, π, σ)
- every orbit is separately Hermitian and cubic-covariant, so symmetry fixes no sign
- no plane-basis convention flips ρ or π : only $+1$ keeps the 144 agreed records
- FINDING: the cold kernel carries the same orientation character, so its flips are real
- the tier collapse is one identity: $D = -6u + 3u^2$ with $u^2 = 2u$
- $u^2 = 2u$ is the cross term of a perfect square

Three amplitudes computed from outside both pipelines

- the engine’s primitives plus textbook second-order theory return $t_3 S_sq$, the C-even hop and the leakage
- FINDING: the chain amplitude is $u = X_QUANTUM$ exactly, on the coplanar and the bent chain; the cold kernel’s $4.13 u$ is wrong
- REPLICATION: a second implementation, with the engine’s PVP assembly and a Krylov resolvent, gives $u = X_QUANTUM$ on three chain types
- the cluster expansion is linked: a plaquette sharing no link with the pair contributes exactly zero
- the lattice fourth-order in-plane nearest-neighbour element, assembled from clusters, is the historical kernel’s π exactly
- the lattice fourth-order normal element, assembled from clusters with the cube-completion term, is the agreed $nu = -5/48 - 4u$ exactly
- the corner cluster’s cumulant agrees between two implementations of the resolvent
- FINDING: the lattice fourth-order rotation element, assembled from clusters, is neither kernel’s ρ

The standing of the records, and a third value

- FINDING: C_shp from the assembled amplitudes is a third value, $-5/96 - u - (\rho + \pi)/2$ with the cluster ρ
- FINDING: $U5$ is falsified – the epsilon sector of $\rho + \pi$ is $-55/6936$, not $-25/512$

What cannot decide it

- FINDING: the marked-cluster engine emits the Gamma scalar only — a completed 609-sweep cannot decide C_shp

The tier collapse, as two integers

- RETRACTED: the vertex count does NOT forbid B_shp and D_shp
- FINDING: the tier collapse is two integer cancellations at record level
- the vanishing coefficients are exactly the degree-3 ones

A near- Γ statement that does not adjudicate

- Jordan bound $q(k) \geq (4/\pi^2)|k|^2$ holds on the whole zone
- the criterion survives C2: K depends on the kernel only through $\sqrt{W_4}$
- the statement is non-vacuous only below an explicit coupling

Rigorous strong-coupling particles and effective operators

- Hamer’s 1+- series matches the C-odd Gamma-point coefficients through x^3
- Hamer’s 0++ series matches the C-even Gamma-point coefficients through x^3
- the $m_n = 2^{(n-1)} a_n$ bridge is the $x = 2u$ conversion
- the Hamer table metadata is pinned; no fourth-order agreement is used
- the KPS 1980 string-tension table equals the certified sigma series EXACTLY
- $\sigma_n^{\text{phys}} = (-1)^n \sigma_n^{\text{raw}}$, and C5 is the $n = 3$ case
- the ratio and sigma series reproduce E_{flat} exactly
- the errata-resolved Euclidean series is doubly sourced, transcription for transcription
- FINDING: Munster’s 1985 table shifts his 1982 erratum at eighth order
- the overlap obstruction was published in 1988, and it scales
- a cross-regime paper never supplies a value

Scope

- $\Delta_L = 4 \tau(u) \sin^2(\pi/L)$ is positive and falls as L^{-2}

Channel weights in closed form

- the four channel weights follow from dimension and Casimir

The Weingarten index sums

- the shared-link weights are Weingarten, not an isotropy assumption
- $SU(3)$ Weingarten values follow from the general formula
- the fourth moment integral $|U_{11}|^4 = 1/6$ at $N = 3$

The $SU(3)$ coefficient ledger

- the manuscript’s $SU(3)$ ledger is this registry, value by value

The charge-even ledger

- one assembly formula gives every C-even value at both orders
- FINDING: the certificate key ‘bandmin’ holds the band MAXIMUM, at both orders

Finite-volume chain calculation

- $d_2 d_3 = 0$ on the built complex
- $\text{rank } d_3 = L^3 - 1$ on the built complex

The two-cube geometry

- the $B=4$ comparator on the same geometry gives $-1/12$ with the reversed spectrum

The second-order ledger at every rank

- the second-order on-site scalar is closed form at every rank: same-face route $+ 1/C_F + 12 \text{ ell}_N$, and L cancels
- the charge-even second-order symbol is ell_N times the unsigned incidence: the vacuum route cancels the doubly-excited route on every link-disjoint pair

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Claim	Status	Evidence in this edition
Signed incidence spectrum and $L^3 + 2$ carrier	Proven	Analytic integer topology; exact replay
Adjacent all-rank coefficient t_N	Proven	Weingarten derivation; channel algebra proof-checked
Global order- u^2 torus assembly	Proven	Retained-shell theorem; process-completeness lemma; explicit projector matrix element
Second-order scalar ledger, all ranks	Proven	Same-face route, bubble count, process census; joined at $N = 3$ to the register and to an independent engine
Charge-even second-order symbol	Proven	Pair-by-pair cancellation of the vacuum route; unsigned incidence; Hamer 0^{++} at x^2
Uniform first electric spectral window	Proven	Spin-network support census; all-rank Casimir shelf; trivial one-form flux
$SU(3)$ nonperturbative finite-volume Riesz island	Proven from cited periodic-boundary version, existential domain	Kinematic Yarotsky enclosure; exact additive spectrum; proved gauge-sector descent
Two-cube six-channel census	Reported numerical	Rational reconstructions; shared amplitude lineage; assembly test
Open-prism radius-two Ritz extension	Reported numerical	Detached replay of pinned parents; no odd Q_3 enclosure
Unsigned-incidence cubic and range	Proven algebraically	Exact characteristic polynomial and finite-grid replay
Charge-even identification at third order	Conditional	Third-order coefficients from the supplied domino ledger
$SU(3)$ factorisation through u^3	Upstream cold-exact	251/251 exact gates reported by the frozen authority; raw replay not bundled here
Historical fourth-order pencil	Exact for saved kernel	Hash-pinned payload and cold fixed-kernel checks
Hodge normal form of the saved kernel	Exact for saved kernel	Six orbit amplitudes; one operator outside the Laplacian algebra, weight C_{shp}
Two-hop, in-plane and normal fourth-order amplitudes	Reproduced from outside	Cluster expansion from the engine's primitives; two implementations
Complete physical fourth-order kernel	Disputed/open	Three recorded values; the rotation amplitude agrees with neither record; the corner cluster decides
Exact G17 reduction	Proven reduction; estimate open	Differentiated inhomogeneous partition function; missing uniform one-plaquette bound
Infinite-volume particle or continuum gap	Open	No thermodynamic Wilson frame, transfer identification, or continuum bridge

Table 1: Truth status and evidence level are separate. “Exact for a saved kernel” does not establish that kernel’s physical identification.